

SAPIENZA
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A Hybrid Mixture-of-Experts Architecture for Sensor-Attack Detection in Water Distribution Networks

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Abstract

Sensor attacks in water distribution networks differ in their temporal patterns, and missing measurements can hinder detection. This dissertation investigates how general and specialized detectors can be combined while controlling false alarms.

The proposed architecture has three main branches: General, Drift and Noise. General combines five internal tree-based components using residuals from a learned reference that excludes the target sensor group. Drift and Noise supply specialized temporal evidence under a declared three-hour decision delay. Routing, feedback and verification regulate specialist acceptance while preserving General's immediate decision path. In L-Town, 42 features derived from actually received pressure history extend General's representation.

Evaluation uses simulated Modena and L-Town scenarios with pressure missing probability fixed at 0.50. The selected detectors use received pressures and their availability masks, and detection scores concern observed pressure endpoints. Flow-based features were examined during development but were not retained. Each selected system is assessed across ten model seeds and six separately generated scenario batches, separating fitting variation from variation in generated data. Mean pooled F1 reaches 0.8669 in Modena and 0.8950 in L-Town; family macro F1 reaches 0.9045 and 0.9083. Clean-period false-positive rates are 0.0584% and 0.0550%, respectively. All five Modena family means exceed 0.80. Four L-Town family means exceed 0.89, while replay remains at 0.7312.

A separate three-seed factorial with downstream refitting finds higher pooled F1 with iterative reweighting in both networks, but no consistent benefit from target-group exclusion.

The matched integration change substantially reduces false positives in Modena, while adding received history improves the family means in L-Town. At the original operating points, the hybrid provides stronger replay, drift and noise coverage than the three-hour single-classifier control calibrated for pooled F1, at the cost of more clean false alarms. Supplementary threshold comparisons show that these gains depend on the calibration objective and false-positive allowance. The simpler control remains competitive, particularly under strict false-alarm constraints.

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Chapter 1

Introduction

1.1 The architectural problem

A change in measured pressure may come from demand variation, measurement error or deliberate manipulation. Distinguishing these causes requires information about the surrounding network and recent observations. Missing observations make this distinction harder: the detector must reason from an incomplete view of the network, and the absence of supporting measurements can weaken an otherwise useful comparison. A detector that treats every deviation in the same way may therefore miss a persistent disturbance or raise an alarm on a legitimate change.

This dissertation studies the architecture of a sensor-attack detector under these conditions. The central question is how to organize general anomaly evidence and specialized temporal evidence so that the complete system handles several attack mechanisms without an excessive false-alarm cost. The work develops from spatial and temporal graph models to a hybrid expert architecture and evaluates the selected systems on two water distribution networks, Modena and L-Town.

A specialist can detect its assigned family well yet add little beyond General's existing alarms, while introducing false positives elsewhere. Its contribution therefore has to be measured in the complete system. The evidence supplied to each component, its training population and the rule for accepting its output all affect that contribution.

The BATADAL competition assessed cyber-attack detection on simulated SCADA measurements using explicit classification and detection-time criteria [1]. The present work uses a separate simulator-based benchmark to investigate sensor-level decisions under declared observation and attack conditions.

1.2 From model complexity to complementary evidence

The initial approach used graph neighbourhoods and temporal recurrence to exploit relationships among measurements. GraphSAGE provided a way to aggregate node information, while a gated recurrent unit provided a temporal representation. A mixture with six spatio-temporal experts then introduced components associated with clean operation and five attack families. The design assumed that different mechanisms might benefit from different predictors and that a router could learn to combine them.

Specialization does not necessarily improve detection. The general model may already cover the obvious cases, and patterns learned by a specialist may also occur in other attack families. Even a router that identifies the family correctly can select a weaker detector. Controlled ablations and inspection of expert outputs were used to examine these possibilities.

Benchmark conditions also affect detection: an attack that produces a large displacement relative to ordinary variation differs from one near the noise scale. A persistent episode supplies different temporal information from attacks resampled independently at each snapshot. Changes in such conditions can alter the apparent contribution of a graph layer, a recurrent state or an expert mixture. The research therefore separates changes in the evaluation problem from changes in the detector.

The resulting hybrid design has three main branches: General, Drift and Noise. General supplies broad residual evidence, while the other two branches use temporal representations suited to gradual change and excess variability. General contains five internal tree-based components; these are not five additional top-level experts. Routing, feedback and verification regulate the use of specialized outputs, while the general path remains available. Drift and Noise decisions use a declared three-hour delay. The earlier six-expert graph architecture and the final three-branch hybrid are different stages of the research, with different roles and evaluation records [E1].

1.3 Why two networks are necessary

Modena is the principal development and evaluation network, where many architectural choices were first investigated. L-Town extends the assessment to a larger network with different hydraulic characteristics, testing the method beyond its original development setting.

Both networks form the final evaluation, with ten model seeds and six generator sources each. A generator source is a batch of scenarios produced for one network with a specified generator seed; its integer identifier is that seed. Model seeds instead

control the stochastic fit. The evaluation distinguishes these two forms of variation rather than treating repeated fits on a shared source as independent datasets [E2].

Both implementations use the General/Drift/Noise organization. In L-Town, General also uses features derived from received pressure history. Models and operating points are fitted separately for each network. The comparison is thus an evaluation of the architectural framework and its declared network-specific implementation, not zero-shot transfer of one fitted model or a test using identical feature sets.

1.4 Research questions

The following questions guide the presentation of the completed research. The chronology of the individual frozen protocols is retained in the experiment records.

1.4.1 RQ1: organization of expert evidence

How should general anomaly detection and specialized temporal evidence be organized within a hybrid expert architecture to address different attack mechanisms?

Historical ablations and expert diagnostics connect component roles to the evidence they contribute.

1.4.2 RQ2: integration and operational trade-offs

What trade-offs arise from routing, feedback and verification when combining expert outputs, in terms of attack-family detection, pooled performance, clean-period false alarms and declared decision delay?

The complete alarm path is assessed against family coverage, background errors and the time needed to obtain evidence.

1.4.3 RQ3: consistency across the two networks

How consistently does the selected hybrid architecture perform across attack families, model seeds and evaluation sources in Modena and L-Town, and what limitations emerge in the larger network?

The final evaluation examines all five families, aggregate errors and variation across fits and sources.

1.5 Contributions and boundaries

Controlled studies of topology, recurrence and mixtures informed the representation and integration choices. Rejected alternatives are part of the experimental record and help explain how the architecture developed.

The resulting hybrid detector combines a general residual mixture with specialized Drift and Noise evidence, using a guarded decision process. The method specifies both the information that can produce an alarm and the time at which it becomes available.

A repeated-fit evaluation on Modena and L-Town reports family and aggregate performance, together with variation across sources and all ten fits per network [E2]. Supplementary single-classifier controls at zero and three hours assess the performance gained from the compound detector and its associated trade-offs [E9].

The evaluation uses simulated pressure observations and their availability masks, with pressure missingness fixed at 0.50. Decisions concern observed pressure endpoints. Flow-based development candidates and the generator's shared family bookkeeping are described in Chapters 3 and 4. The specialist delay remains part of the task. Field deployment and control-system recovery require separate evidence.

Replay exposed weaknesses in representation and fusion and provides a focused case study of architectural refinement. The L-Town history design is assessed alongside the other families and the full system's false-alarm behaviour within the two-network evaluation.

1.6 Organization of the dissertation

Chapter 2 provides the conceptual background, and Chapter 3 defines the task and benchmark. Chapter 4 explains the design decisions, followed by the formal architecture in Chapter 5. Chapter 6 presents the experimental protocol and results. Chapter 7 relates the findings to the research questions and discusses their limitations; Chapter 8 concludes the dissertation. Appendix A contains the evidence register, supporting comparisons and reproducibility records. Appendix B reports supplementary operating-point and component results, while Appendix C presents the reference-factorial results.

Chapter 2

Background and Related Work

2.1 Water-network monitoring as an inference problem

A water distribution model describes the connections between hydraulic components. The monitoring system provides measurements from that network. The graph specifies where components are connected; an observation mask specifies which readings are available at a particular time. A dense physical network can still be poorly observed. Conversely, many simultaneous readings need not provide independent information when they reflect common hydraulic variation.

In this project, pressure observations are interpreted relative to other measurements and to their own temporal context. The detector is not given the clean simulator state as a prediction input. It must instead distinguish normal variation, measurement noise and deliberate corruption from quantities that could actually have been received. Simulator truth can be used to evaluate predictions, but it is unavailable to the detector.

WNTR is an open-source Python framework for analysing the resilience of water distribution systems, developed by Sandia in collaboration with the US Environmental Protection Agency [2]. Klise et al. describe its simulation and resilience-analysis framework through an earthquake case study [3]. EPANET provides extended-period hydraulic simulation, including node pressures and link flows [4]. The project uses a hydraulic simulation workflow to generate the states from which observations and attack scenarios are constructed. The simulator supplies repeatable scenarios and known labels under the observation and attack assumptions specified in Chapter 3.

Measurement attacks alter the observation stream after hydraulic simulation. They can therefore change what the detector sees while leaving the underlying physical trajectory unchanged. This differs from the epanetCPA toolbox, which

simulates the hydraulic response to cyber-physical attacks [5]. The present generator applies its own observation corruptions and does not use epanetCPA. This distinction determines the reading-level task formulated in the next chapter.

2.2 Relevant water-network benchmarks

BATADAL provides a precedent for explicit cyber-attack detection evaluation on a water network. Its C-Town challenge used simulated SCADA observations and compared several detection approaches, including data-driven, model-based and rule-checking methods [1]. It is relevant here as an example of defining the task and its evaluation criteria together. Its dataset, attacks and reported scores are not used as the numerical baseline for this dissertation.

Physical testbeds provide another basis for security studies. SWaT is a water-treatment testbed introduced by Mathur and Tippenhauer [6]; Goh et al. document a dataset collected from it during normal operation and attacks, containing both process measurements and network traffic [7]. WADI extends this experimental setting to water distribution [8]. These testbeds distinguish the physical installation from the particular data collection and attack protocol used in a detection study.

BattLeDIM addresses a different problem: leakage detection and isolation. The competition introduced a virtual water distribution setting with pressure and flow measurements, and compared submitted methods under its own evaluation design [9]. L-Town supplies the network model used in the present project, but the cyber-attack scenarios are generated separately. Chapter 3 defines the project's observation process and attack task.

Modena has a separate origin in water-network design optimisation. The supplied network input file credits Bragalli, D'Ambrosio, Lee, Lodi and Toth (2008). Their later journal account, published in 2012, describes the network within a mixed-integer nonlinear programming approach to network design [10]. Network properties reported in this thesis are taken from the supplied file.

The same network can support several different learning problems. A leakage event changes the hydraulic process. A forged reading changes what the detector sees. Their effects may overlap in some measurements, but their labels and decision targets are not interchangeable. Benchmark provenance therefore requires more than naming a network: it includes the simulator configuration, observation process, attack construction, sampling interval and evaluated population.

2.2.1 Recent detection studies in water systems

Earlier BATADAL studies illustrate two ways to construct a reference for attack detection. Taormina and Galelli use autoencoders to detect and localize attacks through inconsistencies in sensor readings [11]. Housh and Ohar combine a detailed hydraulic model with anomaly detection to identify attacks that hydraulic rules alone may miss [12]. Both are evaluated on BATADAL, but the reference comes from learned measurement relationships in the first approach and hydraulic modelling in the second. The present thesis learns a pressure reference and supplies its residuals to supervised classifiers.

Chandy et al. use a deep generative model with variational inference to learn normal behaviour from water-system observations and identify attacks through reproduction probabilities [13]. They also report a need to reduce false alarms through better threshold selection. This separates two design questions that recur in the present work: how to represent normal behaviour and how to convert departures from it into alarms.

Moazeni and Khazaei study supervised neural detection of false data injected into tank-level measurements, including sensitivity to measurement noise [14]. Their formulation connects detector design to the measured hydraulic quantity and the corruption process. The present benchmark extends this concern to five pressure-corruption families under fixed missingness, where a shared detector must preserve coverage across mechanisms.

Deep H2O, by Sikder et al., develops an attention-based temporal graph convolutional model and a high-confidence autoencoder for cyber-attack detection on BATADAL data [15]. Its forecasting path combines prediction errors with a robust Mahalanobis-distance rule, while the autoencoder supplies a reconstruction-based alternative. The present thesis also separates the representation of discrepancies from the alarm rule, using a broad residual path and selectively admitted temporal specialists.

These studies differ in their measured variables, supervision and alarm definitions. Comparing their detection scores with this thesis would require evaluating the methods under the same protocol.

2.3 Spatial representations and graph aggregation

GraphSAGE constructs node representations through learned aggregation of neighbouring node features rather than assigning an independent embedding parameter to every node [16]. A generic layer computes the neighbourhood aggregate in

Equation (2.1) and the node update in Equation (2.2):

$$m_i^{(\ell)} = \mathrm{AGG}_\ell\{h_j^{(\ell-1)} : j \in \mathcal{N}(i)\}, \tag{2.1}$$

$$h_i^{(\ell)} = \sigma\left(W_\ell[h_i^{(\ell-1)} \| m_i^{(\ell)}]\right), \tag{2.2}$$

where $\mathcal{N}(i)$ denotes a neighbourhood and $\|$ denotes concatenation. This aggregation principle underlies the historical graph models.

For the earlier models in this project, the graph provided a way to include neighbouring sensor information before predicting a reading or its anomaly score. The relevant empirical question was whether those messages carried useful information beyond what a node could learn from its own features. The topology-free control therefore retained the model's temporal machinery while disabling message passing. Keeping the temporal path fixed makes the contribution of graph messages easier to assess than a comparison in which the inputs, training conditions and parameter budgets also change.

A matched topology ablation tests the predictive value of graph messages. Hydraulic relationships also depend on pumps, valves, demand and operating conditions, so the learned representation combines adjacency with the observed variation in the training data.

Missing readings need an explicit representation: an unavailable value cannot safely be treated as an ordinary zero pressure. The model must know whether the input is present, or a numerical fill value may acquire a false physical meaning. In the historical neural models, observation masks accompany the numerical inputs. In the final reference-based system, visibility controls which measurements enter a prediction. Both approaches acknowledge missingness, but they enforce the information constraint differently.

Water-specific graph models also address incomplete monitoring. Xing and Sela use graph neural networks to estimate hydraulic states at unmonitored locations from limited measurements, considering supervised and semisupervised learning [17]. Truong et al. reconstruct pressures from sparse observations and train with varying sensor locations to accommodate changes in monitoring coverage [18]. These are state-estimation tasks. Attack detection also requires deciding whether an available measurement can be trusted, which motivates the exclusion and residual checks used here.

Deng and Hooi's Graph Deviation Network (GDN) learns sensor relationships and uses graph attention to forecast observations from historical windows [19]. Its SWaT and WADI experiments connect graph learning directly to attack detection in water-system testbeds. GDN learns the dependency graph from data; the historical

GraphSAGE controls here examine supplied network topology. The common question is whether relationships among sensors improve the reference used to judge a measurement.

2.4 Temporal recurrence and observation windows

A temporal model can use how observations change, not only their current values. The gated recurrent unit introduced by Cho et al. uses gates to regulate the contribution of past state and new input [20]. One common convention defines the reset and update gates in Equations (2.3) and (2.4), the candidate state in Equation (2.5), and the updated state in Equation (2.6):

$$r_t = \sigma(W_r x_t + U_r h_{t-1}), \tag{2.3}$$

$$z_t = \sigma(W_z x_t + U_z h_{t-1}), \tag{2.4}$$

$$\widetilde{h}_t = \tanh(W_h x_t + U_h(r_t \odot h_{t-1})), \tag{2.5}$$

$$h_t = z_t \odot h_{t-1} + (1 - z_t) \odot \widetilde{h}_t. \tag{2.6}$$

Biases are omitted for readability. The equations explain the recurrent mechanism; implementation conventions may exchange the roles of z_t and $1 - z_t$.

In the historical architecture, spatial representations were processed over a six-step observation window. This input horizon is separate from the six predictive components in the expert mixture.

Recurrence can exploit both episode persistence and ordinary short-range correlation. The episode and temporally permuted controls in Chapter 4 examine these contributions to predictive performance. These controls help determine how much of a temporal gain comes from persistent attacks and how much comes from other correlations.

Window construction also affects which endpoints can be evaluated. A six-step model needs a history that a one-step model does not. If each is evaluated on its own available endpoints, a performance difference may partly reflect different populations. The controlled historical studies used the common endpoint range and recalibrated the relevant operating points on the corresponding validation subset.

A long history and a delayed decision should likewise be distinguished. A model using only observations at or before time t can be causal at t even if its history is long. A model allowed to use observations after t to decide about the endpoint at t has a later decision time. The final system makes that allowance explicit for Drift and Noise. The observation horizon and the decision time must both be specified.

TranAD uses transformer encoders, reconstruction-based self-conditioning and adversarial training for multivariate anomaly detection, with experiments including SWaT and WADI [21]. It offers an alternative to recurrence for representing temporal context. In this thesis, received-history comparisons and delayed specialist windows specify which observations are available. Comparing such approaches requires matching the observation horizon and decision time, as well as the model's capacity to represent a sequence.

2.5 Mixtures, expert roles and routing

Adaptive mixtures of local experts combine component predictions through an input-dependent gating mechanism [22]. For binary component scores $s_k(x)$ and non-negative weights $w_k(x)$, a soft mixture has the form

$$s(x) = \sum_{k=1}^{K} w_k(x) s_k(x), \qquad \sum_{k=1}^{K} w_k(x) = 1. \tag{2.7}$$

The decomposition in Equation (2.7) allows different predictors to contribute in different regions of the input space; component diagnostics establish where specialization is useful.

Jordan and Jacobs extend learned gating to a tree-structured hierarchy of experts and derive an expectation–maximization training procedure [23]. Their formulation establishes hierarchical expert combination; the mixtures examined here use their own training and calibration procedures.

Table 2.1 distinguishes the roles of specialization. An expert produces a score, and an internal mixture weights those scores. A separate system-level rule decides which specialist alarms to accept. General uses the first form of integration, a soft mixture; the Drift and Noise alarms pass through the second, a guarded combination.

Hard top-1 routing retains only the score associated with the largest routing weight. That operation may remove useful evidence present in the other components.

Table 2.1. Different meanings of specialization in this dissertation. These are architectural roles, not claims that each component is optimal for its assigned family.

Level	Predictive role	Integration role
Historical neural MoE	Six spatio-temporal experts	Internal soft mixture or routing diagnostic
Final General branch	Five residual tree components	Internal weighted combination
Final full system	General, Drift and Noise branches	Guarded use of specialized alarms

An unconditional logical union has the opposite problem: it accepts every component alarm and can accumulate false positives. Neither rule can be judged solely by an expert's isolated F1. Their effect depends on the joint pattern of component predictions over the evaluated population.

Shazeer et al. use input-dependent sparse gating to select a small subset of experts in large neural networks [24]. This makes selective routing an established alternative to evaluating a dense mixture. Here, the routing comparisons concern detection errors and component coverage within a much smaller system.

A simple set view clarifies the issue. Let A_G be the endpoints already alarmed by General and A_D those alarmed by a Drift specialist. Only $A_D \setminus A_G$ changes the union. Within that set, some endpoints are new true positives and others are new false positives. If all Drift true positives already lie in A_G, the union cannot improve recall, even if the Drift component has a high standalone score. The relevant contribution is therefore the detections and errors added outside General's existing alarm set.

The same reasoning applies to feedback. Suppressing a specialized alarm can help when it removes a false positive but harm when it removes a true positive not covered by General. General retains its own alarms when feedback suppresses a specialist. The resulting benefit depends on the balance of removed false positives and lost detections beyond General's coverage.

Huang et al.'s Graph-MoE uses experts to combine representations from different GNN layers and memory-augmented routers to weight them using historical context [25]. Its evaluation includes SWaT and WADI. Graph-MoE combines graph representations at different depths; the final hybrid here combines residual classifiers within General and controls the alarms added by mechanism-oriented Drift and Noise branches. The thesis examines this organization and its measured trade-offs.

2.6 Residual references and prediction independence

A residual detector compares an observed value with an estimate of what is expected. Write

$$e_{i,t} = y_{i,t} - \widehat{p}_{i,t}, \qquad u_{i,t} = e_{i,t}/\sigma_i, \tag{2.8}$$

In Equation (2.8), σ_i is a scale estimated from training observations. In this project, neither $\widehat{p}_{i,t}$ nor σ_i is the simulator's clean test value. They are learned reference quantities.

The reference should not be able to reproduce a suspicious reading merely by seeing that same reading as an input to its own prediction. Noise2Self formalizes a

related prediction-independence idea for denoising, using functions whose predictions for a partition do not depend on noisy measurements within that partition [26]. Its assumptions concern the noise and signal structure. The thesis uses target-group exclusion as a design principle, not the denoising theorem as an attack-detection guarantee.

The project's normal reference partitions sensors into fixed groups. When predicting a target group at an endpoint, it uses observed sensors outside that group. Training observations determine a low-rank representation and a robust scale; the robust variant reduces the influence of measurements that disagree strongly with the fitted reference. Target-group exclusion prevents direct copying; corrupted supporting sensors can still distort the estimate.

The interpretation of a large residual may depend on how many usable predictors support the reference. A small residual can also occur during an attack if the attacked observation remains close to the expected value. Availability, support and temporal consistency are useful complements to residual magnitude. They let the classifier learn distinctions that a single absolute-error threshold cannot express.

The hybrid separates reference construction from classification. This exposes the expected value and its supporting evidence for inspection, while tree models learn how to combine the resulting discrepancies and reliability features. Chapter 4 relates this representation choice to the observed expert errors and the controlled comparisons.

2.7 Tree models for structured evidence

Gradient-boosted trees offer a flexible way to combine heterogeneous numerical features. LightGBM is a boosting framework designed to improve the efficiency of tree learning through techniques including gradient-based one-side sampling and exclusive feature bundling [27]. The saved estimator configuration specifies the techniques enabled in this implementation.

The hybrid uses tree models on residual and temporal features because the experiments shifted toward explicitly represented evidence: current deviations, reference support, local variability and seasonal comparisons. Such features have different scales and interactions. A classifier can learn, for example, that a deviation should be interpreted differently under weak support or that a persistent directional change is more informative than an isolated spike.

With the representation fixed, subsequent comparisons can examine component scores, integration rules and operating points separately.

Additional features can repeat existing information or shift errors between families. Candidate representations therefore require both a detection benefit and protection of the wider system. Chapter 4 applies this criterion to the history experiments.

2.8 Evaluation as part of architectural design

Under class imbalance, the false positives associated with a detector must be considered alongside its true positives. Saito and Rehmsmeier explain why precision–recall views are important when interpreting binary classification under an excess of negatives [28]. The present thesis therefore reports thresholded F1 together with pooled performance and clean-period false alarms. A ranking metric and an operating-point metric answer different questions.

For a fixed evaluated population, Equation (2.9) defines precision and recall:

$$P = \frac{TP}{TP + FP}, \qquad R = \frac{TP}{TP + FN}, \tag{2.9}$$

and their harmonic mean is

$$F_1 = \frac{2TP}{2TP + FP + FN}. \tag{2.10}$$

Each metric also requires a defined population over which TP, FP and FN are counted. The operational family-conditioned score counts endpoints within that family's periods; pooled F1 includes the whole evaluated corpus. Clean FPR concerns periods labelled clean. Chapter 3 makes these scopes explicit.

Model selection creates a second concern. Cawley and Talbot show that optimizing a finite-sample selection criterion can itself overfit and bias subsequent performance evaluation [29]. In this project, that concern motivates separating training diagnostics, calibration and frozen confirmation. A completed calibration search is not an additional test, and a source used to evaluate one design cannot later be described as unseen evidence for a revised design selected using its results.

Scoring a baseline and a candidate on the same source controls some source-specific variation in their difference. Comparing a candidate on a new source with an older baseline mean does not have that property. Likewise, multiple model seeds on one generated source do not reproduce the variation obtained by generating new sources. The final campaign retains both levels, while its extension adds model fits on the same evaluation sources.

Kim et al. show that point adjustment can inflate time-series anomaly scores and alter method rankings [30]. Under that convention, detecting one point in a

labelled anomalous interval can count the whole interval as detected. This differs from scoring each observed sensor endpoint. A fair comparison must therefore align label granularity and scoring rules, in addition to inputs and calibration. High reported F1 values alone cannot resolve those differences.

2.9 Position of this dissertation

The recent literature establishes several ways to exploit sensor relationships, temporal context and expert integration. Table 2.2 relates these approaches to the present design. The thesis concentrates on their interaction at the final alarm: whether specialized evidence adds useful detections under incomplete observations, what false positives it introduces, and how the resulting balance differs across Modena and L-Town. Matched internal comparisons support these architectural findings; external-method superiority would require a separate comparison under the same protocol.

Table 2.2. Selected post-2020 studies and their relationship to the thesis. This is a comparison of method and evaluation scope; published performance values are not ranked across datasets.

Study	Main approach	Connection to the present work
Moazeni and Khazaei (2022) [14]	Supervised detection of injected tank-level readings	Hydraulic measurement and noise assumptions define the task
Deep H2O (2023) [15]	Temporal graph forecasting and autoencoding on BATADAL	Representations require a calibrated alarm rule
GDN (2021) [19]	Learned sensor graph and attention-based forecasting	Cross-sensor relationships inform expected readings
TranAD (2022) [21]	Transformer-based anomaly detection	Temporal representation and available context are distinct choices
Graph-MoE (2025) [25]	Experts across GNN depths with memory-based routing	Expert integration is established; component roles differ here
Kim et al. (2022) [30]	Analysis of point-adjusted evaluation	Match scoring and decision units before comparing F1

Chapter 3

Problem Formulation and Benchmark Design

3.1 The prediction target

The final detector assigns an attack decision to an observed pressure reading at a sensor and time. Inputs and errors are defined at this sensor-time level.

Let $\mathcal{G} = (V, E)$ denote the network model. At time t, the simulator supplies a pressure vector p_t and a flow vector q_t. For each channel, the observation process produces a binary availability mask. The mask takes value $m^p_{i,t} = 1$ when a pressure is received and $m^p_{i,t} = 0$ otherwise; $m^q_{e,t}$ has the analogous meaning for flow. The symbol $y^p_{i,t}$ refers to a received pressure, and its numerical value is meaningful to the detector only when the corresponding mask is one.

A binary label $a_{i,t}$ records whether the pressure observation at (i, t) was actually modified by the attack process. It is not inferred from how far the value lies from the clean simulator pressure. Large natural deviations retain negative labels, while a small injected change can carry a positive label. Numerical no-ops are not labelled positive. Missing observations also do not receive positive observed-endpoint labels in the generator [E3].

The complete evaluated endpoint set is restricted by observation availability and feature warm-up. If $\mathcal{I}$ denotes the set retained by the feature construction, then

$$\mathcal{I} \subseteq \{(s, i, t) : m^p_{s,i,t} = 1\}, \tag{3.1}$$

In Equation (3.1), s identifies a scenario. Scores over $\mathcal{I}$ therefore describe available pressure endpoints, rather than all physical nodes at every hour. A missed observation

is a condition of the experiment, not a false negative produced by the classifier. Table 3.1 collects the main notation.

3.2 Modena and L-Town

The study uses two supplied EPANET network models. Their component counts in Table 3.2 were obtained directly from the input files preserved with the project. The node total includes junctions and boundary/storage nodes. Modena's 272 nodes comprise 268 junctions and four reservoirs.

Figure 3.1 shows the two network layouts. Modena provides the principal development setting; L-Town extends the assessment to a larger network with different hydraulic characteristics. Each network has its own fitted reference, classifiers and calibrated operating points.

L-Town originates from the BattLeDIM network setting [9]. This project defines its own observation and corruption process on that network model.

Network size is not isolated experimentally: topology, hydraulics and measurement distributions also differ. The comparison therefore characterizes performance in the two settings.

Generator sources and scenarios. A *generator source* is one execution of the scenario generator for a specified network and generator seed. Its integer identifier is that seed. Each source produces a batch of scenarios sharing the network model, sensor ordering and generator settings. A source is distinct from a sensor and from a model seed, which controls a repeated model fit.

The generator reloads the network for each scenario and draws a new set of junction-demand multipliers. The demand cycle and the parameter ranges remain fixed. Observation masks and measurement noise vary by snapshot; attack targets

Table 3.1. Main notation and its experimental meaning. Availability, attack labels and mechanism identity are different quantities.

Symbol	Meaning
$p_{i,t}, q_{e,t}$	Simulated physical pressure and flow; not detector inputs as clean truth
$m^p_{i,t}, m^q_{e,t}$	Observation availability indicators
$y^p_{i,t}, y^q_{e,t}$	Received values after observation noise and any eligible corruption
$a_{i,t}$	Binary pressure-corruption label for an observed endpoint
$c_{s,t}$	Stored family code for evaluation and supervised training bookkeeping
$\widehat{p}_{i,t}$	Prediction from the fitted normal reference
$\mathcal{I}$	Retained observed pressure endpoints after feature construction
Δ	Declared additional decision delay for specialized evidence

Table 3.2. Network components counted from the project's input files. Counts describe the network models; observation availability is defined separately.

Component	Modena	L-Town
Junctions	268	782
Reservoirs	4	2
Tanks	0	1
Total nodes	272	785
Pipes	317	905
Pumps	0	1
Valves	0	3
Total links	317	909

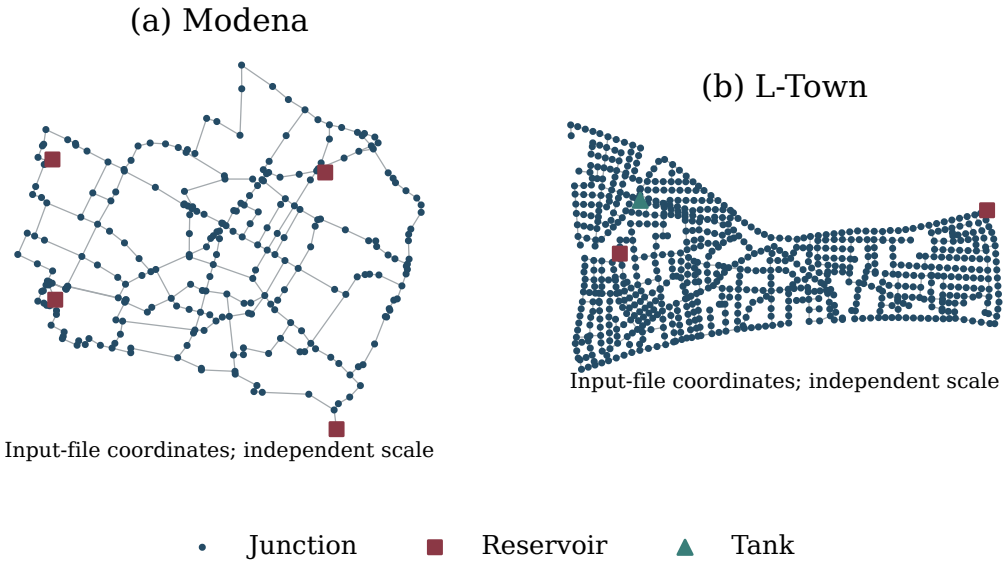


Figure 3.1. Network layouts from the coordinates and connectivity in the preserved input files. The panels have independent coordinate scales. Grey lines show links; blue points show junctions; larger markers identify reservoirs and tanks. The plots illustrate structural differences and do not represent the final observation mask.

and perturbation parameters vary by episode. Within a source, these draws follow continuing random-number streams rather than newly seeded streams for each scenario. The shuffled attack-family queue also continues across scenario boundaries. The observation/noise, attack-schedule and attack-perturbation streams use the generator seed and two fixed seed offsets, respectively [E3].

3.3 Observation process and fixed operational settings

The final operational configuration generates 168-hour scenarios at a 60-minute sampling interval. Missing probabilities are fixed at 0.50 for pressure and 0.50 for flow. The separate random streams described above allow the attack schedule to change without redefining the base observation process [E3].

For an available pressure before attack, Equation (3.2) gives the observation model:

$$y_{i,t}^{p,0} = p_{i,t} + \epsilon_{i,t}^{p}, \qquad \epsilon_{i,t}^{p} \sim \mathcal{N}(0, \sigma_p^2), \tag{3.2}$$

with $\sigma_p = 0.1$ m. Flow uses the analogous process with $\sigma_q = 10^{-4}$ m^3/s. Superscript 0 distinguishes the pre-tampering observation stream from the final received stream. It does not indicate clean hydraulic truth: observation noise is already present.

Independent missingness frequently leaves the detector with only a subset of supporting readings. Correlated communication failures and extended outages require a different observation model. The realized missing fraction in a finite scenario can differ from exactly one half. Attack eligibility can further reduce the number of observations that are modified.

Table 3.3 lists the fixed generator settings. The nominal target fraction determines how many locations are selected for an episode. The positive-label rate in the final corpus also depends on which of their observations can be modified. Clean gaps, missing observations, episode truncation and family-specific eligibility all affect the observed prevalence. For example, a replay modification requires both a current reading and an available historical reading. Reporting a five-percent target fraction as five-percent positive endpoints would therefore misstate the experiment.

The generator creates corrupted pressure and flow observations, but the selected detector inputs and endpoint labels concern pressure. Received flow values and flow masks enter only the development candidates described in Section 4.9. Flow missingness remains fixed at 0.50 in the preserved benchmark. The stored family code is shared by a network snapshot: it is nonclean when either a pressure or a flow observation is modified. A flow-only modification can therefore place otherwise negative pressure endpoints in an attack-family period. This convention affects the clean-period subset and the normal snapshots used to fit the pressure reference; it does not make those pressure endpoints positive. Within each paired comparison, both systems retain the same stored codes, pressure labels and endpoint population [E3].

Table 3.3. Fixed operational generator settings shared by the selected Modena and L-Town campaigns. Values describe the synthetic experiment, not parameters estimated from observed malicious incidents.

Quantity	Setting
Scenario duration; reporting interval	168 h; 60 min
Pressure / flow missing probability	0.50 / 0.50
Pressure measurement-noise standard deviation	0.1 m
Flow measurement-noise standard deviation	10^{-4} m^3/s
Nominal target fraction	0.05 of channel locations
Requested attack duration	6–18 h
Clean gap between scheduled episodes	48–96 h
Pressure bias magnitude	0.5–2.0 m
Flow bias magnitude	5×10^{-4}–2×10^{-3} m^3/s
Drift ramp duration	6–18 h
Replay lag	2–6 h
Added attack-noise factor	3–8 times channel noise scale
Demand variation; pattern amplitude	0.2; 1.0

Pressure and flow also consume shared observation and perturbation random-number streams within the generator, so removing the flow-generation steps and regenerating the data would not necessarily preserve the pressure realizations.

3.4 Five attack families

The operational generator schedules persistent episodes rather than independently choosing a new family at every snapshot. The family queue cycles through a shuffled list, and each episode retains its selected targets. The requested duration is bounded by the remaining scenario length. Attack counts therefore use the event ledger, which records any episode shortened by the scenario boundary.

Table 3.4 summarizes the transformations. The equations below describe pressure observations; the generator applies analogous channel-specific transformations to flow. The historical implementation label `stealthy` corresponds to the gradually increasing bias called *drift* in the final reporting.

3.4.1 Abrupt and targeted corruption

For a selected sensor with a current observation, the random family applies the signed additive offset in Equation (3.3):

$$y^{p}_{i,t} = y^{p,0}_{i,t} + b_i, \qquad |b_i| \in [0.5, 2.0] \text{ m}. \tag{3.3}$$

Table 3.4. Operational attack mechanisms and the information relevant to their detection. These descriptions concern the final generator; historical regimes used different corruption settings.

Family	Observation transformation	Potentially useful evidence
Random	Persistent signed additive bias at selected locations	Current deviation and local/reference consistency
Replay	Substitution from an available historical pre-tampering observation	Cross-time and cross-sensor consistency
Drift	Signed bias rising toward its specified magnitude	Directional persistence and temporal comparison
Noise	Additional zero-mean random disturbance	Local variability and changes in residual dispersion
Targeted	Additive bias at locations selected using a degree-based priority	General deviation with a different target distribution

The sign and magnitude are chosen for the episode. The word *random* refers to the sampled corruption and targets, not to an independent bias being redrawn every hour. This distinction affects which temporal signatures are available.

Targeted corruption uses the same additive form but changes the target-selection pool. In the code, pressure locations are prioritized by graph degree, and link priority is based on endpoint degrees. This degree-based heuristic defines the targeted family; it measures connectivity rather than hydraulic damage.

Contemporaneous residual discrepancies can reveal both additive families. Their scores also test whether refinements for temporal mechanisms preserve the detector's broad coverage.

3.4.2 Gradual drift

For drift, the episode bias is scaled with age. If r is the sampled ramp duration in steps and k is the zero-based age of an endpoint within the episode, then

$$y_{i,t}^{p} = y_{i,t}^{p,0} + b_i \min\left(\frac{k+1}{r}, 1\right). \tag{3.4}$$

The perturbation in Equation (3.4) can therefore begin below its eventual magnitude. An instantaneous residual threshold may detect the late portion and miss the early portion. Temporal evidence can help distinguish a sequence of small, similarly directed discrepancies from an isolated deviation, but its benefit must be assessed against the time required to observe that sequence.

Attack and ramp durations are sampled separately, so an episode can end before reaching the full bias. The event record retains the applied trajectory.

3.4.3 Excess noise

For the noise family, the generator adds a sampled perturbation whose standard deviation is a factor $\alpha \in [3, 8]$ times the ordinary channel noise scale:

$$y_{i,t}^{p} = y_{i,t}^{p,0} + \eta_{i,t}, \qquad \eta_{i,t} \sim \mathcal{N}(0, (\alpha\sigma_p)^2). \tag{3.5}$$

Equation (3.5) describes an additional disturbance on top of the base observation noise. The factor describes the injected component, not the total resulting standard deviation. Some individual noise draws will be small even in an episode with a large factor, which makes a sequence's variability relevant.

The detector must also avoid confusing normal reference uncertainty with attack noise. Missing support may produce unstable residuals without an injected perturbation. Availability and reference-quality features help the detector distinguish this uncertainty from attack noise.

3.4.4 Replay and its information assumptions

Equation (3.6) replaces an eligible current value with the corresponding value from the historical pre-tampering observation stream:

$$y_{i,t}^{p} = y_{i,t-\ell}^{p,0}, \qquad \ell \in \{2, \ldots, 6\}, \tag{3.6}$$

and requires $m_{i,t}^{p} = m_{i,t-\ell}^{p} = 1$. The historical reading is already noisy and masked. It is not the simulator's clean hydraulic value. The generator keeps its historical base records immutable when applying an attack.

The attacker draws replay values from the pre-tampering stream, whereas the detector receives the final observation stream. If an earlier received observation was itself corrupted, the detector's history contains that corrupted value. It is not allowed to recover the generator's unmodified base history as a feature. This distinction becomes particularly important when interpreting received-history features in the L-Town refinement.

A replayed pressure can remain within the normal range when legitimate pressure changes slowly. Detection may require disagreement with current cross-sensor conditions or temporal relationships. Replay consequently provided a useful diagnostic of the evidence available to the architecture.

3.5 Benchmark development and comparability

The project contains several benchmark generations. Earlier snapshot experiments examined graph and temporal models under corruption sampled at individual times. Subsequent work examined coherent episodes and matched controls. The operational generator then made physical units, persistent targets, missingness and event provenance explicit. Table 3.5 records their different roles.

Each benchmark stage asks a different question. Absolute scores across stages cannot measure an architectural gain; that requires matched inputs, endpoints, operating-point selection and decision delay.

Copying a clean simulated value can create different statistical cues from copying an observed noisy value. Temporal persistence also changes the task: attacks independently resampled at each step differ from a sustained bias on fixed targets. The final operational campaign fixes these assumptions rather than selecting easier ones after seeing a weak-family result.

3.6 Training, calibration and evaluation sources

The final campaign separates parameter fitting, operating-point selection and assessment. Modena's training corpus comprises 110 scenarios from six generator sources. Its calibration corpus comprises 99 scenarios from five sources. Source 811 contributes disjoint scenario subsets to these two corpora. Modena therefore has scenario separation with partial source overlap between training and calibration. Source-held fitting within training excludes the held source when generating out-of-fold evidence.

L-Town training comprises 120 scenarios from five sources and calibration comprises 96 scenarios from four different sources. Both networks' final evaluation collections contain six sources with 24 scenarios each. Table 3.6 describes the corpus sizes; source identifiers are retained in the experiment records [E1, E2].

Model seeds 701–710 identify repeated fits. Each fit is evaluated separately on all six sources, and the final summary gives equal weight to the resulting 60 model/source cells. Predictions from the ten models are not combined into an ensemble.

The first five seeds retain their original confirmation results. The five added seeds refit the frozen recipe and use the same evaluation sources, measuring further fitting variation. All added models and rules were frozen before extension evaluation. Figure 3.2 summarizes this separation of fitting, calibration and evaluation.

Table 3.5. Roles of the benchmark stages in the research. They are not a sequence of directly comparable absolute F1 scores.

Stage	Main purpose	Interpretation boundary
Snapshot and difficulty studies	Examine how regime and replay construction affect component measurements	Different corruption assumptions from the final campaign
Coherent episode controls	Assess topology, recurrence and mixture contributions under matched conditions	Historical training and evaluation protocol
Operational development	Build residual and temporal experts with explicit units and masks	Development/calibration evidence must be labelled
Frozen final evaluation	Assess selected Modena and L-Town systems across sources and fits	Fixed design and distribution; network-specific fitted systems

Table 3.6. Scenario counts in the final campaign. Source counts are not additional model seeds. Modena source 811 contributes disjoint training and calibration scenarios.

Network	Corpus	Sources	Scenarios
Modena	Training	6	110
Modena	Calibration	5	99
Modena	Final evaluation	6	144
L-Town	Training	5	120
L-Town	Calibration	4	96
L-Town	Final evaluation	6	144

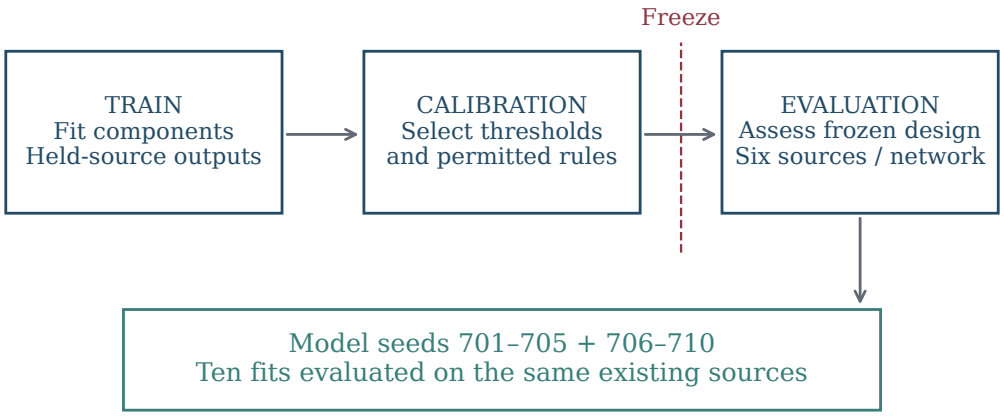


Figure 3.2. Evidence flow in the final campaign. Training supports model fitting and source-held diagnostics; calibration selects permitted operating points. Evaluation follows freezing. The five added model seeds reuse the existing evaluation sources, so the extension adds fitted realizations rather than six new datasets.

3.7 Information time and decision delay

For an immediate decision about endpoint (i, t), only received observations at times up to t may be used. Equation (3.7) gives a compact information-set notation:

$$\mathcal{F}_t = \sigma\Big(\{y^p_{j,u}, m^p_{j,u} : u \le t\}\Big), \tag{3.7}$$

with unavailable values excluded by their masks. A historical feature is causal at t when it is computed from this information and parameters fitted before evaluation.

The General path requires no additional decision delay. Drift and Noise decisions are allowed the declared interval $\Delta = 3$ hours. A decision assigned to endpoint time t may thus use specialized evidence available by $t + 3$. That is legitimate under the declared delayed task but must not be reported as if the later evidence were available at t. The delay is not a runtime benchmark and does not describe how long model prediction takes on the computer. Figure 3.3 illustrates the distinction between observation history and decision delay.

Timestamp matching matters when observations are missing. The next received observation in a stored array need not occur one hour later. Temporal operations must remain within the same source, scenario and sensor series and match the intended hour offsets. Otherwise a three-hour feature may silently become a variable-horizon feature, or information from another series may enter the calculation. The feature implementations explicitly retain identifiers and offsets for these operations.

3.8 Metric scope and uncertainty

The family-conditioned F1 for family f uses endpoints whose stored family code equals f. False positives within that scope are alarms on unmodified sensors during those family-coded periods. Long clean periods are not included in that family F1. For this reason, a strong set of family scores does not determine the pooled score.

Pooled F1 applies Equation (2.10) to all retained endpoints in the corpus. Clean FPR is the fraction of alarms among endpoints with clean family code. It excludes the unmodified sensors within attack periods that would enter an FPR over all negative endpoints.

The family macro score is the unweighted arithmetic mean of the five family F1 values,

$$F_{1,\text{macro}} = \frac{1}{5} \sum_{f \in \mathcal{A}} F_{1,f}, \qquad \mathcal{A} = \{\text{random}, \text{replay}, \text{drift}, \text{noise}, \text{targeted}\}. \tag{3.8}$$

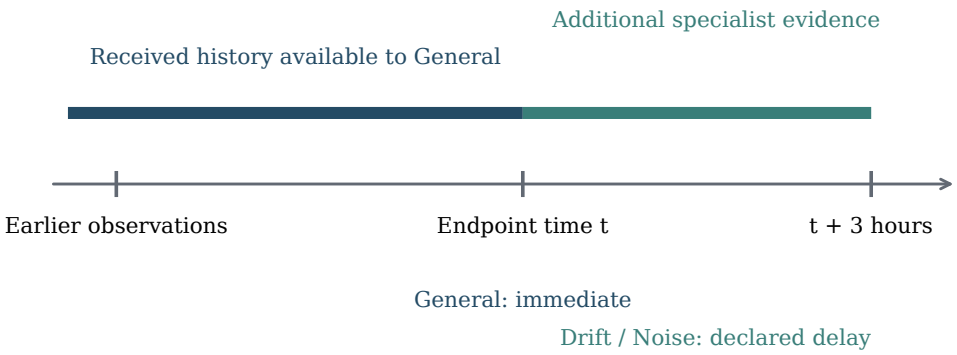


Figure 3.3. Conceptual distinction between observation history and decision delay. General uses information available at endpoint time t. Specialized evidence may accumulate until $t + 3$ for the decision about that endpoint. The diagram is an information-time schematic, not a measured response-time trace.

Equation (3.8) gives equal importance to the reported families, but it does not restore the clean-period errors excluded from their individual scopes. Pooled F1 and clean FPR therefore remain necessary even when macro F1 is high.

Consider a hypothetical example of the difference between family and pooled F1. A detector might produce 100 true positives, 10 false positives within attack-family periods and 20 false negatives. Its F1 on that restricted population is $200/(200 + 10 + 20) \approx 0.870$. If the same evaluated corpus contains another 100 false positives in clean periods, pooled F1 becomes $200/(200 + 110 + 20) \approx 0.606$. The gap comes entirely from the clean-period errors; the attack-period predictions are unchanged.

Model seeds measure fitting variation conditional on the data; generator sources vary the scenarios and observations. Repeated endpoints within an episode are correlated, and model seeds evaluated on the same source share those observations. Reporting 60 model/source cells as 60 independent datasets would conceal this structure. The final records retain model means, source means and cell ranges so that the result can be examined at the appropriate level.

3.9 What the benchmark can establish

The benchmark supports paired architectural comparisons on generated, observed pressure endpoints under declared corruption and observation conditions. It measures family balance and false-alarm trade-offs across two networks. Extending these findings to operational use requires field telemetry and incident-level evaluation. The controlled comparisons in Chapter 4 preserve the benchmark conditions within their respective protocols.

Chapter 4

Research Development and Design Decisions

4.1 From component design to complementary evidence

The early graph and temporal experiments showed that relationships among observations helped detection. Inspection of the expert outputs then exposed weaknesses in the reference and routing. The later work addressed these problems through residual modelling, temporal specialists and checks on the alarms they added to the system.

The experiments served different purposes. Historical controls compared components within their own benchmark; operational development and calibration guided the design and operating points. Frozen confirmation then evaluated the selected designs. Appendix A reproduces the quantitative evidence and identifies the accompanying experiment records for each level.

Figure 4.1 summarizes the questions that guided the architectural development. Rejected alternatives are discussed in the following sections. Drift, noise and clean-period false alarms drove the main architectural changes; replay later tested the adequacy of General's representation in L-Town.

4.2 Starting with graph and temporal models

The initial neural design encoded sensor information on the network graph and processed it over time. A shared spatial backbone represented each snapshot; recurrence combined representations across a window. Reconstruction outputs and anomaly heads then used the resulting features. The architecture was intended to use neighbouring observations and temporal context to interpret a pressure reading.

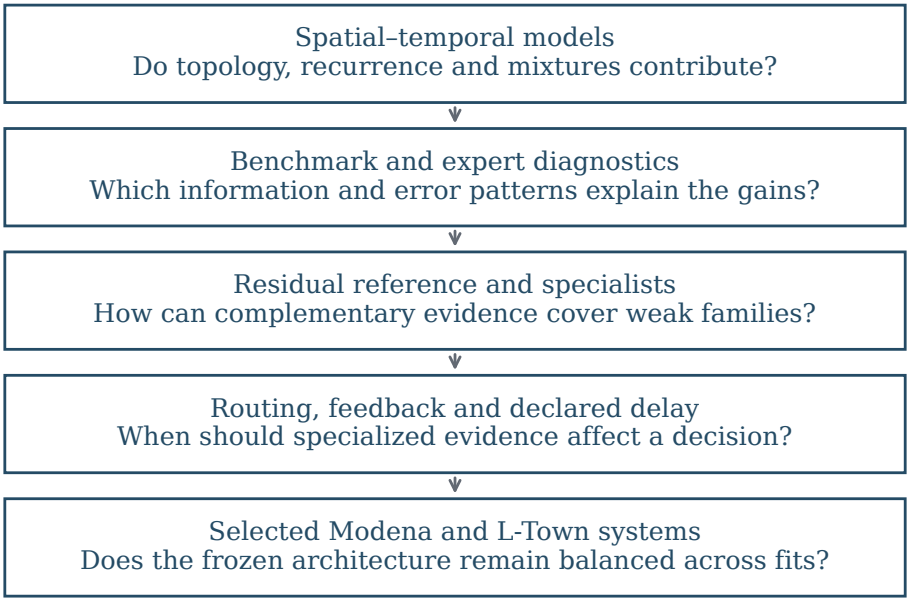


Figure 4.1. Conceptual development of the research. Each stage identifies a question that changed the design. Arrows indicate the reasoning sequence, not a controlled increase in performance across identical datasets. Both Modena and L-Town belong to the final evaluation.

Matched controls separated spatial and temporal contributions. The topology-free model disabled graph messages while preserving the temporal path. The one-step temporal ablation used endpoints and validation conditions aligned with those of the longer-window model.

The historical architecture in Figure 4.2 extended this design by assigning components to clean operation and the five attack families. Each expert retained a full spatio-temporal predictor, and the router combined outputs. The six-expert mixture therefore added six full predictors, not just small output heads. The comparison with a single temporal model tested whether this extra capacity supplied useful information and whether the router retained it.

The historical episode ablation preserved ten fits for each of four configurations on one generated Modena dataset. The stored results showed that removing topology or temporal context had a larger F1 cost than removing the mixture. Because all forty fits shared one underlying dataset, the next confirmation varied data seeds as well as model seeds [E4].

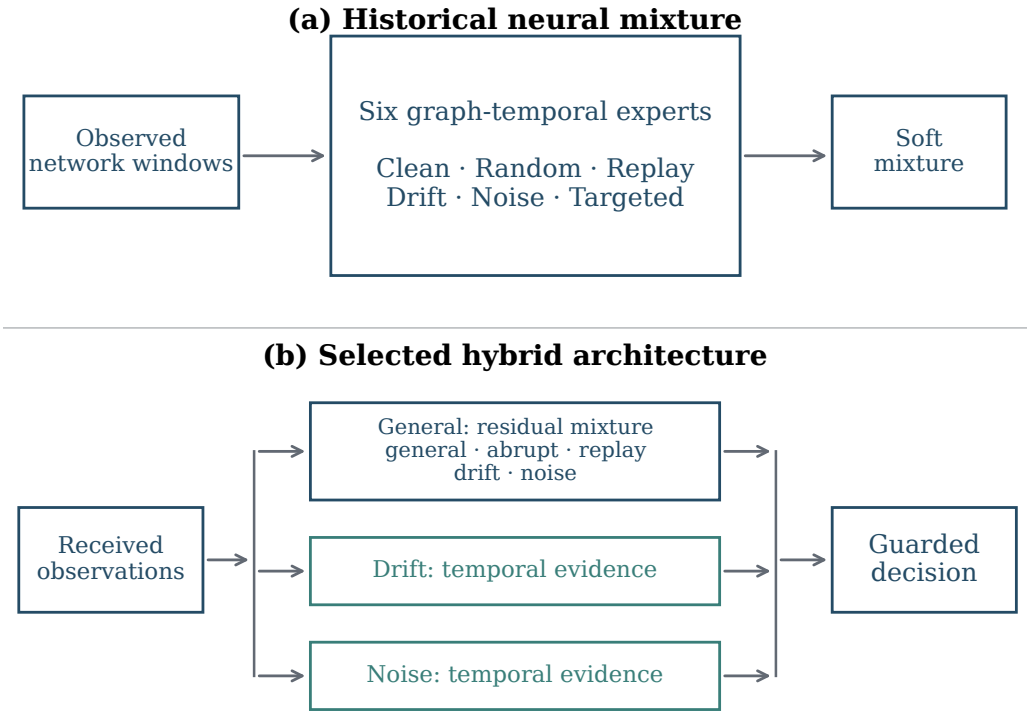


Figure 4.2. Two architectural stages at different levels of abstraction. The historical system mixes six graph-temporal experts. The selected hybrid has three top-level branches, with five internal components inside General. The diagram describes organization, not an isolated same-data performance comparison.

4.3 What the controlled episode experiments established

The spatial–temporal confirmation compared matched spatial and temporal models over three data seeds, with three model seeds nested within each. The mean AUPRC gain of the temporal model was positive in both networks. The topology comparison likewise retained temporal modelling while disabling messages in the control. Its positive differences supported the use of relational information in those episode experiments [E4].

Topology and temporal context contributed in both networks under the controlled episode protocol. The mixture added a smaller positive average gain in Modena. The intervals summarize three data-seed means; the table's scope note identifies the absent L-Town mixture comparison. Figure 4.3 displays the individual data-seed means.

Table 4.1. Historical controlled episode comparisons. Entries are mean paired AUPRC differences, not final hybrid scores. Reported 95% intervals use three data-seed means; each data seed contains three paired model fits. Source: preserved episode confirmation summaries, reproduced in Appendix A.

Comparison	Network	Mean ΔAUPRC	95% interval
Temporal minus spatial	Modena	+0.0798	[0.0707, 0.0889]
Temporal minus spatial	L-Town	+0.0890	[0.0514, 0.1267]
Topology-aware minus free	Modena	+0.0850	[0.0395, 0.1305]
Topology-aware minus free	L-Town	+0.0820	[0.0432, 0.1208]
MoE minus temporal	Modena	+0.0335	[0.0043, 0.0626]

Scope note. The retained matched MoE–temporal confirmation covers Modena only (three data seeds, three model seeds each). A corresponding L-Town confirmation is absent from this campaign; no L-Town MoE effect is estimated here.

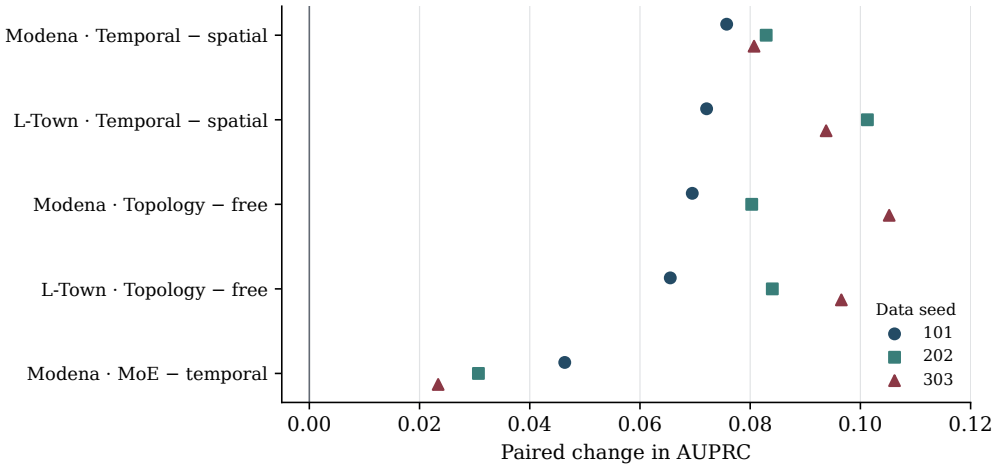


Figure 4.3. Data-seed mean AUPRC differences in the historical controls. Each point averages paired model fits for one data seed; the horizontal axis is a paired change. The comparisons share the same scale. The figure covers the matched historical controls.

The matched temporal-permutation control examined the role of coherent episode structure. In Modena, the mean temporal AUPRC gain was approximately 0.0757 on episodes and 0.0401 in the matched IID control, with a mean interaction of 0.0357. Persistence thus contributed to the measured advantage, while the positive IID gain showed that recurrence also exploited other temporal information.

This distinction motivated explicit representations of temporal change in the later design: the useful signal and its observation window could then be inspected directly.

4.4 Expert labels did not settle expert quality

The neural mixture required its own diagnostic beyond the aggregate MoE comparison. The Modena specialization analysis compared soft mixing with hard top-1 routing and a diagnostic using true family information. The mean soft-mixture AUPRC was 0.5637, while mean router accuracy was 0.4014. The family-owning expert ranked first in 40 of 45 family-by-seed comparisons, but specialization was incomplete and hard top-1 performance was weaker than soft mixing [E4].

The diagnostics suggest useful expert information that the router does not reliably isolate. True-family routing helps investigate this mismatch, but requires labels unavailable at inference; it is neither a deployable candidate nor a universal upper bound, since even the family-owning expert can make endpoint errors.

Correctly recognizing a family does not ensure that the router selects the best detector for a reading. An expert may also discriminate its own family well while producing false positives elsewhere. Reporting only its assigned-family performance would miss these problems.

Specialization was therefore assessed by the true detections it added beyond General and the errors it introduced elsewhere. This criterion guided the subsequent Drift and Noise branches.

4.5 Changing the normal reference

In the earlier reconstruction path, a reading could enter the representation used to predict its own expected value. This allowed corruption to affect the reference against which it was judged.

The blind-reference prototype prevented this direct copying by predicting each target group from observed sensors outside it. It fitted a low-rank representation to noisy observed pressures in normal training episodes. The target group also remained excluded during robust reweighting [E5].

The prototype supplied tree classifiers with normalized residual magnitude and temporal summaries such as slope, variability and persistence, together with observation support. It also introduced five internal components: general, abrupt, replay, drift and noise. Random and targeted corruption shared the abrupt mechanism in this internal decomposition, while their evaluation labels remained separate. This organization is the predecessor of the final General branch.

In the small development validation comparison, the blind-reference mixture performed strongly on random and targeted attacks, but drift and noise F1 were

only about 0.2222 and 0.2985. The aggregate improvement justified further work on the reference, while these weak families required a different approach.

The prototype changed both the representation and estimator, so their separate contributions cannot be inferred from this gain. The earlier matched neural controls provide the evidence for graph and temporal contributions.

A later control replaced the reference-derived inputs while holding the fitted dynamic-feature classifier and its thresholds fixed. Pooled F1 changed from 0.8181 to 0.8335 at the overall-priority threshold, but from 0.7325 to 0.7115 at the family-balanced threshold. This comparison evaluates the multi-view, noise-scaled and leverage-adjusted reference as a package. Both references already excluded the target group and used robust reweighting, so the comparison does not isolate the effects of exclusion, robustness or their interaction. The thresholds were retained from the earlier input distribution; the comparison does not estimate the performance of separately recalibrated systems [E5]. Section 6.10 defines a separate factorial comparison of target-group exclusion and iterative reweighting, with downstream refitting and a common calibration policy. It holds whitening and median aggregation fixed, so its scope is narrower than robustness as a whole.

4.6 Developing Drift and Noise evidence

Gradual bias and excess noise posed different temporal problems. For drift, the early part of a ramp can remain close to the reference, so evidence may need to accumulate across time. For noise, individual values can still look ordinary even when their local distribution has changed. General residual magnitude alone did not give the desired coverage, and similarly structured family experts were not reliably better than the general component.

The next experiments represented directional consistency, dispersion, reference support and daily context explicitly. They led to the seasonal and fast-response components. Daily comparisons reflect the generator's demand pattern, so their usefulness depends on that temporal context.

Other robust-reference experiments changed classifier training, weighting or score combination at the same time. Their results concern those combined changes, unlike the fixed-classifier reference control above.

Alarm persistence caused a further problem. Retaining a positive state could help detect weak episodes but also prolong alarms after an event ended, even when family F1 looked favourable. Candidate temporal mechanisms were therefore checked for clean-period false positives and their ability to return to normal after an event.

Drift and Noise became separate branches because they required temporal evidence and treatment distinct from the broad residual path. General retained its role in detecting abrupt, targeted and other observable discrepancies.

4.7 Making latency part of the task

The delay study allowed three hours for specialist evidence to accumulate. It recorded both the endpoint time and the latest observation time permitted for that decision.

An early screening study used the maximum of existing specialist scores over a bounded future window. Its purpose was to determine whether later evidence could improve the weak-family ranking under a fixed latency allowance. In that TRAIN-based hindsight screen, the forward maximum improved the reported drift and noise F1 relative to the zero-delay score. A learned delayed head sometimes improved detection scores but also kept producing false positives after an event ended, beyond the accepted limit [E6].

Following the TRAIN hindsight screen, a Modena study compared separately calibrated zero-hour and three-hour systems on six held evaluation sources, comprising 144 scenarios and 3,015,103 observed pressure endpoints. Both operating points were frozen on calibration before evaluation, under the same clean-FPR ceiling of 0.005. Table 4.2 reproduces that comparison from the retained records [E6].

Table 4.2. Historical Modena latency comparison: one fitted model configuration, six shared evaluation sources. F1 is computed over the concatenated endpoints, rather than averaged across the final campaign's model/source cells. Clean FPR is shown as a percentage; its difference is in percentage points.

Metric	Zero hours	Three hours	Difference
Random F1	0.9359	0.9427	+0.0068
Replay F1	0.7955	0.7886	−0.0069
Drift F1	0.7618	0.8368	+0.0750
Noise F1	0.7346	0.8437	+0.1091
Targeted F1	0.9542	0.9605	+0.0064
Pooled F1	0.6812	0.6867	+0.0054
Clean FPR (%)	0.3126	0.3847	+0.0721

The delayed configuration improves drift and noise substantially, while replay declines slightly and clean false alarms increase. General remains immediate in both configurations. At three hours, the specialists blend a forward maximum with a delayed head; the zero-hour control uses causal scores. Calibration also selects different integration rules. This comparison therefore measures the benefit and cost

of permitting a delayed configuration, including its recalibration, rather than the isolated effect of waiting three hours. It supports the development decision to use later evidence for weak temporal families, but does not establish that three hours is optimal or provide a latency curve for the final ten-seed systems.

The retained design therefore keeps General immediate and finalizes specialist decisions on the three-hour clock. Chapter 5 specifies how the bounded observations enter that decision.

4.8 From expert outputs to guarded decisions

Accepting every specialist alarm also admitted errors caused by other mechanisms. Evidence routing and local feedback were developed to control those errors across the evaluated corpus.

Routing describes the mechanism suggested by network-level evidence. Local feedback and verification then assess whether a specialist alarm should be retained. Their contribution is measured through final detection errors, with General's decision path preserved.

An earlier evaluation of the guarded system is recorded as EVAL-3 in the project archive. Its matched comparisons established the value of regulating specialist decisions in that evaluation population. The subsequent ten-model-seed campaign assessed the later selected implementation on its own declared sources; Appendix A keeps the two records separate.

The subsequent matched integration campaign (Stage-E in the project records) compared the baseline with both a threshold-only alternative and a candidate adding verification and early-warning evidence. This separates the benefit available from recalibration alone from that of the larger intervention [E1].

Section 6.9 examines the conditional contribution of each branch and guard to the final alarm rule. Those supplementary removals retain the fitted models and original thresholds, so their effects differ from those of retraining or recalibrating a reduced architecture.

The decision heads also required out-of-fold training. A verifier trained only on in-sample expert outputs could learn from unrealistically clean score patterns. Source-held expert predictions provide a more relevant training signal for the integration layer. In the final extension, the stochastic components were refitted for each added model seed rather than varying only the last threshold or router parameter. Table 4.3 summarizes the design decisions and their evidence boundaries.

Table 4.3. Design decisions and the observations that motivated them. The evidence column distinguishes controlled comparisons from development studies; the latter explain the choices without isolating causal effects.

Observation	Architectural response	Evidence boundary
Topology and recurrence contributed under episode controls	Retain spatial/temporal lessons in the design rationale	Historical matched comparisons
A named expert was not always the best detector	Inspect component and mixture outputs separately	Specialization diagnostics
A target observation could influence its own reconstruction	Use target-group-excluding reference predictions	Reference prototypes and controls
Drift/noise remained weak under broad residual evidence	Develop specialized temporal representations	Operational development
Unconditional specialization could add false positives	Retain General and regulate specialist alarms	Matched integration comparisons
Later evidence helped weak-family discrimination	Declare specialist decision latency	TRAIN screen followed by calibrated evaluation

4.9 A bounded refinement for a difficult family

L-Town replay tested whether the established architecture needed a different operating point or a more informative representation. Acceptance still required balanced performance across the complete system.

A symmetric recalibration of the internal mixture router improved replay in its fresh paired confirmation, from mean F1 0.6461 to 0.6757, while pooled F1 increased from 0.8799 to 0.8841. A preliminary attempt to force a larger false-alarm budget had produced little replay benefit and a large pooled loss. These outcomes justified examining representation rather than assuming that a more permissive threshold would solve the remaining problem [E7].

The first shared-history candidate added 56 features summarizing histories of existing residual evidence. On the held-source TRAIN diagnostic, General-mixture replay F1 fell from 0.7173 to 0.6954, leading to rejection.

The subsequent representation used differences between actually received pressures at common lags, with explicit availability indicators. Fourteen lags supplied signed change, absolute change and availability, giving 42 new features. The same bank was made available to the five internal General components and their router; it did not add a new top-level expert or a replay-specific inference rule. Differences were placed in 0.1-reference-scale bins to reduce reliance on exact floating-point equality [E8].

The initial TRAIN pilot improved General-mixture replay F1 to 0.7432, but it contained only two replay events. The broader TRAIN replication improved replay in all 20 primary source-held fold/seed comparisons while failing the declared noise-protection gate. Its mean noise change was slightly below the permitted limit, and one source showed a larger loss. The candidate was therefore rejected at this stage.

A separate protected full-system experiment subsequently assessed the representation with the external specialized branches retained. It compared predeclared General-update variants on calibration, selected full history and froze the design before fresh confirmation. The five-model-seed confirmation improved replay from its paired baseline of 0.7064 to 0.7303, with protection checks passing. Appendix A supplies all 30 paired replay values and their calculation. This comparison uses its own paired baseline, distinct from the earlier 0.6757 measured on other confirmation sources [E8].

Flow-based development candidates. Flow information was tested in two development stages. An earlier Modena study used source 811 to compare local pressure features, pressure-context features and joint pressure/flow context for the weak-family experts. These three feature candidates used matching classifier hyperparameters, with references fitted inside their training folds. Adding flow did not improve normal-reference accuracy: mean absolute error increased from 0.1892 to 0.2043 m on calibration and from 0.1678 to 0.1810 m on the reused development validation set. The joint candidate reached pooled validation F1 of 0.8484, but drift and noise F1 were 0.2593 and 0.3429. Calibration selected the pressure-context candidate; the reused validation set did not trigger reselection. That set contained only one event for each weak family, limiting the finding to this historical development comparison [E10].

The later L-Town replay study compared the received-pressure-history reference with three alternatives: internal probability calibration and routing correction, received-flow features, and their combination. The flow representation added 44 inputs to the existing 158: a rank-16 masked flow representation, a regularized regression from that representation to normal pressure, pressure discrepancies, availability measures, and pressure/flow change consistency at 1, 3, 6, 12 and 24 hours. All references and teachers excluded the relevant held sources. Five source folds and five model seeds produced 25 paired General-branch diagnostics.

Mean TRAIN replay F1 was 0.704881 for the reference, 0.703202 for probability fusion, 0.718817 for flow and 0.706480 for their combination. The three candidates improved on the reference in 12, 23 and 14 of the 25 comparisons, respectively.

Eligibility required at least 20 wins, mean replay gain of at least 0.01 and a positive mean gain in every source. Only the flow candidate qualified, and it also passed the limited pressure-feature rounding checks. At full-system calibration, however, none of the five fits had an operating point satisfying the protection constraints. At the rejected maximum-worst-family settings, all five seeds failed the strict requirement that replay F1 exceed its paired reference (deficits of 0.002410–0.003376). Seeds 702 and 705 also violated the no-loss pooled-F1 and no-increase clean-FPR constraints; seed 702 additionally exceeded the 0.005 permitted F1 loss in each of the other four attack families, whereas seeds 701, 703 and 704 passed the remaining aggregate protections. The rejected settings that maximized worst-family F1 averaged replay F1 of 0.702639, compared with 0.705392 for their paired reference. The candidate did not proceed to fresh confirmation [E10].

A subsequent L-Town candidate combined the same 44 flow inputs with 140 received-trajectory additions, giving 342 inputs in total. Its General-branch TRAIN replay F1 increased from 0.704881 to 0.738154, with improvement in all 25 comparisons. Full-system calibration satisfied the family, pooled-performance and false-alarm protections, but replay increased only from 0.705392 to 0.709936. The paired gain of 0.004544 fell short of the predeclared 0.02 progression requirement, preventing fresh confirmation. These calibration values use a different corpus from the confirmed 0.730307 history result and cannot be added to it. The confirmed pressure-history system remained the selected L-Town design [E10].

4.10 Settling the two-network architecture

Modena retains the confirmed Stage-E system; L-Town retains its received-pressure-history version of General. Their shared three-branch framework combines broad evidence, temporal specialization and controlled specialist acceptance. Chapter 5 specifies both implementations, and Chapter 6 evaluates their performance.

The final extension increased model seeds from 701–705 to 701–710 on each network. It preserved the attack distribution and selection procedure, freezing the added fits and calibration rules before evaluation on the existing six sources [E2].

Chapter 5

Proposed Hybrid Architecture

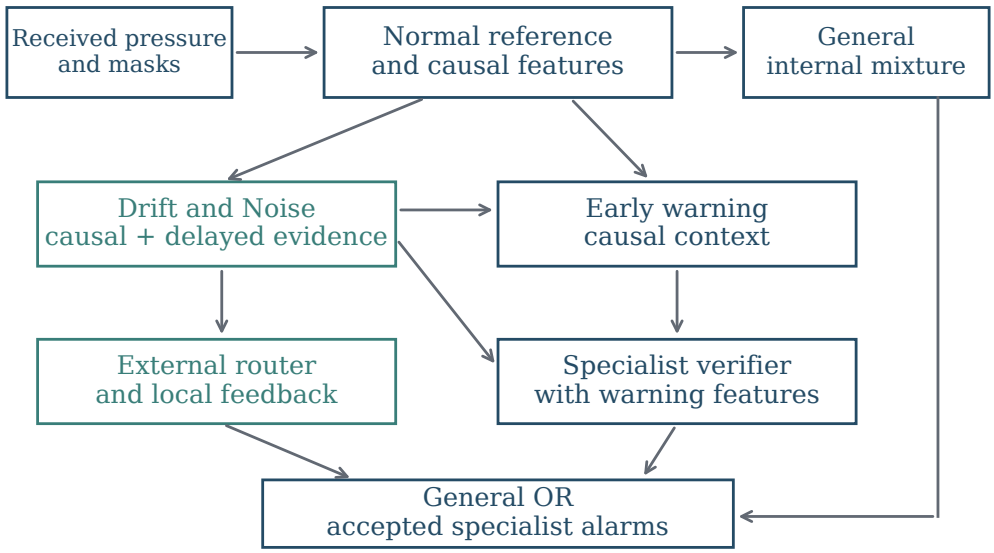
5.1 From received observations to an alarm

The detector has three branches: General, Drift and Noise. General evaluates broad deviations using a mixture of residual classifiers. Drift and Noise combine causal temporal evidence with observations collected during a declared three-hour decision interval. A network-level evidence router, local feedback heads and a verifier regulate the specialist alarms. The final prediction is the union of General's alarm and the specialist alarms that survive these checks.

Feedback and verification can reject specialist additions, but every General alarm remains in the final output. The experiments assess whether restricting those additions improves the balance of detections and errors.

Figure 5.1 shows the information paths and the two routers. General's internal router weights its five component predictions at one observed endpoint. The external evidence router describes the likely mechanism at a network-time group and informs specialist rejection. The early-warning head is another network-time predictor, but uses causal inputs only. Its outputs are inputs to the verifier, not a separate branch of sensor alarms.

For an observed pressure endpoint (i, t), let $x_{i,t}$ denote the common causal feature vector. Source and scenario identifiers separate trajectories during feature construction and are excluded from the classifier inputs. Training labels and event identities determine supervision and sample weights. At inference, the prediction path consumes received observations, masks, time alignment and fitted parameters. The source-code and configuration map in Appendix A identifies the implementation behind the equations in this chapter.



L-Town adds received-history features inside General.

Figure 5.1. Information flow in the selected architecture. Arrows show information dependencies. Only causal specialist scores enter the early-warning head. General remains available independently of those checks. The received-history extension enters General in L-Town. Both networks retain the three-branch organization.

5.2 A target-group-excluding normal reference

5.2.1 Fitting the normal representation

The reference estimates a pressure reading from other observed sensors. It is fitted on normal TRAIN observations, including their measurement noise and missingness. For sensor i, the base fit stores an observed-data mean μ_i and a scale a_i representing its normal variation. It standardizes available values as $(y_{i,t} - \mu_i)/a_i$ and performs iterative low-rank completion. At each iteration, the observed entries remain fixed and only missing entries are replaced by the low-rank reconstruction.

The retained basis $B \in \mathbb{R}^{N\times 16}$ represents normal cross-sensor variation. Twelve completion iterations are used. Sensors are assigned to four groups by their fixed index modulo four. When the reference predicts a group, every member of that group is excluded from the supporting observations. This also excludes the target reading from the robust reweighting calculations used for its own prediction.

The base reference first estimates a prediction-error scale $\sigma_i^{(0)}$ from normal observed residuals. The calculation uses the median absolute prediction error divided by 0.67448975, with a positive floor. This quantity scales prediction discrepancies; it differs from the wider hydraulic-variation scale a_i. The robust wrapper subsequently

recalibrates a final error scale σ_i on normal training observations. These fitted quantities remain fixed during calibration and evaluation.

5.2.2 Robust inference from visible sensors

For a target group g, the robust reference uses three fixed views of its permitted supporting sensors. One includes every sensor outside g; two retain random subsets with an 80% inclusion probability, generated with seed 821. A subset with insufficient support for the rank falls back to the full permitted set. The same view masks are reused across prediction times.

Equation (5.1) defines the whitened design row and observed response:

$$d_j = \frac{a_j B_{j,:}}{\sigma_j^{(0)}}, \qquad r_{j,t} = \frac{y_{j,t} - \mu_j}{\sigma_j^{(0)}}. \tag{5.1}$$

For view v, the latent vector solves the weighted ridge problem in Equation (5.2):

$$\widehat{z}_{g,v,t} = \arg\min_z \left\{ \sum_{j \notin g} w_{j,v,t}(r_{j,t} - d_j z)^2 + \lambda \|z\|_2^2 \right\}, \qquad \lambda = 10^{-3}. \tag{5.2}$$

A missing or excluded observation has zero weight. Each view performs six solves. Between solves, the implementation reduces the weight of observations with large leverage-adjusted discrepancies, initially with a bounded inverse-error weight and subsequently with a bounded biweight rule. The later rule uses cutoff 4.0 and retains a minimum relative weight of 0.01 for visible observations. Group exclusion remains in force throughout.

Each view gives an estimate $\mu_i + a_i B_{i,:} \widehat{z}_{g,v,t}$ for a target sensor. The final reference takes the median across the three estimates. The median absolute disagreement among these estimates provides a reliability feature. A separate support feature records the proportion of sensors outside the target group that are currently observed. Support is measured before robust downweighting. Figure 5.2 illustrates target exclusion and the three reference views.

The resulting normalized residual is

$$u_{i,t} = \frac{y_{i,t} - \widehat{p}_{i,t}}{\sigma_i}. \tag{5.3}$$

Equation (5.3) measures the difference between the observation and its reference in units of the fitted error scale. Robust reweighting reduces the contribution of inconsistent supporting readings. The disagreement and support channels record variation across reference views and the availability of supporting readings.

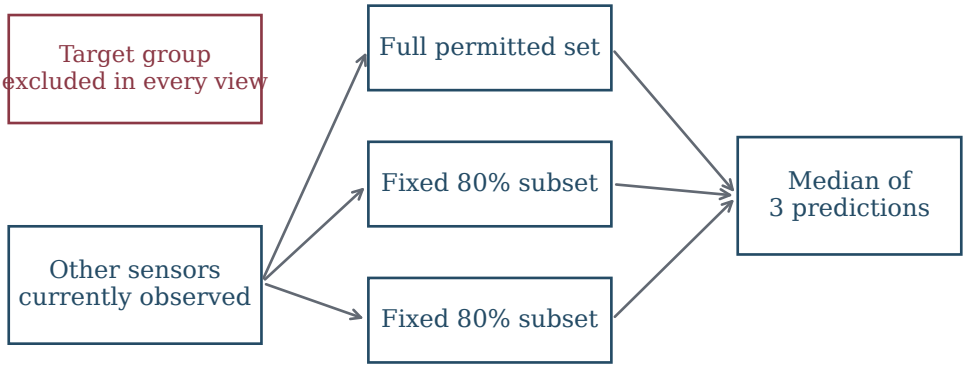


Figure 5.2. Schematic of reference prediction for one target group. Every view excludes the complete target group. Missingness reduces the visible support within each view, and the median combines the three predictions. The group and view masks are illustrative; the numerical fit uses the frozen masks and rank specified in the text.

5.3 The common evidence bank

The shared causal bank contains 116 columns. Its first 109 combine residual summaries, adaptive dynamic evidence, reference disagreement and sequential mechanism features. Seven daily comparison features complete the bank. The construction retains observed endpoints from hour 15 onward, providing the initial history required by the base residual window. An absent pressure reading produces no evaluated endpoint at that hour.

Current residual magnitude captures an immediate discrepancy. Windowed means, variation, slopes and sign consistency describe how that discrepancy behaves over time. Dynamic and sequential channels summarize innovations, accumulated directional evidence and excess energy. They let a classifier distinguish a large isolated deviation from a sustained pattern and interpret either in relation to observation support. Appendix A links to the ordered feature manifests and construction modules in the repository.

The daily comparisons use actually received pressures at lags 24 and 48 hours. In this bank their normalization is based on the declared observation-noise scale $\sigma_p = 0.1$ m:

$$d^{\text{seasonal}}_{i,t,\ell} = \frac{y_{i,t} - y_{i,t-\ell}}{\sqrt{2}\sigma_p}, \qquad \ell \in \{24, 48\}. \tag{5.4}$$

Each lag in Equation (5.4) supplies a signed difference, its magnitude and a support indicator. If the historical reading is absent, the difference is filled with zero and

support is zero. The seventh feature combines the signs and magnitudes of the two supported comparisons:

$$c_{i,t}^{\text{seasonal}} = \text{sign}(d_{24}d_{48}) \min(|d_{24}|, |d_{48}|). \tag{5.5}$$

The signed differences and their combination in Equation (5.5) exploit the daily context of the generated demand process. Their implementation requires a contiguous hourly scenario grid and checks the alignment of the flattened observed endpoints.

The final General mixture uses the 116-column bank, with feature subsets selected internally. The L-Town history extension increases its input to 158 columns. The specialist causal bank remains at 116 in both networks. Table 5.1 lists the feature dimensions at each interface.

5.4 General and its five internal components

5.4.1 Component roles and training populations

General contains five histogram gradient-boosted classifiers: general, abrupt, replay, drift and noise. The first uses all available columns and learns binary corruption detection across every attack family. The other four use mechanism-oriented feature profiles. Abrupt shares current residual and short-window evidence across random and targeted attacks. Replay includes lag-related and longer-window evidence. Drift emphasizes coverage, means, slopes and sign consistency; noise emphasizes coverage and dispersion. Common residual, support and gap features are available to all profiles.

The fit retains every positive TRAIN endpoint and samples at most 60,000 negative endpoints. A sampled negative receives a population-correction weight equal to the total number of negatives divided by the sampled count. The general component preserves this weighted prevalence. Each internal specialist trains on its own mechanism's positives and the sampled negatives, with the two binary classes rebalanced. Other mechanisms' positive endpoints are excluded from that specialist's binary fit.

The internal router uses five classes with a different interpretation from five attack labels. Class zero covers unmodified endpoints; random and targeted positives share the abrupt class, and replay, drift and noise positives occupy the remaining classes. The router is trained on the same sampled population with mechanism-balanced weights. Its class-zero probability weights the general component, which can still

Table 5.1. Dimensions of the selected feature interfaces. A component's internal feature profile may use a subset of its available bank. Graph-time features summarize the observed sensors at one time.

Interface	Columns	Construction
Common causal bank	116	109 residual/dynamic/sequential columns plus 7 seasonal columns
L-Town General bank	158	Common bank plus 42 received-history columns
Delayed specialist input	175	Common bank plus 48 offset values and 11 summaries
External router input	135	Mean, standard deviation and maximum of 45 local channels
Local feedback input	50	45 local channels plus 5 external-router probabilities
Early-warning input	160	Causal graph-time aggregates, sensor count and past differences
Verifier input	57	37 raw channels, 8 score channels and 12 early-warning channels

assign a high anomaly score. The router probability determines a component's weight; the component score estimates corruption.

All five components and the internal router use the same fixed estimator recipe: 120 boosting iterations, at most 15 leaves, a minimum of 30 samples per leaf, learning rate 0.08 and L2 regularization 10. Early stopping is disabled. The model seed changes the stochastic realization and sample selection; calibration chooses the operating point after fitting.

5.4.2 Internal mixture and symmetric calibration

Let $s_k(x)$ be the five component scores and $\pi_k(x)$ the internal-router probabilities. Equation (5.6) defines the basic General score:

$$g(x) = \sum_{k=0}^{4} \pi_k(x) s_k(x). \tag{5.6}$$

Modena retains this direct mixture in its selected Stage-E system. L-Town uses the saved symmetric router transformation. With temperature $T > 0$ and uniform

shrinkage $\rho \in [0, 1]$,

$$\widetilde{\pi}_k(x) = \frac{\max(\pi_k(x), 10^{-12})^{1/T}}{\sum_j \max(\pi_j(x), 10^{-12})^{1/T}}, \tag{5.7}$$

$$w_k(x) = (1 - \rho)\widetilde{\pi}_k(x) + \rho/5, \tag{5.8}$$

$$g(x) = \sum_{k=0}^{4} w_k(x) s_k(x). \tag{5.9}$$

Temperature in Equation (5.7) controls concentration, and shrinkage in Equation (5.8) retains a uniform contribution from all components in the score of Equation (5.9). The same transformation applies to every class. The selected values and General threshold are stored for each fitted realization. No family label is supplied to this calculation at inference.

5.4.3 The received-history extension in L-Town

The extension adds the same 42 columns to every General component's profile and to its internal router. Equation (5.10) gives the lag set:

$$\mathcal{L} = \{1, 2, \ldots, 12, 24, 48\} \text{ hours}. \tag{5.10}$$

For an available historical reading, Equation (5.11) defines the quantized difference:

$$h_{i,t,\ell} = 0.1 \operatorname{round}\left(\frac{y_{i,t} - y_{i,t-\ell}}{0.1\sigma_i}\right). \tag{5.11}$$

Each lag contributes $h_{i,t,\ell}$, $|h_{i,t,\ell}|$ and an availability indicator. Missing historical differences remain missing numerical values, with the indicator set to zero. This encoding differs from the seasonal bank's zero-plus-support convention and uses the fitted reference-error scale rather than $\sqrt{2}\sigma_p$.

Matching uses the exact historical hour within the same scenario and sensor. The retained history is the received stream, so previous corruption remains part of what the detector knows. Quantization removes access to arbitrarily precise equality through these new columns. The extension changes the learned representation inside General; the external specialist pipeline is preserved. The development and confirmation evidence for selecting it is documented in Chapter 4 and Appendix A.

5.5 Drift and Noise: causal scores and delayed evidence

5.5.1 Complementary causal components

The Drift branch combines a full seasonal classifier with a tuned residual classifier. The Noise branch combines a fast seasonal classifier with a full seasonal classifier. The fast profile emphasizes recent changes and energy rather than long-lived accumulated states, helping its score return toward normal after a disturbance ends.

The combination operates in log-odds space. Define $L(p) = \log(p/(1-p))$, with probabilities clipped to $[10^{-6}, 1-10^{-6}]$, and let sigmoid be its inverse. The causal branch scores are

$$c_D = \text{sigmoid}\left(0.90L(s_{D,\text{seasonal}}) + 0.10L(s_{D,\text{tuned}})\right), \tag{5.12}$$

$$c_N = \text{sigmoid}\left(0.95L(s_{N,\text{fast}}) + 0.05L(s_{N,\text{full}})\right). \tag{5.13}$$

The implementation clips the combined logits in Equations (5.12) and (5.13) to $[-30, 30]$. The coefficients are part of the frozen recipe, while the component classifiers are refitted for each model seed.

The seasonal classifiers use LightGBM. Full models have 400 trees and at most 31 leaves; the fast noise model has 350 trees and at most 23 leaves. Their training target is an altered endpoint of the assigned family. Training weight is divided between own-family positives (0.45), unmodified endpoints in that family's events (0.30), clean endpoints (0.15) and other-family endpoints (0.10). Weights are further balanced across events or scenarios within each category. Early positive endpoints receive a multiplier of two before within-event normalization. This weighting trains the specialist to detect its assigned family while rejecting competing mechanisms.

5.5.2 A bounded observation interval

For endpoint (i, t), the delayed feature constructor gathers 16 specified causal columns at offsets $t+1$, $t+2$ and $t+3$. These include residuals, rates, support, innovations, short-window drift/noise evidence and seasonal comparisons. Exact offset matching yields 48 values. Eleven additional summaries describe the available residual trajectory over $[t, t+3]$: extrema, mean, dispersion, rise, slope, absolute-value summaries, support and sign agreement.

A missing offset stays missing. Summary statistics use available values, and slope uses their actual hour offsets. If there is no temporal variation in the available offsets, slope remains unavailable. The current observed endpoint always anchors

the window. Near the end of a finite scenario, the operator uses only the permitted observations present in that scenario; it never crosses into another trajectory.

Separate delayed Drift and Noise classifiers consume the 175-column augmented bank. Each uses the full seasonal LightGBM recipe and family-weighting procedure. In parallel, a forward maximum pools the causal branch scores over available endpoints in the same bounded interval:

$$m_b(i,t) = \max_{u \in [t,t+3]:\,(i,u)\ \text{available}} c_b(i,u), \qquad b \in \{D, N\}. \tag{5.14}$$

The final specialist score combines the learned delayed score d_b and this maximum with equal logit weights:

$$q_b(i,t) = \text{sigmoid}\left(\tfrac{1}{2}L(d_b(i,t)) + \tfrac{1}{2}L(m_b(i,t))\right). \tag{5.15}$$

The delayed classifier summarizes the local trajectory, while Equation (5.14) retains the strongest causal score within the allowance. Equation (5.15) combines these two scores. Both finalize the decision about t on the $t+3$ clock.

5.6 External routing and local feedback

The external router receives 37 designated raw evidence channels and eight specialist score channels: causal, forward-maximum, delayed and final scores for each branch. For each source/scenario/time group, the 45 local channels are aggregated by mean, population standard deviation and maximum across the retained observed endpoints. The resulting 135-dimensional vector produces five probabilities for clean, abrupt, replay, drift and noise evidence.

Because these inputs include delayed scores, this router runs on the specialized decision clock. Its probabilities are broadcast to the observed endpoints in that group. Two feedback classifiers then combine the 45 local channels with the five router probabilities. Each estimates whether the endpoint belongs to its own specialist's positive target. Training includes own-family positives, positives from other mechanisms and sampled negatives, including high-scoring negatives. It uses source-held router probabilities to construct the feedback training inputs.

The selected veto rejects specialist evidence when replay is more likely and local feedback does not support the specialist mechanism. Let r_R, r_D and r_N denote the external probabilities for replay, drift and noise, and let f_b denote local feedback.

Then

$$V_b(i,t) = \mathbf{1}\{r_R > r_b + 0.10\}\,\mathbf{1}\{f_b(i,t) < 0.10\}, \qquad b \in \{D, N\}. \tag{5.16}$$

Both conditions in Equation (5.16) must hold: the network context must favor replay and local feedback must reject the specialist mechanism. The guard addresses overlap between replay and specialist evidence. The received-history extension retains this rule.

The veto can remove a specialist alarm but cannot generate an alarm or cancel General. Feedback is a learned check on expert evidence, with parameters fixed during evaluation.

5.7 Causal early warning and specialist verification

5.7.1 A separate network-time signal

The early-warning head uses the same 37 raw evidence channels and only the two causal specialist scores. Their three graph-time aggregates give 117 values. The observed-sensor count adds one, and differences in 13 selected mean channels at past lags 1, 2 and 3, with three availability indicators, complete the 160-column input.

The head predicts the five evidence classes using a histogram gradient-boosted classifier. A confidence threshold permits abstention. In the training procedure, the first three hours of an event receive extra weight, and source-held predictions determine the abstention setting under a clean-warning budget. Event onset is a training/evaluation annotation; prediction uses the causal feature vector.

A warning describes the network-time group; it does not identify which sensor is corrupted. For final sensor decisions, warning probabilities, confidence, abstention, persistence and one- and two-hour Drift/Noise probability lags become twelve verifier inputs. Lack of a warning is not implemented as a mandatory rejection rule, although the learned verifier can use the warning features when judging a specialist candidate.

5.7.2 Verification on the candidate region

Each verifier head receives 57 columns: 37 raw channels, eight specialist scores and twelve early-warning channels. Its training population is the high-score candidate region defined from clean TRAIN out-of-fold scores. The pre-threshold corresponds to the upper two-percent region of that branch's clean score distribution. This selects the region in which specialist precision matters most.

The target is an endpoint modified by an attack from the verifier's assigned family. Other-family positives therefore serve as negatives for that branch-specific verification task, as do unmodified endpoints. A class-balanced histogram boosting model with 200 iterations and at most 23 leaves fits each head. The region boundary and fitted model are then frozen.

Let $v_b(x)$ be the verifier's keep probability within its region. Outside that region the implementation returns one, leaving those scores unaffected. At a calibrated cutoff κ_b, the verification mask is

$$K_b(i,t) = \mathbf{1}\{v_b(i,t) \geq \kappa_b\}. \tag{5.17}$$

A cutoff of zero in Equation (5.17) disables the check for that branch. General is never passed through a verifier. The selected cutoff therefore determines whether each fitted verifier head is active.

5.8 The final decision and its calibration

Given fitted scores and frozen thresholds τ_G, τ_D, τ_N, define

$$A_G(i,t) = \mathbf{1}\{g(i,t) > \tau_G\}, \tag{5.18}$$

$$A_b(i,t) = \mathbf{1}\{q_b(i,t) > \tau_b\}\,(1 - V_b(i,t))K_b(i,t), \quad b \in \{D,N\}, \tag{5.19}$$

$$\widehat{y}_{i,t} = A_G(i,t) \vee A_D(i,t) \vee A_N(i,t). \tag{5.20}$$

The immediate path in Equation (5.18) can emit A_G at t. Specialist acceptance in Equation (5.19) can add an alarm for that endpoint after the declared interval; Equation (5.20) takes their union. Relative to the same General score and threshold, the final positive set always contains General's positive set. Guards and verification only restrict the specialist additions. Changes to General's representation or threshold require a separate paired comparison because they change this reference set. Table 5.2 summarizes the endpoint calculation.

Threshold selection is performed on calibration. The Stage-E search allocates a clean false-alarm budget of 0.005 across the three raw branch thresholds, then evaluates their guarded union. The selected candidate maximizes pooled F1 subject to the recorded family protections, with worst-family F1 used in its ordering of tied candidates. Verifier cutoffs are chosen on the declared grid. Calibration may use less than this budget. The ceiling applies to calibration FPR and does not guarantee the evaluation rate.

Table 5.2. Logical evaluation for endpoint (i, t). History/state construction is shared across endpoints; the steps specify information dependencies rather than a required software execution order.

Step	Operation
1	Build the target-excluding reference and the causal feature vector from received observations and frozen parameters.
2	Compute the five General component scores, internal weights and immediate alarm A_G. Include the 42 received-history columns for L-Town.
3	Retain causal specialist and early-warning outputs; collect permitted specialist evidence through $t + 3$.
4	Form forward maxima, delayed scores and final specialist scores q_D, q_N.
5	Evaluate external routing, local feedback and the two verifier probabilities. Apply each veto and verifier once.
6	Return the union in Equation (5.20), retaining the endpoint timestamp and declared decision time.

L-Town's protected-history selection holds the specialized alarm path fixed while choosing General's symmetric router settings and threshold. Its protection checks compare the complete candidate with its paired baseline, including source-specific checks. The resulting operating points are retained separately for each model seed. Appendix A gives an example and links to the complete configuration records in the repository.

Source-held fitting supports the integration layers. Each TRAIN source is held out in turn; the reference and base experts used for its out-of-fold predictions are fitted without that source. These out-of-fold scores provide training inputs for the warning, feedback and verifier stages. Delayed classifiers are fitted within each fold alongside the base experts. Final classifiers are then fitted for the selected recipe, and calibration selects decision settings. The normal reference and the deterministic feature caches are shared across model-seed repetitions, while the stochastic classifiers and their relevant samples are refitted.

5.9 The two selected implementations

Modena retains the confirmed Stage-E candidate: the 116-column General mixture, the two specialized branches and the guarded decision layer with early-warning-assisted verification. L-Town retains the confirmed full-history General with 158 available columns, symmetric internal-router calibration and the preserved specialized pipeline. Both use the same functional separation between broad detection, temporal specialization and controlled admission of specialist alarms.

Chapter 6

Experimental Protocol and Results

6.1 Evaluation objectives and selected systems

The final evaluation examines the complete detector on Modena and L-Town. It measures attack detection and false alarms across fitted models and generated sources, including the families on which performance remains weaker. Both networks contribute to the result: Modena evaluates the selected architecture in its principal development setting, while L-Town tests a network with a larger and different hydraulic structure.

Modena retains the confirmed Stage-E system; L-Town retains the confirmed history extension within General. Chapter 5 specifies these network-specific implementations. The following comparisons assess their final decisions.

The core campaign contains three comparisons. The original Modena confirmation compares the previous guarded system, a threshold-only alternative and the Stage-E candidate on the same fresh sources. The L-Town history confirmation compares the selected history system with its matched no-history baseline. The final extension adds five model seeds to each selected design and summarizes all ten fits. Their baselines and source sets differ, and the tables identify the relevant population at each step [E1, E8, E2]. A subsequent single-classifier comparison addresses complexity under the retained benchmark in Section 6.8.

6.2 Frozen experimental procedure

6.2.1 Training, calibration and assessment

The final benchmark retains the observation and attack settings defined in Chapter 3. Scenarios span 168 hours with hourly observations, pressure and flow missing probabilities remain 0.50, and the generator distribution, attack severity and scenario splits are preserved. The prediction target is an actually received pressure endpoint after feature warm-up. Missing readings are excluded from the decision population through the availability masks.

Training fits the normal reference and the supervised components. Source-held predictions supply the integration heads with expert outputs from sources excluded from the corresponding expert fit. Calibration selects operating points using the permitted search procedure and protection criteria. Evaluation follows the freezing of the selected models and rules. This separation is especially relevant to the history comparison, where a feature update changes the score distribution and therefore requires a calibrated operating point.

The two networks use different training and calibration collections. Modena has 110 training and 99 calibration scenarios; L-Town has 120 and 96, respectively. The corpus definitions, including Modena's shared source 811 with disjoint scenario subsets, are given in Table 3.6. Evaluation uses six source datasets per network, each containing 24 scenarios. Thus each fitted model is assessed on 144 scenarios, and the same scenarios recur across model seeds within that network.

The original Modena selection uses the Stage-E calibration procedure. In L-Town, the original combination of clean-FPR and absolute family-floor constraints was infeasible. Before L-Town evaluation, the campaign amended the procedure to retain the clean-FPR budget of 0.005 and relative family protection while reporting the unmet absolute floors. The selected L-Town rule satisfies this amended procedure; the original replay floor remains unmet [E1].

The initial research goal for L-Town replay F1 was 0.80; this was revised to 0.75 during development when the original goal proved difficult to attain. These values specify research performance goals rather than alarm-score thresholds.

The later protected-history protocol adds its own explicit checks. It requires a replay improvement while protecting pooled performance, clean FPR and the other families, with specified tolerances at model and source levels. The external specialist pipeline is preserved. The full-history variant is selected on calibration and frozen before its original fresh confirmation. The failed earlier General-only

noise-protection gate remains a development result, as described in Chapter 4; the full-system result concerns a separately specified intervention [E8].

6.2.2 From five fits to ten

The original confirmation uses model seeds 701–705. The extension adds seeds 706–710, refitting the stochastic components with the selected recipe. It retains the network-specific calibration algorithms and freezes all added models and rules across both networks before extension evaluation. The added seeds use the existing six evaluation sources. This measures sensitivity to fitting randomness conditional on those sources, while preserving the original confirmation chronology [E2].

For Modena, the evaluation sources are 40811, 41811, 42811, 43811, 44811 and 45811. For the selected L-Town history system they are 110811, 111811, 112811, 113811, 114811 and 115811. The *original locked test* denotes four whole scenarios (0, 1, 8 and 20) reserved from the initial 24-scenario Modena source 811. The original split was selected using scenario-level family coverage and observed replay-positive availability, independently of detector scores. Those scenarios remained excluded from fitting, calibration and the reported evaluation. The later campaign defined separate generator-source collections for confirmation and repeated-fit assessment, so the results reported here concern those collections; no performance estimate on the original reserved split is claimed. All ten fits are retained in the final result, including their source-level variation; the reported system is evaluated as individual fitted realizations, without combining their predictions into an ensemble.

The reported endpoint scores include specialist decisions made by $t+3$, alongside General's immediate decisions at time t. Early warning supplies network-time context to verification.

6.3 Aggregation and uncertainty

Let $M_{r,s}$ denote a metric for model seed r and source s, within one network. The final reported mean is

$$\overline{M} = \frac{1}{10 \cdot 6} \sum_{r=1}^{10} \sum_{s=1}^{6} M_{r,s}. \tag{6.1}$$

For pooled F1, each $M_{r,s}$ is computed from the true-positive, false-positive and false-negative counts over all retained endpoints in that cell. Those F1 values are then averaged. Since F1 is nonlinear, this differs from first merging counts across all fits and computing one F1. Reusing an endpoint under ten fitted models also does not create ten independent observations of the underlying hydraulic process.

Metric populations follow Chapter 3: family scores cover family-coded periods, pooled F1 covers all retained endpoints, and clean FPR covers clean periods. Family macro F1 gives equal weight to the five families. Tables express clean FPR as a percentage; a rate of 0.000584 appears as 0.0584%.

Equation (6.2) defines two marginal summaries of the variation:

$$\overline{M}_{r,\cdot} = \frac{1}{6}\sum_{s} M_{r,s}, \qquad \overline{M}_{\cdot,s} = \frac{1}{10}\sum_{r} M_{r,s}. \tag{6.2}$$

The sample standard deviation of the ten model means describes fit-to-fit variation after averaging over sources. The sample standard deviation of the six source means describes differences between generated collections after averaging over fitted models. Neither is the standard error of 60 independent datasets. Cell ranges and counts below reference levels complement these summaries by showing outcomes hidden by the overall mean.

For matched interventions, the reported difference is

$$\overline{\delta} = \frac{1}{|\mathcal{P}|}\sum_{(r,s)\in\mathcal{P}} \left(M_{r,s}^{\text{candidate}} - M_{r,s}^{\text{baseline}}\right), \tag{6.3}$$

where $\mathcal{P}$ is the set of evaluated pairs. Modena's original confirmation contains 30 pairs, and the final L-Town history comparison contains 60. Positive-pair counts describe the consistency of the observed intervention. The analysis reports these descriptive quantities with the source structure visible, rather than treating repeated endpoints or model fits as independent replications for a significance claim. Differences are computed from unrounded metrics and may therefore differ slightly from differences between the displayed values.

6.4 Final results on both networks

Table 6.1 reports the selected systems. Modena achieves mean pooled F1 of 0.8669 and family macro F1 of 0.9045. L-Town achieves 0.8950 and 0.9083, respectively. Their clean FPRs are 0.0584% and 0.0550%. Expressed as normalized reading-level rates, these correspond to approximately 58.4 and 55.0 alarms per 100,000 clean evaluated endpoints.

The family profiles differ substantially despite their similar macro scores. Modena is strongest on random and targeted corruption, followed by replay. Drift and noise are its weaker families. L-Town is strongest on random and targeted corruption

Table 6.1. Final selected-system means over 60 model/source cells per network. F1 is on the 0–1 scale; clean FPR is expressed as a percentage.

Metric	Modena	L-Town
Random	0.9549	0.9920
Replay	0.9283	0.7312
Drift	0.8274	0.8990
Noise	0.8439	0.9279
Targeted	0.9680	0.9918
Family macro F1	0.9045	0.9083
Pooled F1	0.8669	0.8950
Clean FPR (%)	0.0584	0.0550

and retains high drift/noise scores, while replay is substantially lower. Figure 6.1 compares the five family scores in each network.

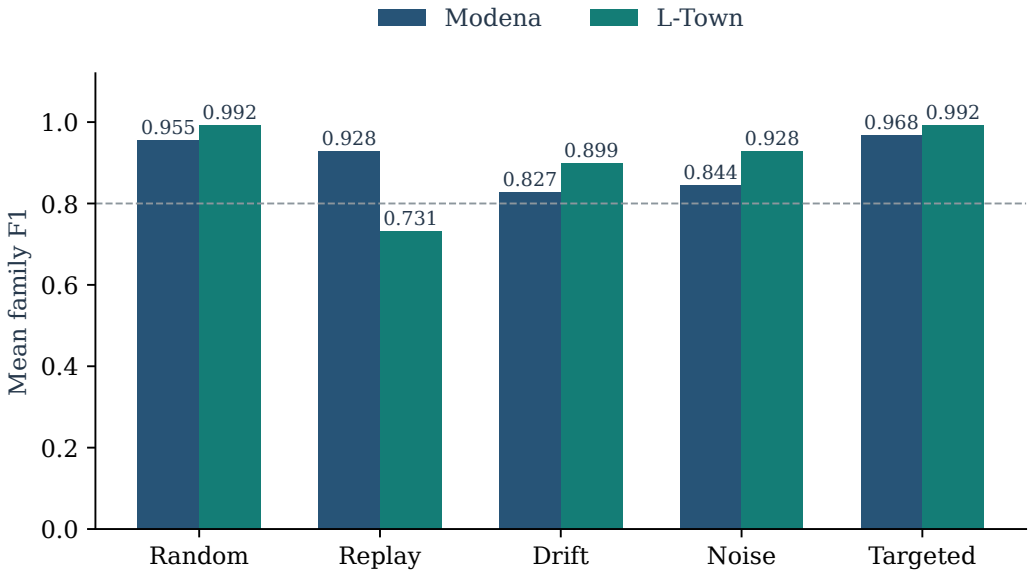


Figure 6.1. Mean family-conditioned F1 of the selected systems, each evaluated across ten model seeds and six source datasets. The dashed reference marks 0.80. Values summarize the 60 model/source cells within each network.

6.5 Modena: improved precision with a measured trade-off

6.5.1 Matched confirmation of the integrated system

The original Stage-E confirmation provides the controlled evidence for the major pooled-F1 improvement. All three arms use the same five model seeds and six fresh sources. Table 6.2 reports their mean scores. The candidate increases pooled F1 from 0.6863 to 0.8672, a paired gain of 0.1809, whereas the threshold-only alternative reaches 0.6998. The candidate outperforms the baseline in pooled F1 in all 30 cells [E1].

The confusion counts explain the scale of this improvement. Summed across the repeated evaluations, baseline true positives are 79,809, false positives 62,722 and false negatives 10,156. The candidate yields 76,532 true positives, 9,910 false positives and 13,433 false negatives. False positives decrease by 84.2%, while true positives decrease by 4.1%. The resulting precision gain more than compensates for the reduced recall in pooled F1. These sums describe repeated model evaluations on shared sources; they are used here to explain the error balance, not as counts of distinct physical incidents.

The family-level effects are more selective. Random improves by 0.0029, replay by 0.0250 and targeted by 0.0069. Drift decreases by 0.0071 and noise by 0.0051. These losses fall within the recorded 0.01 mean-protection tolerance, but they are systematic: drift declines in 22 of the 30 cells and noise in 25. The improved precision comes with small but frequent losses in the specialized families.

The earlier EVAL-3 pooled F1 of 0.6515 explains why background alarms became a development priority. Its evaluation population differs from the fresh Stage-E comparison. The matched gain attributed to the later intervention is 0.8672

Table 6.2. Original Modena confirmation: five model seeds and six fresh sources, with all three arms scored on the same 30 cells. The final ten-seed selected-system mean is reported separately in Table 6.1.

Metric	Baseline	Threshold only	Candidate	Δ candidate
Random	0.9513	0.9551	0.9542	+0.0029
Replay	0.9030	0.9071	0.9280	+0.0250
Drift	0.8333	0.8494	0.8262	-0.0071
Noise	0.8506	0.8428	0.8455	-0.0051
Targeted	0.9605	0.9634	0.9674	+0.0069
Pooled F1	0.6863	0.6998	0.8672	+0.1809
Clean FPR (%)	0.4262	0.3997	0.0574	-0.3687

minus 0.6863. The threshold-only arm further shows that a new operating point on its own accounts for only a small part of that change. The larger intervention combines the selected calibration procedure with verification and early-warning context; the experiment evaluates that package, rather than separately identifying the contribution of each added head.

6.5.2 The final ten-seed Modena profile

With ten model seeds, Modena's pooled mean is 0.8669, close to the original five-seed value of 0.8672. Random, replay and targeted retain means of 0.9549, 0.9283 and 0.9680. Drift and noise average 0.8274 and 0.8439. The added fits reproduce both the strong aggregate result and the weaker drift and noise performance.

All family means exceed 0.80, but the source-level pattern is uneven. Drift falls below 0.80 in 19 of the 60 cells and noise in ten. Their minima are 0.7652 and 0.7811. In contrast, every random, replay and targeted cell remains above 0.80. The family averages therefore conceal some weaker outcomes on individual sources and fits.

These weaker scores are structured by source. Section 6.7.2 reports their source averages alongside the corresponding L-Town profile.

6.6 L-Town: protected history improves the complete detector

6.6.1 Matched full-system effects

The selected L-Town intervention supplies received-pressure history to all five internal General components and its internal router, while preserving the external specialist pipeline. In its original fresh confirmation, mean replay F1 increases from 0.7064 to 0.7303 and pooled F1 from 0.8895 to 0.8951. All 30 replay differences are positive, and the predeclared protection checks pass. These results establish the original confirmation of the history design [E8].

The final ten-seed comparison extends the same selected recipe across model randomness. Table 6.3 reports the corresponding 60-cell matched baseline and history means. Replay increases by 0.0247, from 0.7065 to 0.7312. Pooled F1 increases by 0.0088, from 0.8862 to 0.8950. Both differences are positive in all 60 cells. The candidate's stable pooled mean across the original and added seeds is reported in Table 6.5; the changed paired gain reflects the fitted baseline as well as the candidate.

All four other family means also improve. Random gains 0.0026, drift 0.0014, noise 0.0021 and targeted 0.0027. Mean clean FPR decreases from 0.0627% to

Table 6.3. Final matched L-Town history comparison, averaged over 60 model/source cells. Both systems use the preserved external specialist pipeline. FPR changes are percentage points.

Metric	No-history baseline	Full history	Difference
Random	0.9893	0.9920	+0.0026
Replay	0.7065	0.7312	+0.0247
Drift	0.8976	0.8990	+0.0014
Noise	0.9258	0.9279	+0.0021
Targeted	0.9890	0.9918	+0.0027
Pooled F1	0.8862	0.8950	+0.0088
Clean FPR (%)	0.0627	0.0550	-0.0077

0.0550%, approximately a 12.3% relative reduction. The replay gain is accompanied by improvements in the other family means and the overall error balance.

Some individual comparisons show small losses. Drift improves in 49 cells and targeted in 58; random and noise improve in all 60. The largest individual other-family loss is approximately 0.0009 F1. Clean FPR decreases in 40 cells, increases in 19 and is unchanged in one, even though its overall mean falls. The protection criteria apply to the aggregate and source-level results; they allow some losses in individual fits.

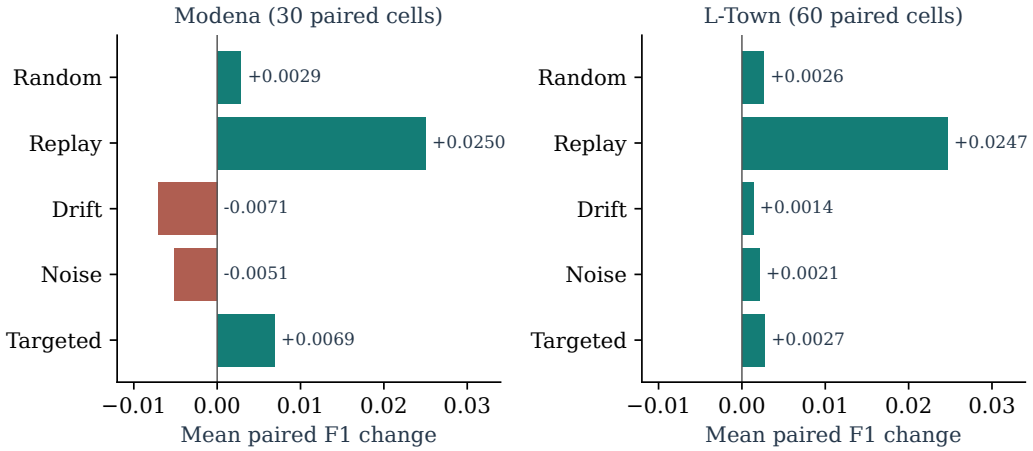


Figure 6.2. Mean paired family-F1 changes for the original Modena Stage-E confirmation and the final L-Town history comparison. The panels use their own matched baselines and contain 30 and 60 cells, respectively. Green bars indicate gains and red bars losses.

Figure 6.2 contrasts the interventions: Modena's large precision gain carries drift/noise losses, whereas L-Town's smaller pooled gain accompanies positive means for every family.

6.6.2 The remaining replay limitation

L-Town replay is improved but remains the limiting family. Its final mean is 0.7312, below the later 0.75 objective and the earlier 0.80 reference. Individual cell scores range from 0.6947 to 0.7657. Fifty of the 60 cells are below 0.75, and all are below 0.80. In contrast, random, drift, noise and targeted exceed 0.80 in every final cell.

Section 6.7.2 reports replay performance separately for each evaluation source.

6.7 Variation across models and sources

6.7.1 Descriptive spread

Table 6.4 separates model and source variation. For Modena pooled F1, the model-mean SD is 0.0034 and the source-mean SD is 0.0118. For L-Town they are 0.0013 and 0.0109. In both networks the spread between source means is larger than the spread between model means. These marginal summaries describe the observed collections; they are not an additive decomposition into independent variance components.

Figure 6.3 displays the individual fitted-model means for pooled F1 and replay. The horizontal lines give the final overall means, and the division after seed 705 identifies the extension. Each point already averages six source datasets. Modena's replay varies relatively little across these model averages, while its pooled performance is more sensitive to the fitted realization. L-Town's pooled model means are tightly grouped, with more visible variation in replay. The distinction is consistent with replay remaining the difficult family even when the aggregate result is stable.

Table 6.5 compares the two five-seed groups directly. Modena pooled F1 changes from 0.8672 to 0.8667; L-Town changes from 0.8951 to 0.8949. The largest family shift between the two groups is Modena noise, which decreases from 0.8455 to 0.8422. L-Town replay increases from 0.7303 to 0.7320. The extension therefore preserves the broad family profile and pooled performance of the originally selected systems.

6.7.2 Source-dependent performance

Table 6.6 gives source-level replay, drift, noise and pooled means; Figure 6.4 shows every family. Modena drift averages 0.7747 on source 44811 and 0.7922 on 42811, compared with 0.8879 on 41811. Noise is weakest on 43811 at 0.7882. L-Town replay ranges from a source mean of 0.7029 on 110811 to 0.7539 on 115811. These differences remain after averaging ten fits.

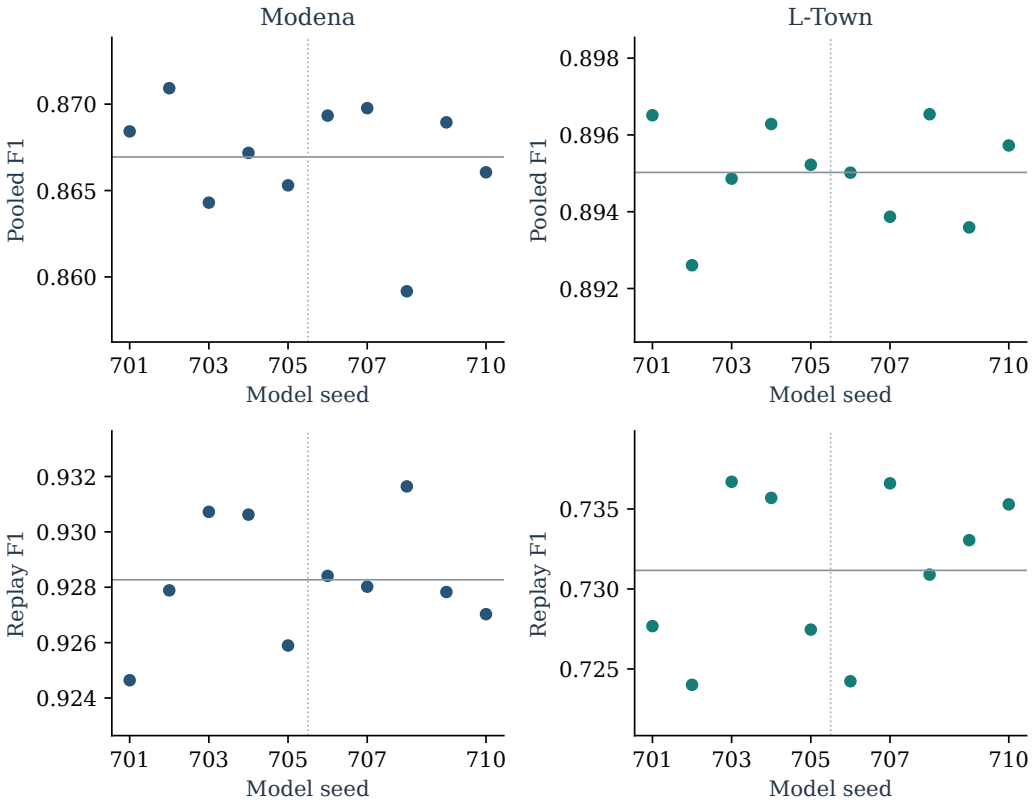


Figure 6.3. Selected-system model means, each averaging the same six sources within its network. The vertical dotted line separates the original five seeds from the added five. Horizontal lines mark the overall means. Panel axes use different restricted ranges to make fitting variation visible.

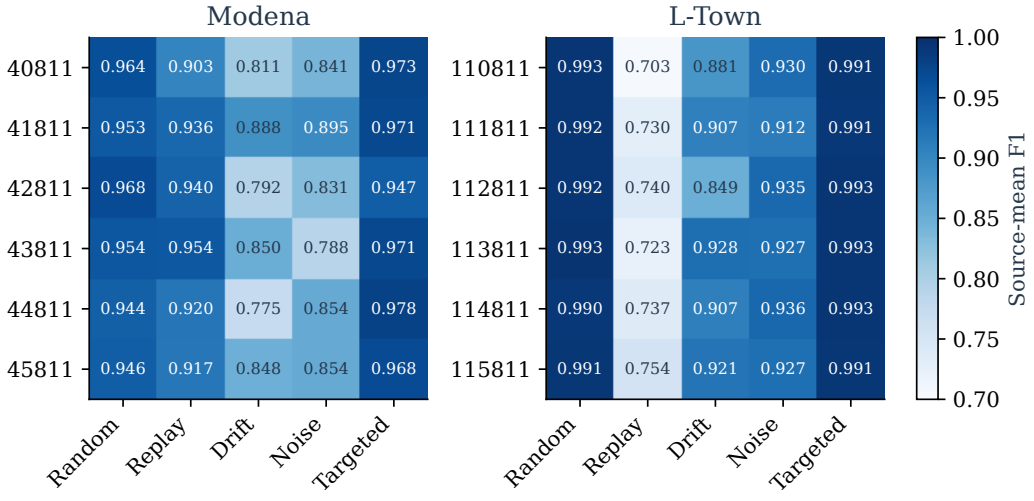


Figure 6.4. Family F1 averaged over ten model seeds for each source. The common color scale spans 0.70–1.00; printed values give the individual source means. Source identifiers distinguish scenario collections, and matching row positions across networks do not denote matched hydraulic scenarios.

Table 6.4. Descriptive variation of the selected systems. Model SD is the sample standard deviation of ten six-source means; source SD is that of six ten-model means. The range covers all 60 cells. FPR entries use percentage units.

Network	Metric	Model SD	Source SD	Cell range
Modena	Random	0.0041	0.0095	0.9366–0.9748
	Replay	0.0022	0.0184	0.8971–0.9605
	Drift	0.0056	0.0422	0.7652–0.8967
	Noise	0.0032	0.0349	0.7811–0.8992
	Targeted	0.0025	0.0109	0.9394–0.9804
	Pooled F1	0.0034	0.0118	0.8419–0.8896
	Clean FPR (%)	0.0084	0.0077	0.0429–0.0941
L-Town	Random	0.0002	0.0011	0.9893–0.9941
	Replay	0.0050	0.0173	0.6947–0.7657
	Drift	0.0021	0.0291	0.8436–0.9317
	Noise	0.0012	0.0085	0.9039–0.9388
	Targeted	0.0002	0.0011	0.9891–0.9940
	Pooled F1	0.0013	0.0109	0.8785–0.9081
	Clean FPR (%)	0.0024	0.0026	0.0470–0.0619

Table 6.5. Selected-system means for the original five model seeds and the added five. Each group averages 30 cells on the same six sources within its network. Clean FPR is expressed as a percentage.

Metric	Modena 701–705	706–710	L-Town 701–705	706–710
Random	0.9542	0.9557	0.9920	0.9919
Replay	0.9280	0.9286	0.7303	0.7320
Drift	0.8262	0.8286	0.8986	0.8993
Noise	0.8455	0.8422	0.9282	0.9275
Targeted	0.9674	0.9686	0.9918	0.9918
Pooled F1	0.8672	0.8667	0.8951	0.8949
Clean FPR (%)	0.0574	0.0594	0.0548	0.0551

Source-mean pooled F1 ranges from 0.8508 to 0.8851 in Modena and from 0.8821 to 0.9063 in L-Town. These values average ten fits; the cell ranges also reflect fitting variation.

Table 6.6. Source-level means across ten fitted models. Source identifiers refer to separate generated scenario collections within each network.

Network	Source	Replay	Drift	Noise	Pooled F1
Modena	40811	0.9025	0.8110	0.8411	0.8693
Modena	41811	0.9357	0.8879	0.8950	0.8851
Modena	42811	0.9405	0.7922	0.8308	0.8508
Modena	43811	0.9535	0.8504	0.7882	0.8708
Modena	44811	0.9201	0.7747	0.8544	0.8579
Modena	45811	0.9173	0.8481	0.8538	0.8677
L-Town	110811	0.7029	0.8814	0.9304	0.8821
L-Town	111811	0.7297	0.9067	0.9121	0.8914
L-Town	112811	0.7403	0.8492	0.9346	0.8831
L-Town	113811	0.7229	0.9277	0.9272	0.9047
L-Town	114811	0.7372	0.9075	0.9358	0.9025
L-Town	115811	0.7539	0.9214	0.9271	0.9063

6.8 Supplementary comparison with a single classifier

6.8.1 Question and comparison protocol

A supplementary experiment tests whether a simpler classifier can reproduce the selected hybrid's balance of family detection and background errors. It was designed after the primary hybrid results were known. The comparison reuses the final benchmark's training, calibration and evaluation collections, while preserving the selected hybrid models. It tests what the compound classifier and alarm path add when the representation is retained [E9].

Each control contains one LightGBM binary classifier trained jointly on the five attack families. The normal reference is retained, so "single classifier" describes the detection stage, rather than a completely unfitted preprocessing pipeline. The controls use all 116 causal features in Modena and the same features plus the 42 received-pressure-history columns in L-Town. A zero-hour control uses those causal inputs alone. A three-hour control also receives the 59 bounded-window features used by the delayed heads, giving 175 and 217 columns, respectively. Temporal features are computed from complete received series before training endpoints are sampled. Neither control receives expert predictions, routing outputs or family labels at inference.

The three-hour control is the principal complexity comparison because it shares the hybrid's maximum information allowance. It finalizes each endpoint with one delayed score and threshold; the hybrid also retains General's immediate path. The observation types and maximum deadline are aligned, while the hybrid has learned

intermediate scores and aggregated evidence. Consequently, the comparison assesses the complete designs under a common allowance; it is not an isolated ablation of the router.

All ten model seeds are fitted for both controls in both networks. Each fit retains every positive TRAIN endpoint and samples up to 60,000 negative endpoints using the existing General sampling convention. The loss gives half its weight to positive endpoints, equally across families and events, and half to negatives, equally across sampled scenarios. The two controls share the sampled rows and weights within a seed. The fixed LightGBM recipe uses 400 trees, 31 leaves, learning rate 0.03, minimum leaf size 20, L1/L2 penalties of 1 and 20, and feature fraction 0.85. These settings follow the existing full temporal classifier recipe; no hyperparameter search is added.

Calibration searches the distinct score thresholds under the existing clean-FPR ceiling of 0.005. Two operating points are mandatory: maximum pooled F1, and maximum worst-family F1. Each breaks ties using the other F1 criterion and then lower clean FPR. The pooled objective defines the primary comparison; the family-balanced objective checks whether its result reflects a threshold trade-off. These single-threshold rules are explicit alternatives to the hybrid's recorded integration and protection searches.

All 40 classifiers and their 80 operating points are frozen before any supplementary evaluation. Each is then assessed on the same six source datasets used for its network's final hybrid result. Means and paired differences use the 60 model/source cells, following Equations 6.1 and 6.3. Both time allowances and both calibration objectives are reported. Reusing previously assessed sources makes this a supplementary controlled comparison, rather than a fresh independent confirmation.

6.8.2 Performance and complexity trade-offs

Table 6.7. Modena: single-classifier controls versus the frozen hybrid, each averaged over ten model seeds and six shared sources. Single-classifier thresholds maximize calibration pooled F1 under the common clean-FPR ceiling; the hybrid retains its recorded operating points. Clean FPR is a percentage.

Metric	Single, 0 h	Single, 3 h	Hybrid
Random F1	0.9672	0.9740	0.9549
Replay F1	0.9278	0.9060	0.9283
Drift F1	0.7015	0.7627	0.8274
Noise F1	0.6135	0.6742	0.8439
Targeted F1	0.9731	0.9795	0.9680
Family macro F1	0.8366	0.8593	0.9045
Pooled F1	0.8259	0.8509	0.8669
Clean FPR (%)	0.0374	0.0317	0.0584

In Modena (Table 6.7), the three-hour single classifier reaches pooled F1 of 0.8509, compared with 0.8669 for the hybrid. Its drift and noise means are 0.7627 and 0.6742; the hybrid reaches 0.8274 and 0.8439. Replay also improves from 0.9060 to 0.9283 in the hybrid. These gains raise family macro F1 from 0.8593 to 0.9045 and retain every hybrid family mean above 0.80. The simpler classifier has its own advantages: higher random and targeted means, and a clean FPR of 0.0317%, compared with 0.0584% for the hybrid. The benefit of the additional structure is stronger temporal-family coverage and pooled performance, with a higher clean-alarm rate at these operating points.

Family-balanced calibration exposes the single classifier's threshold trade-off in Modena. At three hours it raises drift and noise to 0.8311 and 0.8141, but replay falls to 0.7772 and pooled F1 to 0.7787; clean FPR rises to 0.1921%. Thus the two prespecified single-threshold objectives recover different parts of the family profile. The hybrid preserves replay and pooled performance while retaining strong drift/noise means. Table A.6 supplies all results for the alternative calibration objective.

Table 6.8. L-Town: single-classifier controls versus the frozen hybrid, each averaged over ten model seeds and six shared sources. Single-classifier thresholds maximize calibration pooled F1 under the common clean-FPR ceiling; the hybrid retains its recorded operating points. Clean FPR is a percentage.

Metric	Single, 0 h	Single, 3 h	Hybrid
Random F1	0.9927	0.9950	0.9920
Replay F1	0.6927	0.7084	0.7312
Drift F1	0.7949	0.8698	0.8990
Noise F1	0.7706	0.8422	0.9279
Targeted F1	0.9935	0.9961	0.9918
Family macro F1	0.8489	0.8823	0.9083
Pooled F1	0.8685	0.8977	0.8950
Clean FPR (%)	0.0192	0.0184	0.0550

L-Town shows a different balance (Table 6.8). The three-hour single classifier reaches pooled F1 of 0.8977, slightly above the hybrid's 0.8950, and a lower clean FPR of 0.0184% versus 0.0550%. The hybrid is stronger on replay, drift and noise: 0.7312 versus 0.7084, 0.8990 versus 0.8698, and 0.9279 versus 0.8422, respectively. These differences increase family macro F1 from 0.8823 to 0.9083. Random and targeted favor the single classifier by 0.0030 and 0.0043. The hybrid detects the weaker families better but produces more background alarms; pooled F1 is similar.

The family-balanced L-Town threshold raises the single classifier's drift and noise means to 0.8800 and 0.8653. Replay remains at 0.7085, pooled F1 is 0.8929, and clean FPR is 0.0325%. It narrows some differences but retains lower replay, drift and noise scores than the hybrid. Across the two frozen threshold objectives, the single classifier remains an option when pooled F1 and fewer false alarms matter most. The hybrid has better coverage of replay, drift and noise.

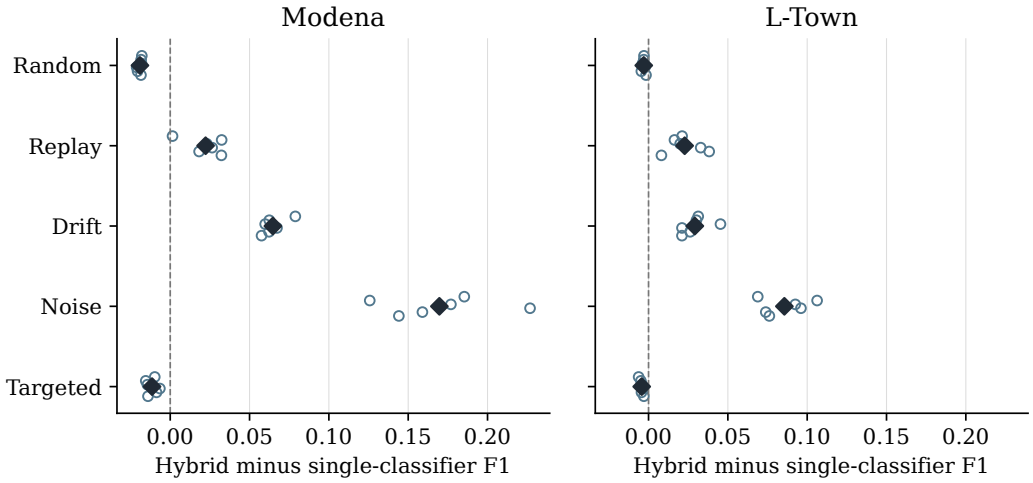


Figure 6.5. Family-F1 differences between the frozen hybrid and the three-hour single classifier at its pooled-F1 operating point. Each circle averages ten model seeds within one evaluation source; diamonds mark the six-source mean. Positive values favor the hybrid. The source spread is descriptive, not a confidence interval.

The paired results support these distinctions across fitted realizations. Relative to the primary three-hour control, the hybrid has higher drift, noise and family macro F1 in every one of the 60 cells in each network. Replay is higher in 54 Modena cells and all 60 L-Town cells. Pooled F1 is higher in 59 Modena cells but only 20 L-Town cells. Clean FPR is higher for the hybrid in every cell of both networks. Figure 6.5 shows the family differences after averaging within sources; the accompanying frozen summary also records model and source variation.

Giving the single classifier three hours also improves its own pooled mean, from 0.8259 to 0.8509 in Modena and from 0.8685 to 0.8977 in L-Town. Drift and noise improve in both networks. Later evidence therefore helps a simple joint classifier as well as specialized branches. Under that allowance, the compound design improves temporal-family coverage, especially in Modena, although it does not lead on every metric. The zero- and three-hour controls are separately fitted and calibrated; their difference concerns the complete information-allowance configurations.

6.9 Operating-point sensitivity and component analysis

This analysis reuses the final fitted models and existing calibration and evaluation collections, without further fitting or data generation [E10]. The three-hour LightGBM control shares the hybrid's maximum information allowance and is the primary comparator; the zero-hour control supplies context. The hybrid also retains

General's immediate alarm path, so the two systems do not have identical decision timing.

A common multiplier λ scales the hybrid's original raw branch clean-tail budgets while their relative allocation remains fixed. The 82 settings comprise zero and 81 logarithmically spaced values from 0.01 to 100. Order-statistic quantiles of clean calibration scores determine the thresholds; $\lambda = 1$ preserves the original thresholds exactly. Alarms require scores strictly above their thresholds, whereas verifier gates retain their at-or-above convention. A zero clean-tail budget can still permit attack-period alarms. Internal routing, verifier cutoffs, the external replay veto and temporal features stay fixed. The verifier's original specialist-score pre-thresholds also stay fixed throughout the λ sweep: endpoints outside each fitted candidate region keep the pass-through value of 1, even when the final alarm threshold is lowered. The path therefore describes threshold sensitivity for the retained branch allocation and guards, rather than an exhaustive frontier or an intrinsic hybrid AUPRC.

The calibration clean-FPR caps are 0, 0.01%, 0.025%, 0.05%, 0.10%, 0.25% and 0.50%. Each fit selects settings on its pooled calibration collections for two objectives: maximum pooled F1 and maximum worst-family F1. Ties use the other F1 criterion, lower clean FPR, then the stricter setting. Hybrid selection searches the specified path; LightGBM searches its distinct tied-score thresholds exactly. All twenty model/network selections were frozen before this evaluation reanalysis. LightGBM's display curves use clean-calibration quantiles at budgets 0.005λ on the finite grid; its cap-constrained search is not restricted to that grid.

Figures 6.6 and 6.7 plot performance against achieved evaluation clean FPR and mark the original operating points. Appendix B gives every cap, objective and achieved calibration/evaluation rate. Means weight the ten-model by six-source cells equally; F1 and nonlinear summaries are calculated within each cell before averaging. Worst-family F1 is consequently the mean of cellwise minima. The protocol was specified after the original evaluation results were known, so these reused sources supply retrospective sensitivity evidence. The original thresholds and research-goal assessment remain unchanged.

Eight diagnostics use the same scores and original thresholds: the full rule, General only, General plus Drift, General plus Noise, specialists only, verifier bypass, external replay-veto removal, and removal of both guards. Remaining branches retain their guards. Removing a specialist's alarms leaves its score channels in the shared router, feedback and verifier inputs; the specialists-only rule does not consume General's output. These are conditional alarm and guard interventions,

not separately retrained architectures or measurements of computational savings. Appendix B reports all diagnostic scores. Their interpretation shares the source-reuse and component-identification limits discussed in Chapter 7.

With pooled-F1 calibration, the hybrid has higher mean pooled F1 at five of seven caps in Modena and six in L-Town. LightGBM leads at the zero cap in both networks and at 0.01% in Modena; it also has higher random and targeted family means at every cap under both objectives. In Modena, the hybrid improves drift and noise at every positive cap. At the 0.025% cap, its pooled F1 is 0.8652 versus LightGBM's 0.8503, with evaluation clean FPRs of 0.02705% and 0.02642%. The false-positive cost depends on the objective: at the 0.1% cap under worst-family-F1 calibration, hybrid pooled F1 is 0.8613 versus 0.8240, while its evaluation FPR is lower, 0.07066% versus 0.10296%. A common calibration cap therefore permits different evaluation rates, including rates above that cap.

L-Town exposes a replay tradeoff. At the 0.025% cap under pooled-F1 calibration, hybrid pooled F1 is 0.9127 versus 0.8977, with evaluation clean FPRs of 0.01950% and 0.01843%. This reverses the original pooled-F1 ordering, but hybrid replay is now lower: 0.6903 versus LightGBM's 0.7084. Hybrid replay remains lower at all seven caps under this objective. Worst-family-F1 calibration restores its replay advantage at caps of 0.05% and above. At 0.05%, hybrid/control replay F1 is 0.7307/0.7091 and pooled F1 is 0.9010/0.8937, while evaluation clean FPR is 0.04675%/0.03094%. The original replay advantage thus depends on the selection objective and false-positive allowance.

The branch diagnostics identify complementary contributions at the original thresholds. General alone has pooled F1 of 0.7200 in Modena and 0.8565 in L-Town, compared with 0.8669 and 0.8950 for the full rule. Adding the specialists raises drift/noise F1 from 0.4979/0.4491 to 0.8274/0.8439 in Modena and from 0.8149/0.7877 to 0.8990/0.9279 in L-Town. Specialists alone retain strong drift/noise scores but reach replay F1 of only 0.0234 and 0.1103. The full rule has higher pooled F1 than every branch-removal variant in all 60 cells of each network. It nevertheless sacrifices L-Town replay performance: General alone reaches 0.7411 and General plus Drift reaches 0.7416, compared with 0.7312 for the full rule. Omitting Noise alarms improves replay in 58 cells and leaves it unchanged in two, while reducing pooled and noise F1 in all 60.

Bypassing the verifier reduces pooled F1 to 0.6737 in Modena and 0.7273 in L-Town. Clean FPR rises from 0.0584% to 0.4578% and from 0.0550% to 0.3409%, respectively; pooled F1 falls and clean FPR rises in all 60 cells of both networks. Modena's family macro F1 changes much less, from 0.9045 to 0.9019, because its

family-conditioned populations omit much of the clean-period background counted by pooled F1. These results establish the verifier's alarm-filtering contribution with its fitted inputs fixed, without isolating the value of its early-warning features.

The external replay veto has a smaller effect while verification remains active. Removing it changes Modena pooled/replay F1 to 0.8663/0.9219 and L-Town to 0.8946/0.7281. L-Town noise F1 instead increases from 0.9279 to 0.9304. Removing both guards lowers pooled/replay F1 further, to 0.6643/0.7927 and 0.7136/0.6518. The larger replay losses after verification is bypassed show that guard effects depend on one another; they cannot be added as independent contributions. General's internal routing is unchanged wherever General alarms contribute. The repeated directions across cells describe the six shared source collections, rather than 60 independent data replications.

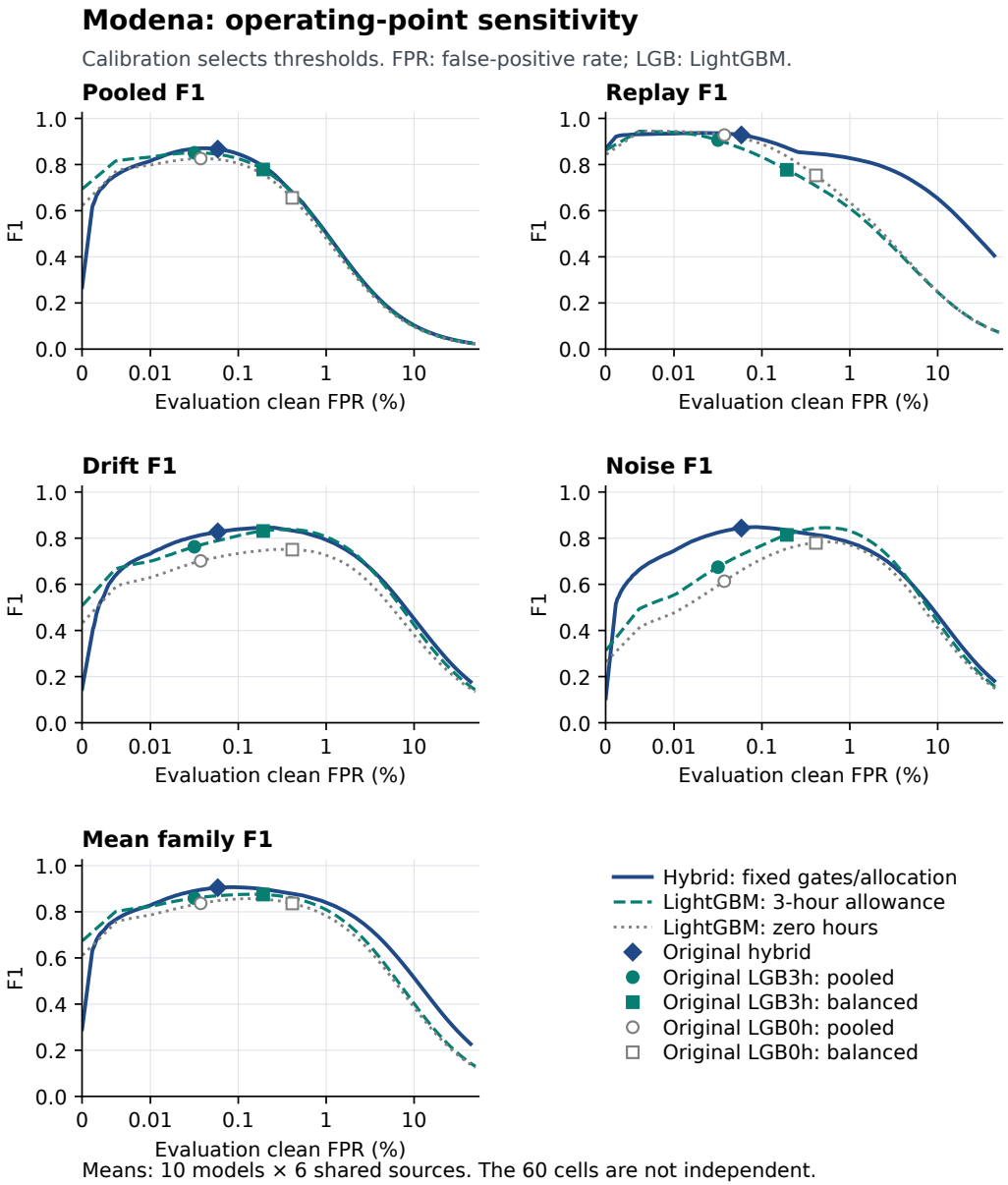


Figure 6.6. Modena operating-point sensitivity on the existing evaluation sources. Curves follow the specified calibration-derived threshold paths; symbols mark original operating points. The horizontal axis shows achieved evaluation clean false-positive rate (FPR) in percent, with a linear region up to 0.01% and logarithmic scaling above it. All points are equal-weight means over 60 shared model/source cells. The hybrid path fixes branch allocation and all guard settings.

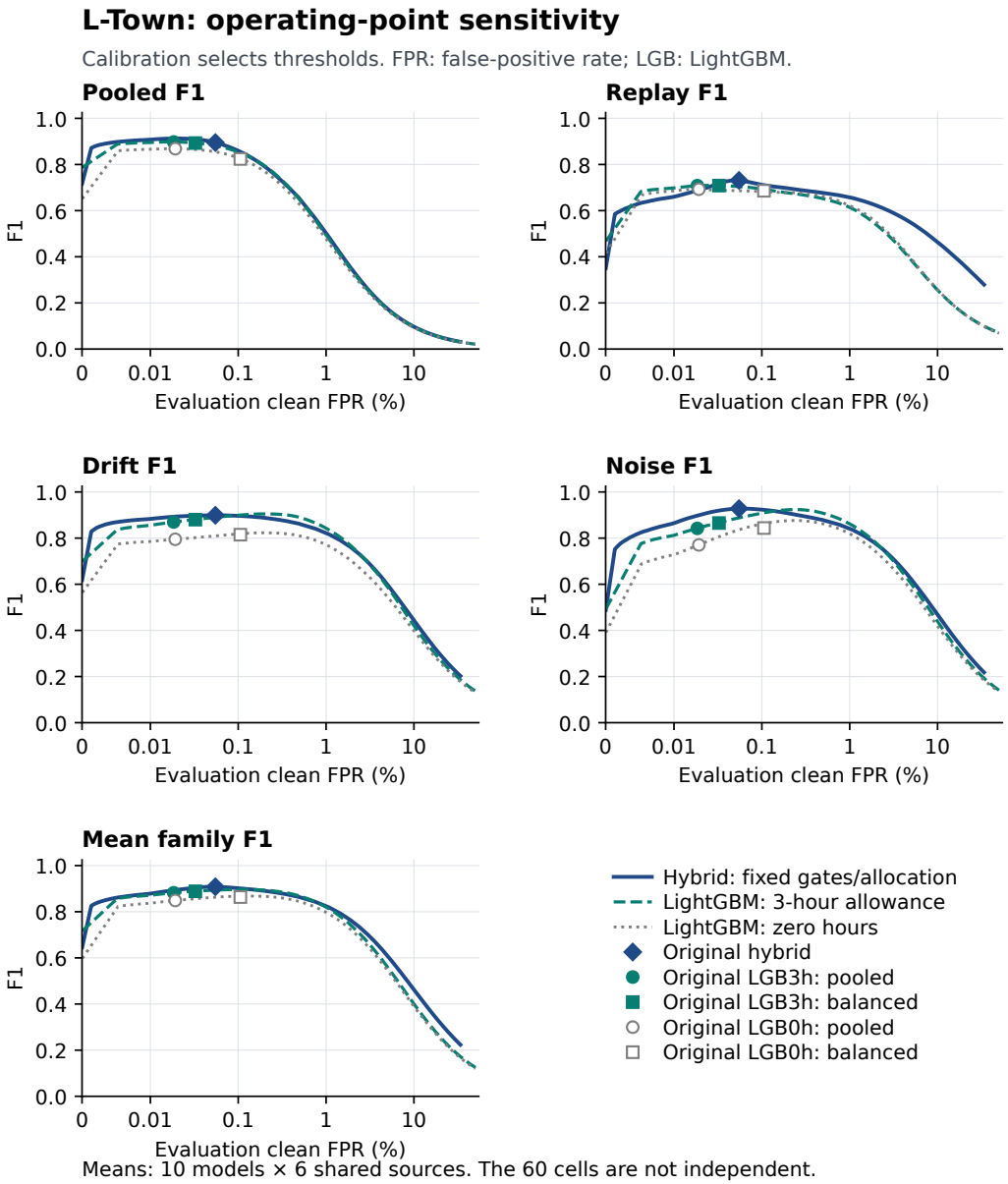


Figure 6.7. L-Town operating-point sensitivity under the same supplementary protocol. Evaluation clean FPR is observed separately from the calibration cap. The three-hour LightGBM control is primary; the zero-hour control supplies timing context. Curves are finite threshold paths, not an exhaustive hybrid frontier. All points average 60 shared model/source cells.

6.10 Exclusion and iterative reweighting in the pressure reference

The historical reference comparison changed several operations together. A separate 2×2 experiment varies target-group exclusion, denoted by E, and iteratively reweighted least squares (IRLS), denoted by R, within the retained final reference. Values of one enable the operation and zero disable it. The experiment uses model seeds 701–703 and the same six existing evaluation sources per network. The three altered conditions refit the downstream detector; $E = 1, R = 1$ reuses the corresponding retained models after reference, feature and score checks. All four conditions select thresholds under the same calibration policy. Their comparison therefore concerns adapted detectors with a shared representation, rather than substituting reference features into an unchanged classifier.

The shared reference retains its TRAIN-fitted mean, rank-16 basis, four sensor groups, ridge parameter, inner whitening scale and outer residual scale. All conditions use three views and median aggregation. Disabling exclusion admits observed target-group readings into the snapshot fit while preserving the original non-target view membership. Disabling IRLS reduces six weighted ridge solves to one solve with binary visibility weights. Whitening and median aggregation remain present, so this intervention isolates iterative reweighting rather than removing every source of robustness. Non-target observation support has the same definition in every condition; prediction and view spread are recomputed. Keeping both scales fixed also preserves the normalization and binning of L-Town's received-history features.

The saved full and source-held references preserve their original fitting populations. Modena's full reference uses its earlier 62-scenario TRAIN population, while the downstream detector uses 110 TRAIN scenarios; L-Town uses 120. Six Modena and five L-Town source-held references exclude their respective held source. Each altered condition rebuilds the reference-derived features and refits General, the causal and delayed specialists, the external router and feedback, the early-warning head and the verifiers. L-Town also refits its full-history General mixture. Estimator settings, seed mappings, sampling algorithms and decision-time allowances are unchanged. Later integration layers retain the original source-held stacking procedure, using assembled base out-of-fold predictions. Upstream models are not rebuilt inside every later-layer fold, so the procedure is not fully nested cross-validation.

Calibration follows the 82-point multiplier path in Section 6.9. Each condition derives its thresholds from its own clean calibration scores, using the original model seed's branch-budget allocation. Final verifier acceptance cutoffs, the external veto

and L-Town's routing transformation remain fixed. Verifier candidate prethresholds and early-warning abstention are learned again from the altered condition's TRAIN out-of-fold predictions. The primary policy maximizes pooled F1 under a calibration clean-FPR cap of 0.05%. Three secondary policies use worst-family F1 at the same cap and both objectives at 0.50%. Ties use the other F1 criterion, lower clean FPR and then the stricter grid setting. The original family-protection constraints are omitted. Evaluation is permitted only after all four policies for all 24 condition/network/model combinations have been fixed.

Metrics are calculated within each model-seed/source cell. The three model-seed values are averaged within each source, followed by equal averaging over six sources. Worst-family F1 is calculated before averaging; descriptive SDs use the six source means. For a metric M_{ER}, the exclusion effects are $M_{11} - M_{01}$ with IRLS and $M_{10} - M_{00}$ without it. The IRLS effects are $M_{11} - M_{10}$ with exclusion and $M_{01} - M_{00}$ without it. Their interaction is $M_{11} - M_{10} - M_{01} + M_{00}$. These contrasts are formed within paired cells before averaging. F1 differences are absolute; FPR differences use percentage points. A common calibration cap does not require equal achieved evaluation FPR.

The protocol was fixed before fitting the altered models, after historical outcomes were already known. The stricter cap was chosen to examine the previously observed low-false-alarm range. The study reuses evaluation sources and three retained model seeds, and its fixed branch allocations and gate settings were selected for $E = 1, R = 1$. These choices limit the comparison to the declared learning recipe and can favor the retained condition. The experiment supplies conditional evidence about exclusion and IRLS; the original selected system and research-goal assessment remain unchanged.

Table 6.9 reports Modena's primary-policy results. Pooled F1 is 0.8709 with both operations and 0.8752 with IRLS alone, compared with 0.8104 with exclusion alone and 0.8168 with neither. IRLS increases pooled F1 by 0.0606 with exclusion and 0.0584 without it; the corresponding replay gains are 0.3052 and 0.3030. These gains accompany higher evaluation clean FPR. With exclusion enabled, IRLS raises clean FPR from 0.0179% to 0.0378%, a difference of 0.0199 percentage points.

The paired effects in Table 6.10 separate these contributions. Exclusion reduces pooled F1 by 0.0042 with IRLS and 0.0064 without it, with replay differences of -0.0091 and -0.0113. The pooled-F1 interaction is 0.0022, small relative to the IRLS effects. Thus, within this learning recipe, Modena supports iterative reweighting but does not show a mean pooled-F1 benefit from target-group exclusion. The result

also distinguishes the reference's information constraint from an empirical guarantee of improved detection.

L-Town shows a smaller pooled-F1 contribution from IRLS (Table 6.11). Pooled F1 is 0.9129 with both operations and 0.9152 with IRLS alone, compared with 0.9017 with exclusion alone and 0.9011 with neither. The paired IRLS effects are 0.0112 with exclusion and 0.0141 without it. With exclusion enabled, IRLS reduces clean FPR by 0.0042 percentage points; without exclusion, the difference is an increase of 0.0002 percentage points.

Replay reveals an interaction that the pooled means partly conceal (Table 6.12). IRLS changes replay F1 by −0.0002 with exclusion and by +0.0188 without it, giving an interaction of −0.0189. Exclusion changes replay by −0.0113 with IRLS and +0.0077 without it. Thus, neither operation has a uniform replay effect across reference conditions. Under the primary policy, IRLS alone has the highest mean pooled and replay F1 among the four conditions in both networks, although the fixed learning recipe and reused sources prevent interpreting this as a newly confirmed system selection.

Appendix C reports all four calibration policies. Relaxing the clean-FPR cap from 0.05% to 0.50% leaves the pooled-objective evaluation summaries unchanged. Changing the objective has a larger effect. For L-Town with both reference operations, worst-family calibration at the stricter cap raises replay F1 from 0.6876 to 0.7293, while pooled F1 falls from 0.9129 to 0.9015 and clean FPR rises from 0.0191% to 0.0462%. These comparisons support a contribution from IRLS to pooled performance, especially in Modena, but do not establish a consistent benefit from exclusion or a replay improvement independent of calibration policy. The complete counts, contrasts and frozen implementation are available in evidence record E11.

Table 6.9. Modena: primary reference-factorial results. Conditional evaluation means over three model seeds and six shared sources. The primary policy maximizes pooled calibration F1 under a 0.05% clean-FPR cap. F1 uses the 0–1 scale; achieved evaluation clean FPR is a percentage.

E	R	Pooled	Random	Replay	Drift	Noise	Targeted	Clean FPR (%)
0	0	0.8168	0.9142	0.6395	0.7934	0.7892	0.9283	0.0170
0	1	0.8752	0.9438	0.9425	0.8150	0.8507	0.9577	0.0380
1	0	0.8104	0.9038	0.6282	0.7873	0.7860	0.9279	0.0179
1	1	0.8709	0.9429	0.9335	0.8144	0.8334	0.9573	0.0378

E denotes target-group exclusion and R denotes iterative residual reweighting; 1 is enabled and 0 is disabled. The shared reference representation and scales remain fixed. A calibration cap is not a guarantee of evaluation FPR.

Table 6.10. Modena: primary paired factorial effects. Conditional effects under the primary policy, paired within model-seed/source cells. Means average three model seeds within each of six shared sources; SD is the sample standard deviation of the six source means. F1 effects are absolute differences; clean-FPR effects and their SDs are percentage points.

Effect	Pooled F1		Replay F1		Clean FPR (pp)	
	Mean	SD	Mean	SD	Mean	SD
Exclusion, IRLS on	-0.0042	0.0046	-0.0091	0.0113	-0.0002	0.0051
Exclusion, IRLS off	-0.0064	0.0062	-0.0113	0.0148	+0.0009	0.0010
IRLS, exclusion on	+0.0606	0.0150	+0.3052	0.0299	+0.0199	0.0043
IRLS, exclusion off	+0.0584	0.0110	+0.3030	0.0249	+0.0210	0.0069
Interaction	+0.0022	0.0065	+0.0022	0.0200	-0.0011	0.0053

Simple effects are enabled minus disabled for the named factor, with the other factor fixed. Interaction is $M_{11} - M_{10} - M_{01} + M_{00}$. Positive F1 differences indicate higher F1; positive FPR differences indicate more false alarms. SD is descriptive, not a standard error.

Table 6.11. L-Town: primary reference-factorial results. Conditional evaluation means over three model seeds and six shared sources. The primary policy maximizes pooled calibration F1 under a 0.05% clean-FPR cap. F1 uses the 0–1 scale; achieved evaluation clean FPR is a percentage.

E	R	Pooled	Random	Replay	Drift	Noise	Targeted	Clean FPR (%)
0	0	0.9011	0.9806	0.6801	0.8778	0.8921	0.9784	0.0177
0	1	0.9152	0.9941	0.6989	0.8903	0.9038	0.9945	0.0179
1	0	0.9017	0.9834	0.6878	0.8859	0.9016	0.9814	0.0234
1	1	0.9129	0.9938	0.6876	0.8900	0.9028	0.9942	0.0191

E denotes target-group exclusion and R denotes iterative residual reweighting; 1 is enabled and 0 is disabled. The shared reference representation and scales remain fixed. A calibration cap is not a guarantee of evaluation FPR.

Table 6.12. L-Town: primary paired factorial effects. Conditional effects under the primary policy, paired within model-seed/source cells. Means average three model seeds within each of six shared sources; SD is the sample standard deviation of the six source means. F1 effects are absolute differences; clean-FPR effects and their SDs are percentage points.

Effect	Pooled F1		Replay F1		Clean FPR (pp)	
	Mean	SD	Mean	SD	Mean	SD
Exclusion, IRLS on	-0.0023	0.0011	-0.0113	0.0036	+0.0012	0.0005
Exclusion, IRLS off	+0.0006	0.0020	+0.0077	0.0042	+0.0057	0.0008
IRLS, exclusion on	+0.0112	0.0013	-0.0002	0.0070	-0.0042	0.0009
IRLS, exclusion off	+0.0141	0.0019	+0.0188	0.0070	+0.0002	0.0004
Interaction	-0.0029	0.0024	-0.0189	0.0049	-0.0045	0.0009

Simple effects are enabled minus disabled for the named factor, with the other factor fixed. Interaction is $M_{11} - M_{10} - M_{01} + M_{00}$. Positive F1 differences indicate higher F1; positive FPR differences indicate more false alarms. SD is descriptive, not a standard error.

Chapter 7

Discussion and Limitations

7.1 Organization of evidence: answering RQ1

For RQ1, the experiments support combining a broad residual detector with specialists that use different temporal evidence. A specialist is useful when its additional detections justify the errors it introduces into the complete system.

The historical controls and final interventions support different parts of this reasoning. Graph and temporal controls establish that relational context carried predictive information in the episode benchmark. Expert diagnostics expose the limits of family assignment and routing. The later matched interventions assess changes to the selected hybrid. These comparisons explain successive design choices, although they were conducted under different benchmark conditions.

The final branches reflect these choices. General judges broad residual discrepancies, Drift captures directional persistence, and Noise captures excess variability. General's internal tree mixture combines multiple views of a reading; the outer decision process governs the extra alarms supplied by temporal specialists.

Target-group exclusion also gives the reference a clear information constraint: the target reading cannot directly supply its own prediction. Robust fitting, support indicators and temporal context help the detector interpret the resulting discrepancy. Corrupted supporting sensors can still distort the reference. The classifier therefore receives support and reliability information alongside residual magnitude and temporal features. Reference representation and fitting changed together during development, so those historical gains cannot separate their effects. The frozen-rule diagnostics also keep the reference fixed. Section 6.10 addresses target-group exclusion and iterative reweighting in a separate factorial design with downstream refitting and common calibration constraints. Its comparisons are conditional on

the retained representation and learning recipe; whitening and median aggregation remain fixed.

The L-Town history result illustrates this approach within General. Histories of existing residual features failed the first pilot, whereas differences between actually received pressure readings supplied a useful alternative representation. The accepted extension adds these features across the internal mixture and its router. Its full-system evaluation improved replay and all other family means. In this setting, changing what the history represented helped more than extending the history of existing residual features.

The factorial experiment does not show a consistent performance benefit from target-group exclusion. Under its primary calibration policy, iterative reweighting improves pooled F1 in both networks, with a much larger replay contribution in Modena. Exclusion has no consistent pooled-F1 benefit, and its L-Town replay effect changes sign depending on whether iterative reweighting is enabled. The information constraint therefore remains a property of the selected design, while its pooled-F1 benefit is unsupported by this comparison. These results concern refitted detectors under fixed representation and gate settings; they do not identify an independently optimized replacement for the selected system.

7.2 Integration and its costs: answering RQ2

Modena illustrates the precision–sensitivity trade-off in RQ2. The matched integrated candidate raises pooled F1 from 0.6863 to 0.8672 and reduces false positives by 84.2%, while true positives fall by 4.1%. Drift and noise F1 decline by 0.0071 and 0.0051. The reduction in false alarms outweighs the lost detections in pooled F1, despite the small drift and noise losses.

This result explains why specialist quality must be assessed at the final alarm level. An accurate specialist can still contribute redundant true positives and many additional false positives. Conversely, suppressing an alarm can improve precision while losing a true detection outside General's coverage. Preserving General protects its existing predictions, but it does not recover every true positive removed from a specialist. Modena shows both the precision benefit and the lost specialist detections.

The threshold-only Modena alternative reaches pooled F1 of 0.6998, well below the integrated candidate. The larger benefit therefore belongs to the package of verification, early-warning context and calibration. A matched factorial ablation would be required to allocate separate effects to those components.

L-Town demonstrates a different balance. Its final history intervention increases pooled F1 from 0.8862 to 0.8950 while improving all family means and reducing mean clean FPR. Small individual drift and targeted losses remain, but the largest other-family loss is approximately 0.0009 F1. The history change brings a smaller improvement than Modena's integration change, but also preserves the required performance of the existing system.

Family macro F1, pooled F1 and clean FPR describe different aspects of these results. The family average describes balanced coverage; the other metrics expose background errors. Modena demonstrates that excellent family scores alone can coexist with weak pooled precision, while the history comparison shows that a weak-family improvement can also benefit the wider detector.

Decision delay is the other explicit cost. Drift and Noise can use evidence through three hours after the endpoint being judged. The historical Modena comparison in Table 4.2 quantifies the trade-off: drift and noise improve under the delayed configuration, while clean false alarms increase and replay declines slightly. This evidence motivates delayed specialist acceptance, without identifying three hours as an optimal setting for the final architecture. General retains an immediate path, and early warning supplies separate network-time context, but neither removes the delay from the reported specialist task. Whether that allowance is acceptable depends on the response time required by an application. Operational response requires a further evaluation of alarm timing and consequences.

7.3 Consistency across networks: answering RQ3

For RQ3, the repeated fits support stable aggregate performance with distinct family profiles. Modena reaches pooled F1 of 0.8669 and L-Town 0.8950. The original and added five-seed groups produce similar selected-system means, reducing concern that the reported profiles depend on a favorable stochastic fit.

Pooled source-mean SDs are about 0.0118 and 0.0109, compared with model-mean SDs of 0.0034 and 0.0013. The larger variation between sources suggests that further replication should prioritize additional scenario collections. More fits on the existing sources would mainly refine the estimate of training randomness.

The family results also show why both networks are necessary. Modena retains strong replay performance, but drift and noise fall below 0.80 in some source/model cells despite means above that level. L-Town retains strong performance on those specialized families, while replay averages 0.7312 and remains below 0.80 in every cell. The history extension improves replay in all 60 paired comparisons, yet most

scores remain below the later 0.75 objective. The replay gain is repeatable, but it remains too small to meet the target in most cells.

L-Town's larger network tests the approach beyond the original development setting, but it is not uniformly more difficult: several family scores exceed their Modena counterparts. Topology, hydraulic behavior, generated observations and fitted representations all differ. For RQ3, the two implementations produce consistent aggregate results, but their weaker families differ and performance still depends on the evaluation source.

7.4 What the complexity comparison establishes

At the original operating points, the hybrid retains all five Modena family means above 0.80 and exceeds the three-hour control calibrated for pooled F1 in pooled performance. Recalibrating that control for family balance recovers drift/noise performance while sacrificing replay and pooled performance. These original comparisons favor the hybrid when temporal-family coverage is the priority.

In L-Town, the same pooled-F1-calibrated control has slightly higher pooled F1 and fewer clean false alarms, while the hybrid has higher replay, drift and noise F1 in all 60 paired cells. Each single classifier uses one decision clock: immediate for the zero-hour control and delayed for the three-hour control. The hybrid preserves an immediate General path alongside delayed specialists. These performance and information-time differences do not quantify runtime or operator workload.

Section 6.9 shows why these original comparisons cannot establish superiority across operating points. Pooled-F1 calibration can reverse both L-Town orderings, giving the hybrid higher pooled F1 but lower replay F1. Under worst-family-F1 calibration, its replay advantage returns at less restrictive caps, with more evaluation false alarms. The component diagnostics also expose a replay cost within the hybrid: retaining General alone or General plus Drift improves L-Town replay while reducing pooled and noise F1. Specialist alarms and verification therefore make measurable, conditional contributions; the simpler classifier remains competitive when their temporal-family gains do not justify the additional errors or structure.

The control shares the retained reference and feature representation and uses a fixed temporal-classifier recipe. The comparison is limited to that recipe. A wider search over joint classifiers could change the comparison. The present experiment reports its two prespecified calibration objectives in full and preserves the original hybrid selection.

External methods such as GDN, TranAD and Graph-MoE remain outside the completed comparison. The thesis prioritizes matched internal controls because its questions concern representation, specialization and alarm integration within the selected framework. Fair external benchmarking would require adaptation and retraining with common endpoint labels, observation masks, training information, calibration conditions and decision-time allowances. Published scores on other datasets cannot supply that comparison. Superiority over those external detectors remains untested.

7.5 Limits of the evidence and priorities for further work

The main external-validity limitation is the observation and attack model. The evaluation uses simulated telemetry, independent missingness and specified corruption processes. These choices make labels and information access inspectable, but they cover only part of the variability that an operational system may encounter. Correlated outages, changing sensor availability, altered demand regimes and attacks adapted to the detector would require separately defined evaluations. Such studies should fix the evaluation protocol before testing and separate the effects of new data conditions from changes to the model.

A second limitation concerns the decision unit. The detector classifies available pressure readings. Clean-period rates of approximately 58.4 and 55.0 false alarms per 100,000 endpoints are informative about background precision, but operator workload depends on how repeated alarms are grouped across sensors and time. A deployment evaluation would therefore need incident aggregation, event-level recall and time to first actionable alert. The selected detectors use pressure inputs. Chapter 3 explains the indirect role of flow corruption in the stored family codes and clean-period subset.

The replication covers ten fitted models and six evaluation sources per network. These six sources were first used in each network's original five-model confirmation, then reused for the five-seed extension, the ten-model summary and the single-classifier comparison. Operating points were selected on calibration data, but prior knowledge of evaluation outcomes can still influence later study choices. Possible selection bias at the study-design level means that these reused comparisons are not equivalent to an untouched final test. Endpoints within episodes and fits evaluated on the same source remain dependent. Source-level summaries preserve this structure, while larger claims about generalization would benefit from additional

independent collections. The supplementary sensitivity protocol was fixed after the original evaluation outcomes were known. Although its thresholds were selected on calibration data before reanalysis, the reused evaluation sources provide retrospective comparisons rather than fresh confirmation. The original selections and research-goal assessment remain in force. The calibration amendment on L-Town must also remain part of the record: the original absolute family-floor constraints were infeasible, and final acceptance follows the declared amended and protected procedures.

Further architectural work should follow the observed errors. In Modena, the priority is recovering drift/noise detections lost through specialist suppression without restoring the clean-period false-positive burden. In L-Town, it is identifying received-observation relationships that separate plausible replay from normal repetition. Any candidate should protect all families and clean periods before a fresh confirmation.

A deployment study would also require computational cost, memory, end-to-end delay and changing-support measurements. Prediction runtime is separate from the three-hour information allowance; the two-network accuracy comparison supplies no runtime scaling estimate. These measurements are needed to assess whether the detector is practical to deploy.

Chapter 8

Conclusion

This dissertation developed a hybrid detector for sensor attacks under incomplete observations. The experiments showed that strong individual experts do not necessarily make a better combined detector: their detections may overlap while their false positives accumulate. The final design addresses this problem by combining complementary evidence and checking the alarms added by specialists.

General combines five internal residual classifiers using a target-group-excluding reference. Drift and Noise add temporal evidence with a declared three-hour decision delay. Routing, feedback and verification check the specialist alarms while General retains an immediate decision path. This design developed through graph and temporal experiments, inspection of expert errors and the rejection of alternatives that failed the evaluation criteria.

The two networks benefited from different changes. In Modena, guarded integration raised pooled F1 from 0.6863 to 0.8672 and reduced false positives by 84.2%, with small average losses in drift and noise F1. In L-Town, adding received-pressure history to General improved every family mean, pooled F1 and mean clean-period FPR in the final matched comparison.

The final assessment retained ten model seeds and six evaluation sources per network. Modena achieved mean pooled F1 of 0.8669 and family macro F1 of 0.9045, with all five family means above 0.80. L-Town achieved 0.8950 and 0.9083, respectively. Four of its family means exceeded 0.89; replay remained at 0.7312. Clean-period false-positive rates were 0.0584% for Modena and 0.0550% for L-Town. Additional fits preserved the broad performance profiles.

The similar macro scores conceal different weaknesses. Modena's drift and noise performance varies by source. In L-Town, replay remains difficult despite improving in every paired history comparison; the later 0.75 objective was not met. Performance on the other families remains strong in the larger network.

The separate three-seed reference factorial in Section 6.10 isolates iterative reweighting and target-group exclusion within the retained representation. After downstream refitting and common-policy calibration, iterative reweighting improves pooled F1 in both networks and substantially improves Modena replay. Exclusion supplies no consistent pooled-F1 benefit, and L-Town replay depends on the interaction between the two operations. These findings support a narrower account of the reference's contribution: iterative reweighting helps under the tested recipe, while the exclusion constraint has no consistent performance advantage.

At the original operating points and against the pooled-F1-calibrated control with the same maximum three-hour allowance, the hybrid improves pooled F1 and temporal-family coverage in Modena. In L-Town, the simpler classifier has slightly higher pooled F1 and fewer clean false alarms, while the hybrid has higher replay, drift and noise F1 in every paired cell. Supplementary pooled-F1 calibration reverses both L-Town orderings at positive calibration caps, while the zero cap favors the simpler control in both networks. The fixed-rule diagnostics show that specialists improve pooled F1 and their intended families, while verification limits background errors; they also expose a replay cost from L-Town's Noise alarms. The added complexity has conditional benefits, determined by the attack families and false-positive allowance.

These results concern simulated, observed pressure endpoints under fixed attack and missingness conditions. Repeated fits share evaluation sources, and specialist decisions retain their stated delay. The supplementary analyses reuse those sources and do not replace the original selected system or provide fresh confirmation. Incident-level evaluation, wider observation conditions and computational measurements are still needed before operational use.

Appendix A

Evidence Register and Reproducibility Records

A.1 Access to the project evidence

The source code and selected experiment records are available in the project repository at `https://github.com/onurozansunger/wdn-thesis`. The E1–E9 records and their implementation links refer to the following fixed commit:

`49036da8016097f2ebb627b3b58cea1a07432a47`

The identifiers E1–E9 refer to the project records in Table A.1. They are separate from the published literature listed in the bibliography. The linked evidence register maps each identifier to its constituent files. This appendix reproduces the principal numerical comparisons and explains their calculation.

The repository contains 218 byte-identical experimental records under `results/evidence/records/`. Its manifest lists their original relative paths, byte counts and SHA-256 digests. The final result file contains the ten-model-seed evaluation for both networks. Verification scripts check file integrity and recompute selected summaries from the saved scores and confusion counts.

These records support inspection of the reported calculations. They do not include the complete simulation datasets, prediction arrays or fitted model binaries. Reproducing predictions from raw telemetry requires those additional artifacts from the research archive; running the supplied checks does not retrain the detector. Historical paths retained inside the records describe the original experiments. The repository register provides their accessible locations.

Table A.1. Project experiment records. Each identifier links to the corresponding file list in the repository at the fixed commit.

Record	Contents	Scope
E1	Campaign protocol, results and four corpus manifests	Training/calibration organization and matched integration
E2	Ten-seed results, selections and artifact audit	Frozen selected implementations
E3	Generator, configurations and input networks	Observation and label construction
E4	Seven episode/control result files	Historical matched studies
E5	Reference and specialist reports and modules	Representation development
E6	Delay, latency and EVAL-3 reports	TRAIN screen, calibrated 0/3-hour comparison and guarded evaluation
E7	Router confirmation report	Earlier independent recalibration
E8	Pilots, replication and protected-history records	Diagnostic gates and selected confirmation
E9	Protocol, model settings and complete source reports	Supplementary single-classifier controls

A.2 Historical controlled comparisons

Table A.2 gives the data-seed means underlying Table 4.1. Each is an average of three paired model fits. The archived summaries retain both these means and the constituent model-pair differences. The 95% intervals reported in Chapter 4 use the three data-seed means as their inference units. MoE–temporal confirmation is available for Modena only in this retained campaign.

Table A.2. Historical paired AUPRC differences, averaged within each data seed.

Comparison	Network	Data seed	Mean difference
Temporal–spatial	L-Town	101	+0.072093
Temporal–spatial	L-Town	202	+0.101270
Temporal–spatial	L-Town	303	+0.093783
Temporal–spatial	Modena	101	+0.075733
Temporal–spatial	Modena	202	+0.082902
Temporal–spatial	Modena	303	+0.080704
Topology–free	Modena	101	+0.069489
Topology–free	Modena	202	+0.080279

Comparison	Network	Data seed	Mean difference
Topology–free	Modena	303	+0.105228
Topology–free	L-Town	101	+0.065500
Topology–free	L-Town	202	+0.084016
Topology–free	L-Town	303	+0.096546
MoE–temporal	Modena	101	+0.046349
MoE–temporal	Modena	202	+0.030684
MoE–temporal	Modena	303	+0.023379

A.3 History confirmation: the complete original pairing

The baseline and received-history system were evaluated on the same six sources for model seeds 701–705 after design selection. Table A.3 reproduces all thirty replay pairs. For each cell, the repository contains the original true-positive, false-positive and false-negative counts for replay and the other families, together with the pooled and false-alarm measures. These records are listed under [E8]; each model seed has a separate `confirmation.json` file.

A cell's F1 is calculated as $2TP/(2TP + FP + FN)$. The reported comparison averages the thirty cell F1 values with equal weight. It is therefore different from concatenating the confusion counts before calculating F1. The original full-precision means are 0.706383226 and 0.730307195, respectively; the values rounded to four decimal places are 0.7064 and 0.7303. All thirty paired differences are positive.

The older 0.6757 router result uses a different confirmation population and is separate from these paired results. The subsequent extension adds five model fits on the same six history-confirmation sources and is recorded separately in the final ten-seed release.

Table A.3. L-Town replay F1 in the original frozen history confirmation. Values are rounded to six decimals here; full precision and confusion counts are retained in the repository.

Model seed	Source	Baseline	History	Difference
701	110811	0.672897	0.696367	+0.023470
701	111811	0.702670	0.723112	+0.020442
701	112811	0.714070	0.736240	+0.022171

Model seed	Source	Baseline	History	Difference
701	113811	0.693733	0.718251	+0.024517
701	114811	0.706500	0.741714	+0.035213
701	115811	0.738191	0.750378	+0.012188
702	110811	0.682353	0.698565	+0.016212
702	111811	0.701878	0.726426	+0.024548
702	112811	0.706553	0.733569	+0.027016
702	113811	0.692177	0.711750	+0.019573
702	114811	0.707084	0.727732	+0.020649
702	115811	0.733025	0.745995	+0.012971
703	110811	0.678571	0.700960	+0.022389
703	111811	0.706675	0.732869	+0.026194
703	112811	0.710210	0.741269	+0.031058
703	113811	0.708633	0.738754	+0.030121
703	114811	0.709079	0.740694	+0.031615
703	115811	0.753647	0.765661	+0.012015
704	110811	0.683544	0.709591	+0.026047
704	111811	0.701019	0.734694	+0.033675
704	112811	0.712166	0.744962	+0.032796
704	113811	0.698497	0.721088	+0.022592
704	114811	0.707190	0.743494	+0.036305
704	115811	0.746434	0.760331	+0.013896
705	110811	0.679943	0.704206	+0.024263
705	111811	0.701456	0.727480	+0.026024
705	112811	0.709050	0.728551	+0.019501
705	113811	0.694718	0.720613	+0.025895
705	114811	0.705882	0.732958	+0.027076
705	115811	0.733652	0.750941	+0.017288

A.4 Development decisions and their evidence scope

The initial residual-history pilot added 56 columns and reduced replay F1 from 0.7173 to 0.6954 on the held-source TRAIN diagnostic. The subsequent received-history pilot improved it to 0.7432 on that diagnostic. Its remaining-fold replication showed replay gains in all twenty primary paired comparisons but failed the declared noise-protection gate. That failure remains part of the record. The protected full-system experiment was a subsequent design with its own calibration protections and confirmation, as represented in Table A.3.

The repository contains the pilot reports, the replication protocol and outcome report, and the protected-system frozen protocol and selections. The blind-reference and temporal-specialist reports likewise retain their own development populations. The earlier EVAL-3 feedback reports and the later repeated-fit campaign are separate evidence groups. The register identifies which records concern development, failed acceptance checks or frozen evaluation.

A.5 Architecture-to-code map

Table A.4 identifies the principal implementation points for the detector described in Chapter 5. Module paths are relative to `src/wdn/`; experiment scripts are under `thesis_v2/experiments/early_warning/`. Each filename links to its implementation at the fixed commit. Archived source copies and their hashes are also retained with the experiment records. The ordered 116-column bank is in the corpus manifests; the observed-history module defines the additional 42 columns.

Table A.4. Code locations for the formal method.

Component	Module or routine
Base/robust reference	`models/blind_reference.py`; `models/robust_reference.py`
General components and profiles	`probe_residual_experts.py`
Received history	`observed_history.py`; `shared_history.py`
Daily comparisons	`seasonal_pressure_features.py`
Causal specialists	`models/seasonal_family.py`; `latency_deployment.py`

Component	Module or routine
Delayed features and heads	`delayed_decision_features.py`; `models/delayed_decision.py`
External router and veto	`evidence_feedback.py`
Early warning	`early_warning.py`
Candidate verifier	`alarm_verifier.py`
Fitting and score combination	Experiment scripts `run_campaign.py`; `screen_verifier.py`
L-Town protected integration	Experiment script `protected_history_system.py`
Ten-seed release	Experiment script `final_ten_model_seeds.py`
Single-classifier comparison	Experiment script `single_model_baseline.py`

A.6 A concrete frozen operating-point example

Table A.5 gives model seed 701 as an example of how the rules are stored. These values are not recommended universal thresholds. Each fit has its own calibration-selected settings, and the repository contains the operating-point records for all ten model seeds in both networks. Modena uses its Stage-E candidate settings. L-Town uses its selected full-history General settings and preserved specialist thresholds.

Table A.5. Selected rule values for model seed 701; thresholds rounded to six decimals for display.

Parameter	Modena	L-Town
General threshold	0.985178	0.882722
Drift threshold	0.959174	0.928356
Noise threshold	0.873057	0.905837
Drift verifier cutoff	0.000000	0.650000
Noise verifier cutoff	0.500000	0.350000
Internal-router temperature	1 (identity)	2.0
Uniform shrinkage	0	0.2
Veto margin / feedback cutoff	0.10 / 0.10	0.10 / 0.10

A.7 Verifying the repository records

The following commands retrieve the cited version and check its records. The verification scripts use the Python standard library and do not require installation of the model-training dependencies.

```
git clone https://github.com/onurozansunger/wdn-thesis.git
cd wdn-thesis
git checkout --detach 49036da8016097f2ebb627b3b58cea1a07432a47
python3 scripts/verify_release.py
python3 results/evidence/verify.py
```

The first script checks the release manifest, frozen protocol and evaluation hashes, all 120 model/source cells and their network means. It also validates the E1–E9 file map. The second checks all 218 evidence-record hashes, reconstructs the original thirty-pair replay means, verifies all 480 supplementary source/model/operating-point reports and their 80 frozen thresholds, and checks the historical zero- and three-hour counts.

A successful hash check establishes consistency with the cited repository version; it does not establish the correctness of an experiment by itself. The protocols, scoring code, other-family counts and configuration records remain available for direct inspection. The reproduction guide describes the additional data and model artifacts required for a full rerun.

A.8 Supplementary single-classifier evidence

The supplementary comparison retains both calibration objectives for every fitted classifier. Table A.6 reports the alternative thresholds selected to maximize worst-family F1 under the same clean-FPR ceiling. These points were frozen alongside the pooled-F1 points before evaluation; neither objective was chosen using the reported evaluation scores. The full per-source confusion counts, per-model settings, input hashes and frozen rules are available in the repository under evidence record [E9].

Table A.6. Single-classifier means when calibration maximizes worst-family F1. Both time allowances use ten model seeds and six sources per network. Clean FPR is a percentage.

	Modena		L-Town	
Metric	Single, 0 h	Single, 3 h	Single, 0 h	Single, 3 h
Random F1	0.9501	0.9743	0.9868	0.9943
Replay F1	0.7529	0.7772	0.6860	0.7085
Drift F1	0.7503	0.8311	0.8149	0.8800
Noise F1	0.7793	0.8141	0.8437	0.8653
Targeted F1	0.9480	0.9767	0.9866	0.9954
Family macro F1	0.8361	0.8747	0.8636	0.8887
Pooled F1	0.6556	0.7787	0.8239	0.8929
Clean FPR (%)	0.4115	0.1921	0.1057	0.0325

The historical latency comparison in Table 4.2 is documented in its evaluation report and frozen selection. The corresponding evaluation script is also available. That record aggregates one earlier configuration across six sources. The evidence verifier recomputes family F1 from its counts and checks the link to the frozen selection. It does not treat these historical scores as final ten-seed results.

A.9 Additional experiment records

Record E10 covers the flow-development studies in Section 4.9 and the supplementary analysis in Section 6.9. The supplementary evidence guide maps these records to their public repository locations at a separate fixed commit:

`300f7274d1158a4e1929f7f97028ed54f975baec`

The E1–E9 records retain the original commit cited above. The flow comparisons are reported in Chapter 4; Appendix B gives the complete selected-point and component tables for the supplementary analysis.

Within the supplementary archive, original relative paths are retained beneath `results/supplementary_20261008/records/`. The Modena development record is `thesis_v2/WEAK_FAMILY_RESULTS.md`, supported by the split, calibration and trial reports under `runs/operational/weak_family_v1/`. It compares local pressure, pressure-context and joint pressure/flow features on source 811. The two L-Town records are `thesis_v2/outputs/tables/ltown_flow_fusion_experiment.md` and `thesis_v2/outputs/tables/ltown_received_joint_075.md`. Their corresponding `flow_fusion_v1` and `received_joint_075_v1` experiment directories retain the frozen protocols, 25 paired TRAIN reports and the full-system calibration decisions. The first flow candidate failed calibration protection; the joint trajectory/flow

candidate passed protection but failed its progression requirement. Neither received fresh confirmation.

The supplementary operating-point analysis retains a fixed protocol, twenty model/network calibration-selection records and 120 model/source evaluation records. The extracted scores are aligned to the single-classifier prediction metadata by source, scenario, time and sensor. For Modena, the original-point checks reproduce the saved pooled confusion counts and rounded family F1 values; no historical endpoint-decision vector was retained for direct equality checking. For L-Town, they reproduce the saved pooled and family counts and the retained specialist-alarm union. The original fitted models and decision rules are unchanged.

All twenty calibration selections were fixed before scoring the supplementary evaluation. The protocol was written after the earlier evaluation results were known, and the sources are reused. The records therefore support the retrospective comparisons reported here. They include no new model fitting, scenario generation or scoring of the original reserved test split. Repeating score extraction requires the fitted model bundles, feature caches and retained prediction arrays in the research archive.

The supplement's manifest lists 955 archived scientific files with byte counts, SHA-256 digests and original-to-published path mappings. The numerical summary and its constituent calibration and evaluation records are stored under `results/supplementary_20261008/analysis/`. The following commands check the supplement from an existing repository clone, using Python 3.11 or later:

```
git fetch origin
git checkout --detach 300f7274d1158a4e1929f7f97028ed54f975baec
python3 results/supplementary_20261008/verify.py
```

The standard-library checker verifies the published files, calibration/evaluation provenance links and summary arithmetic across all 120 cells. It also checks the retained original-result comparisons. Hashes recorded for omitted models and arrays link the records to the research archive; their bytes cannot be independently verified from this checkout. The supplement reproduction guide describes the required external artifacts and the distinction between checking saved summaries and rerunning inference.

Record E11 covers the exclusion and iterative-reweighting study in Section 6.10 and Appendix C. The factorial evidence guide describes its fixed protocol and reproduction requirements. The design specifies four reference conditions, model seeds 701–703 and six reused evaluation sources per network, with 24 calibration records and 144 condition/model/source cells. Altered conditions refit the downstream

detector; the retained condition reuses its models after parity checks. Whitening and median aggregation remain fixed. This study is separate from the original ten-seed selected-system assessment and the E10 analysis.

The E11 records are fixed at the following repository commit:

```
d50fdcc376d41efa20772af02e2fc9fafd4d206d
```

The E11 manifest maps original paths to published files and records their byte counts and SHA-256 digests. The archive preserves the frozen implementation, protocols, parity checks, calibration paths and confusion counts. The factorial summary records condition means and paired effects. Supporting schema checks and execution records document inputs and scheduling separately from the frozen numerical code.

From a checkout of the E11 publication, the standard-library checker runs with:

```
git fetch origin
git checkout --detach d50fdcc376d41efa20772af02e2fc9fafd4d206d
python3 results/reference_factorial_20261008/verify.py
```

It verifies published bytes, protocol and calibration links, selection arithmetic, count-derived metrics and paired summaries. Raw corpora, model binaries and prediction arrays remain in the research archive. Their recorded hashes support consistency checks but do not verify omitted bytes or reproduce predictions. The evidence guide distinguishes inspection of the archived run from a fresh rerun with those additional artifacts.

Appendix B

Supplementary operating-point and component results

This appendix gives the complete tables for Section 6.9. F1 values and false-positive percentages are rounded to four decimal places. Each network mean averages the same ten fitted models evaluated on six shared source datasets. F1, family macro F1 and worst-family F1 are calculated within each cell before averaging. The worst-family column is therefore the mean of cellwise minima. Shared sources limit the independence of these repeated evaluations, as discussed in Chapter 7.

B.1 Comparison under common calibration constraints

Each cap and objective is applied to both systems using calibration data. Hybrid, LGB3h and LGB0h denote the frozen hybrid, the primary three-hour LightGBM control and the contextual zero-hour LightGBM control. Evaluation FPR is measured after freezing the selected rules and can differ between systems or exceed the calibration cap. The balanced objective maximizes worst-family F1, not family macro F1.

B.1.1 Modena: pooled-F1 objective

Table B.1 reports pooled and individual-family F1 at every cap. Table B.2 reports family balance, achieved calibration clean FPR and achieved evaluation false-positive rates for the same selected rules.

Table B.1. Modena: pooled-F1 objective; detection at each calibration cap

Cal. cap (%)	System	Pooled	Random	Replay	Drift	Noise	Targeted
0.0000	Hybrid	0.2678	0.1544	0.8689	0.1468	0.1061	0.1817
0.0000	LGB3h	0.6944	0.8316	0.8627	0.5083	0.3136	0.8544
0.0000	LGB0h	0.6232	0.7481	0.8441	0.4334	0.2648	0.7686
0.0100	Hybrid	0.8162	0.8510	0.9343	0.7341	0.7444	0.8816
0.0100	LGB3h	0.8337	0.9581	0.9391	0.7027	0.5585	0.9671
0.0100	LGB0h	0.7987	0.9357	0.9439	0.6307	0.4782	0.9471
0.0250	Hybrid	0.8652	0.9278	0.9361	0.8002	0.8163	0.9427
0.0250	LGB3h	0.8503	0.9723	0.9140	0.7538	0.6567	0.9782
0.0250	LGB0h	0.8226	0.9621	0.9352	0.6838	0.5732	0.9685
0.0500	Hybrid	0.8706	0.9438	0.9342	0.8151	0.8329	0.9586
0.0500	LGB3h	0.8509	0.9740	0.9060	0.7627	0.6742	0.9795
0.0500	LGB0h	0.8259	0.9672	0.9278	0.7015	0.6135	0.9731
0.1000	Hybrid	0.8706	0.9438	0.9342	0.8151	0.8329	0.9586
0.1000	LGB3h	0.8509	0.9740	0.9060	0.7627	0.6742	0.9795
0.1000	LGB0h	0.8259	0.9672	0.9278	0.7015	0.6135	0.9731
0.2500	Hybrid	0.8706	0.9438	0.9342	0.8151	0.8329	0.9586
0.2500	LGB3h	0.8509	0.9740	0.9060	0.7627	0.6742	0.9795
0.2500	LGB0h	0.8259	0.9672	0.9278	0.7015	0.6135	0.9731
0.5000	Hybrid	0.8706	0.9438	0.9342	0.8151	0.8329	0.9586
0.5000	LGB3h	0.8509	0.9740	0.9060	0.7627	0.6742	0.9795
0.5000	LGB0h	0.8259	0.9672	0.9278	0.7015	0.6135	0.9731

F1 means over 60 model/source cells. Family balance and achieved false-positive rates appear in Table B.2.

Table B.2. Modena: pooled-F1 objective; family balance and achieved false-positive rates

Cal. cap (%)	System	Macro	Worst	Cal. clean (%)	Eval. clean (%)	Eval. neg. (%)
0.0000	Hybrid	0.2916	0.0643	0.0000	0.0000	0.0001
0.0000	LGB3h	0.6741	0.3136	0.0000	0.0000	0.0001
0.0000	LGB0h	0.6118	0.2648	0.0000	0.0001	0.0001
0.0100	Hybrid	0.8291	0.7144	0.0095	0.0102	0.0105
0.0100	LGB3h	0.8251	0.5585	0.0100	0.0106	0.0134
0.0100	LGB0h	0.7871	0.4782	0.0099	0.0103	0.0120

Continued on next page

Table B.2. Modena: pooled-F1 objective; family balance and achieved false-positive rates (continued)

Cal. cap (%)	System	Macro	Worst	Cal. clean (%)	Eval. clean (%)	Eval. neg. (%)
0.0250	Hybrid	0.8846	0.7866	0.0242	0.0270	0.0295
0.0250	LGB3h	0.8550	0.6567	0.0246	0.0264	0.0332
0.0250	LGB0h	0.8245	0.5732	0.0247	0.0262	0.0298
0.0500	Hybrid	0.8969	0.8039	0.0366	0.0392	0.0440
0.0500	LGB3h	0.8593	0.6742	0.0296	0.0317	0.0396
0.0500	LGB0h	0.8366	0.6134	0.0346	0.0374	0.0422
0.1000	Hybrid	0.8969	0.8039	0.0366	0.0392	0.0440
0.1000	LGB3h	0.8593	0.6742	0.0296	0.0317	0.0396
0.1000	LGB0h	0.8366	0.6134	0.0346	0.0374	0.0422
0.2500	Hybrid	0.8969	0.8039	0.0366	0.0392	0.0440
0.2500	LGB3h	0.8593	0.6742	0.0296	0.0317	0.0396
0.2500	LGB0h	0.8366	0.6134	0.0346	0.0374	0.0422
0.5000	Hybrid	0.8969	0.8039	0.0366	0.0392	0.0440
0.5000	LGB3h	0.8593	0.6742	0.0296	0.0317	0.0396
0.5000	LGB0h	0.8366	0.6134	0.0346	0.0374	0.0422

All FPR columns are percentages. Cal. clean averages ten calibration fits; evaluation columns average 60 cells. Family scores appear in Table B.1.

B.1.2 Modena: worst-family-F1 objective

Table B.3 reports pooled and individual-family F1 at every cap. Table B.4 reports family balance, achieved calibration clean FPR and achieved evaluation false-positive rates for the same selected rules.

Table B.3. Modena: worst-family-F1 objective; detection at each calibration cap

Cal. cap (%)	System	Pooled	Random	Replay	Drift	Noise	Targeted
0.0000	Hybrid	0.2678	0.1544	0.8689	0.1468	0.1061	0.1817
0.0000	LGB3h	0.6944	0.8316	0.8627	0.5083	0.3136	0.8544
0.0000	LGB0h	0.6232	0.7481	0.8441	0.4334	0.2648	0.7686
0.0100	Hybrid	0.8162	0.8510	0.9343	0.7341	0.7444	0.8816
0.0100	LGB3h	0.8337	0.9581	0.9391	0.7027	0.5585	0.9671
0.0100	LGB0h	0.7987	0.9357	0.9439	0.6307	0.4782	0.9471

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Table B.3. Modena: worst-family-F1 objective; detection at each calibration cap (continued)

Cal. cap (%)	System	Pooled	Random	Replay	Drift	Noise	Targeted
0.0250	Hybrid	0.8652	0.9278	0.9361	0.8002	0.8163	0.9427
0.0250	LGB3h	0.8502	0.9724	0.9136	0.7541	0.6572	0.9782
0.0250	LGB0h	0.8227	0.9621	0.9351	0.6840	0.5738	0.9685
0.0500	Hybrid	0.8694	0.9511	0.9306	0.8230	0.8403	0.9656
0.0500	LGB3h	0.8470	0.9768	0.8777	0.7844	0.7184	0.9823
0.0500	LGB0h	0.8238	0.9701	0.9185	0.7144	0.6507	0.9752
0.1000	Hybrid	0.8613	0.9578	0.9231	0.8310	0.8473	0.9698
0.1000	LGB3h	0.8240	0.9789	0.8300	0.8121	0.7713	0.9825
0.1000	LGB0h	0.8029	0.9711	0.8859	0.7345	0.7126	0.9760
0.2500	Hybrid	0.8516	0.9588	0.9157	0.8335	0.8473	0.9699
0.2500	LGB3h	0.7787	0.9743	0.7772	0.8311	0.8141	0.9767
0.2500	LGB0h	0.7272	0.9637	0.8080	0.7507	0.7664	0.9638
0.5000	Hybrid	0.8516	0.9588	0.9157	0.8335	0.8473	0.9699
0.5000	LGB3h	0.7787	0.9743	0.7772	0.8311	0.8141	0.9767
0.5000	LGB0h	0.6556	0.9501	0.7529	0.7503	0.7793	0.9480

F1 means over 60 model/source cells. Family balance and achieved false-positive rates appear in Table B.4.

Table B.4. Modena: worst-family-F1 objective; family balance and achieved false-positive rates

Cal. cap (%)	System	Macro	Worst	Cal. clean (%)	Eval. clean (%)	Eval. neg. (%)
0.0000	Hybrid	0.2916	0.0643	0.0000	0.0000	0.0001
0.0000	LGB3h	0.6741	0.3136	0.0000	0.0000	0.0001
0.0000	LGB0h	0.6118	0.2648	0.0000	0.0001	0.0001
0.0100	Hybrid	0.8291	0.7144	0.0095	0.0102	0.0105
0.0100	LGB3h	0.8251	0.5585	0.0100	0.0106	0.0134
0.0100	LGB0h	0.7871	0.4782	0.0099	0.0103	0.0120
0.0250	Hybrid	0.8846	0.7866	0.0242	0.0270	0.0295
0.0250	LGB3h	0.8551	0.6572	0.0249	0.0267	0.0335
0.0250	LGB0h	0.8247	0.5738	0.0249	0.0263	0.0299

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Table B.4. Modena: worst-family-F1 objective; family balance and achieved false-positive rates (continued)

Cal. cap (%)	System	Macro	Worst	Cal. clean (%)	Eval. clean (%)	Eval. neg. (%)
0.0500	Hybrid	0.9021	0.8122	0.0475	0.0501	0.0570
0.0500	LGB3h	0.8680	0.7183	0.0497	0.0514	0.0630
0.0500	LGB0h	0.8458	0.6484	0.0498	0.0530	0.0592
0.1000	Hybrid	0.9058	0.8207	0.0685	0.0707	0.0804
0.1000	LGB3h	0.8750	0.7643	0.0996	0.1030	0.1210
0.1000	LGB0h	0.8560	0.7025	0.0996	0.1041	0.1146
0.2500	Hybrid	0.9051	0.8227	0.0849	0.0877	0.0987
0.2500	LGB3h	0.8747	0.7677	0.1848	0.1921	0.2177
0.2500	LGB0h	0.8505	0.7354	0.2489	0.2508	0.2745
0.5000	Hybrid	0.9051	0.8227	0.0849	0.0877	0.0987
0.5000	LGB3h	0.8747	0.7677	0.1848	0.1921	0.2177
0.5000	LGB0h	0.8361	0.7254	0.4079	0.4115	0.4462

All FPR columns are percentages. Cal. clean averages ten calibration fits; evaluation columns average 60 cells. Family scores appear in Table B.3.

B.1.3 L-Town: pooled-F1 objective

Table B.5 reports pooled and individual-family F1 at every cap. Table B.6 reports family balance, achieved calibration clean FPR and achieved evaluation false-positive rates for the same selected rules.

Table B.5. L-Town: pooled-F1 objective; detection at each calibration cap

Cal. cap (%)	System	Pooled	Random	Replay	Drift	Noise	Targeted
0.0000	Hybrid	0.7173	0.8877	0.3502	0.6185	0.4884	0.8833
0.0000	LGB3h	0.7870	0.9626	0.4673	0.6988	0.4924	0.9613
0.0000	LGB0h	0.6515	0.8087	0.4147	0.5642	0.3911	0.8098
0.0100	Hybrid	0.9069	0.9919	0.6574	0.8820	0.8608	0.9919
0.0100	LGB3h	0.8954	0.9953	0.6984	0.8555	0.8121	0.9962
0.0100	LGB0h	0.8660	0.9924	0.6854	0.7851	0.7276	0.9932
0.0250	Hybrid	0.9127	0.9938	0.6903	0.8932	0.9000	0.9942
0.0250	LGB3h	0.8977	0.9950	0.7084	0.8698	0.8422	0.9961
0.0250	LGB0h	0.8685	0.9927	0.6927	0.7949	0.7706	0.9935

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Table B.5. L-Town: pooled-F1 objective; detection at each calibration cap (continued)

Cal. cap (%)	System	Pooled	Random	Replay	Drift	Noise	Targeted
0.0500	Hybrid	0.9127	0.9938	0.6903	0.8932	0.9000	0.9942
0.0500	LGB3h	0.8977	0.9950	0.7084	0.8698	0.8422	0.9961
0.0500	LGB0h	0.8685	0.9927	0.6927	0.7949	0.7706	0.9935
0.1000	Hybrid	0.9127	0.9938	0.6903	0.8932	0.9000	0.9942
0.1000	LGB3h	0.8977	0.9950	0.7084	0.8698	0.8422	0.9961
0.1000	LGB0h	0.8685	0.9927	0.6927	0.7949	0.7706	0.9935
0.2500	Hybrid	0.9127	0.9938	0.6903	0.8932	0.9000	0.9942
0.2500	LGB3h	0.8977	0.9950	0.7084	0.8698	0.8422	0.9961
0.2500	LGB0h	0.8685	0.9927	0.6927	0.7949	0.7706	0.9935
0.5000	Hybrid	0.9127	0.9938	0.6903	0.8932	0.9000	0.9942
0.5000	LGB3h	0.8977	0.9950	0.7084	0.8698	0.8422	0.9961
0.5000	LGB0h	0.8685	0.9927	0.6927	0.7949	0.7706	0.9935

F1 means over 60 model/source cells. Family balance and achieved false-positive rates appear in Table B.6.

Table B.6. L-Town: pooled-F1 objective; family balance and achieved false-positive rates

Cal. cap (%)	System	Macro	Worst	Cal. clean (%)	Eval. clean (%)	Eval. neg. (%)
0.0000	Hybrid	0.6456	0.3502	0.0000	0.0000	0.0001
0.0000	LGB3h	0.7165	0.4543	0.0000	0.0000	0.0000
0.0000	LGB0h	0.5977	0.3692	0.0000	0.0000	0.0000
0.0100	Hybrid	0.8768	0.6574	0.0096	0.0094	0.0110
0.0100	LGB3h	0.8715	0.6984	0.0099	0.0100	0.0171
0.0100	LGB0h	0.8367	0.6836	0.0100	0.0099	0.0124
0.0250	Hybrid	0.8943	0.6903	0.0199	0.0195	0.0223
0.0250	LGB3h	0.8823	0.7084	0.0189	0.0184	0.0277
0.0250	LGB0h	0.8489	0.6927	0.0195	0.0192	0.0250
0.0500	Hybrid	0.8943	0.6903	0.0199	0.0195	0.0223
0.0500	LGB3h	0.8823	0.7084	0.0189	0.0184	0.0277
0.0500	LGB0h	0.8489	0.6927	0.0195	0.0192	0.0250
0.1000	Hybrid	0.8943	0.6903	0.0199	0.0195	0.0223
0.1000	LGB3h	0.8823	0.7084	0.0189	0.0184	0.0277
0.1000	LGB0h	0.8489	0.6927	0.0195	0.0192	0.0250

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Table B.6. L-Town: pooled-F1 objective; family balance and achieved false-positive rates (continued)

Cal. cap (%)	System	Macro	Worst	Cal. clean (%)	Eval. clean (%)	Eval. neg. (%)
0.2500	Hybrid	0.8943	0.6903	0.0199	0.0195	0.0223
0.2500	LGB3h	0.8823	0.7084	0.0189	0.0184	0.0277
0.2500	LGB0h	0.8489	0.6927	0.0195	0.0192	0.0250
0.5000	Hybrid	0.8943	0.6903	0.0199	0.0195	0.0223
0.5000	LGB3h	0.8823	0.7084	0.0189	0.0184	0.0277
0.5000	LGB0h	0.8489	0.6927	0.0195	0.0192	0.0250

All FPR columns are percentages. Cal. clean averages ten calibration fits; evaluation columns average 60 cells. Family scores appear in Table B.5.

B.1.4 L-Town: worst-family-F1 objective

Table B.7 reports pooled and individual-family F1 at every cap. Table B.8 reports family balance, achieved calibration clean FPR and achieved evaluation false-positive rates for the same selected rules.

Table B.7. L-Town: worst-family-F1 objective; detection at each calibration cap

Cal. cap (%)	System	Pooled	Random	Replay	Drift	Noise	Targeted
0.0000	Hybrid	0.7173	0.8877	0.3502	0.6185	0.4884	0.8833
0.0000	LGB3h	0.7869	0.9626	0.4673	0.6988	0.4924	0.9613
0.0000	LGB0h	0.6515	0.8087	0.4147	0.5642	0.3911	0.8098
0.0100	Hybrid	0.9069	0.9919	0.6574	0.8820	0.8608	0.9919
0.0100	LGB3h	0.8954	0.9953	0.6982	0.8554	0.8118	0.9962
0.0100	LGB0h	0.8660	0.9924	0.6852	0.7850	0.7272	0.9932
0.0250	Hybrid	0.9125	0.9938	0.6980	0.8946	0.9073	0.9941
0.0250	LGB3h	0.8968	0.9947	0.7093	0.8741	0.8526	0.9958
0.0250	LGB0h	0.8682	0.9927	0.6925	0.7963	0.7776	0.9935
0.0500	Hybrid	0.9010	0.9923	0.7307	0.8986	0.9265	0.9923
0.0500	LGB3h	0.8937	0.9944	0.7091	0.8795	0.8640	0.9954
0.0500	LGB0h	0.8651	0.9923	0.6910	0.8017	0.8005	0.9929
0.1000	Hybrid	0.8966	0.9920	0.7316	0.8989	0.9274	0.9919
0.1000	LGB3h	0.8929	0.9943	0.7085	0.8800	0.8653	0.9954
0.1000	LGB0h	0.8452	0.9901	0.6880	0.8118	0.8366	0.9902

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Table B.7. L-Town: worst-family-F1 objective; detection at each calibration cap (continued)

Cal. cap (%)	System	Pooled	Random	Replay	Drift	Noise	Targeted
0.2500	Hybrid	0.8966	0.9920	0.7316	0.8989	0.9274	0.9919
0.2500	LGB3h	0.8929	0.9943	0.7085	0.8800	0.8653	0.9954
0.2500	LGB0h	0.8281	0.9876	0.6861	0.8148	0.8434	0.9875
0.5000	Hybrid	0.8966	0.9920	0.7316	0.8989	0.9274	0.9919
0.5000	LGB3h	0.8929	0.9943	0.7085	0.8800	0.8653	0.9954
0.5000	LGB0h	0.8239	0.9868	0.6860	0.8149	0.8437	0.9866

F1 means over 60 model/source cells. Family balance and achieved false-positive rates appear in Table B.8.

Table B.8. L-Town: worst-family-F1 objective; family balance and achieved false-positive rates

Cal. cap (%)	System	Macro	Worst	Cal. clean (%)	Eval. clean (%)	Eval. neg. (%)
0.0000	Hybrid	0.6456	0.3502	0.0000	0.0000	0.0001
0.0000	LGB3h	0.7165	0.4542	0.0000	0.0000	0.0000
0.0000	LGB0h	0.5977	0.3692	0.0000	0.0000	0.0000
0.0100	Hybrid	0.8768	0.6574	0.0096	0.0094	0.0110
0.0100	LGB3h	0.8714	0.6982	0.0099	0.0100	0.0171
0.0100	LGB0h	0.8366	0.6833	0.0099	0.0098	0.0123
0.0250	Hybrid	0.8976	0.6980	0.0236	0.0230	0.0262
0.0250	LGB3h	0.8853	0.7093	0.0239	0.0230	0.0333
0.0250	LGB0h	0.8505	0.6925	0.0219	0.0213	0.0277
0.0500	Hybrid	0.9081	0.7307	0.0482	0.0468	0.0547
0.0500	LGB3h	0.8885	0.7091	0.0321	0.0309	0.0425
0.0500	LGB0h	0.8557	0.6910	0.0327	0.0317	0.0404
0.1000	Hybrid	0.9084	0.7316	0.0541	0.0528	0.0623
0.1000	LGB3h	0.8887	0.7085	0.0337	0.0325	0.0443
0.1000	LGB0h	0.8633	0.6880	0.0717	0.0699	0.0823
0.2500	Hybrid	0.9084	0.7316	0.0541	0.0528	0.0623
0.2500	LGB3h	0.8887	0.7085	0.0337	0.0325	0.0443
0.2500	LGB0h	0.8639	0.6861	0.1003	0.0982	0.1118

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Table B.8. L-Town: worst-family-F1 objective; family balance and achieved false-positive rates (continued)

Cal. cap (%)	System	Macro	Worst	Cal. clean (%)	Eval. clean (%)	Eval. neg. (%)
0.5000	Hybrid	0.9084	0.7316	0.0541	0.0528	0.0623
0.5000	LGB3h	0.8887	0.7085	0.0337	0.0325	0.0443
0.5000	LGB0h	0.8636	0.6860	0.1083	0.1057	0.1196

All FPR columns are percentages. Cal. clean averages ten calibration fits; evaluation columns average 60 cells. Family scores appear in Table B.7.

B.2 Frozen-rule component diagnostics

All eight variants use the original thresholds. Branch removals retain the original verifier and external replay veto on remaining specialist alarms. Verifier bypass retains the external replay veto; replay-veto removal retains the verifier, including its original candidate-region behavior. Removing both guards admits raw branch alarms. General's internal mixture routing is unchanged wherever General alarms contribute. All model scores were computed for these diagnostics, but the specialists-only alarm rule does not consume General's output. Removing a specialist's final alarm contribution does not remove its score channels from the shared router, feedback and verifier inputs. These diagnostics describe the contribution of alarm rules to the fitted system; they do not isolate target-group exclusion or reference fitting. The separate factorial study in Section 6.10 varies exclusion and IRLS with downstream refitting, while keeping whitening and median aggregation fixed.

B.2.1 Modena

Tables B.9 and B.10 report the same eight alarm-rule variants, separating detection scores from family balance and false-positive rates.

Table B.9. Modena: frozen-rule diagnostic detection scores

Alarm rule	Pooled	Random	Replay	Drift	Noise	Targeted
Full frozen rule	0.8669	0.9549	0.9283	0.8274	0.8439	0.9680
General alarms only	0.7200	0.8581	0.9279	0.4979	0.4491	0.8832
General + Drift alarms	0.8284	0.9547	0.9287	0.8251	0.5654	0.9681
General + Noise alarms	0.7914	0.8610	0.9282	0.5505	0.8389	0.8853
Drift + Noise alarms	0.6612	0.6236	0.0234	0.8152	0.8396	0.6252
Verifier bypassed	0.6737	0.9470	0.9247	0.8289	0.8522	0.9567
External replay veto disabled	0.8663	0.9549	0.9219	0.8274	0.8440	0.9680
Verifier and external replay veto disabled	0.6643	0.9470	0.7927	0.8289	0.8521	0.9568

Original models and thresholds are retained. Family balance and false-positive rates appear in Table B.10.

Table B.10. Modena: frozen-rule diagnostic balance and false-positive rates

Alarm rule	Macro F1	Worst F1	Clean FPR (%)	All-negative FPR (%)
Full frozen rule	0.9045	0.8167	0.0584	0.0667
General alarms only	0.7232	0.4384	0.0263	0.0290
General + Drift alarms	0.8484	0.5654	0.0517	0.0532
General + Noise alarms	0.8128	0.5505	0.0335	0.0432
Drift + Noise alarms	0.5854	0.0234	0.0355	0.0412
Verifier bypassed	0.9019	0.8233	0.4578	0.4454
External replay veto disabled	0.9033	0.8167	0.0585	0.0677
Verifier and external replay veto disabled	0.8755	0.7793	0.4603	0.4698

Both FPR columns are percentages. Macro and Worst average cellwise summaries. Individual-family F1 appears in Table B.9.

B.2.2 L-Town

Tables B.11 and B.12 report the same eight alarm-rule variants, separating detection scores from family balance and false-positive rates.

Table B.11. L-Town: frozen-rule diagnostic detection scores

Alarm rule	Pooled	Random	Replay	Drift	Noise	Targeted
Full frozen rule	0.8950	0.9920	0.7312	0.8990	0.9279	0.9918
General alarms only	0.8565	0.9920	0.7411	0.8149	0.7877	0.9918
General + Drift alarms	0.8683	0.9920	0.7416	0.8990	0.7886	0.9917
General + Noise alarms	0.8843	0.9920	0.7307	0.8158	0.9274	0.9918
Drift + Noise alarms	0.5409	0.0402	0.1103	0.8929	0.9208	0.0246
Verifier bypassed	0.7273	0.9757	0.7082	0.9084	0.9267	0.9753
External replay veto disabled	0.8946	0.9920	0.7281	0.8993	0.9304	0.9918
Verifier and external replay veto disabled	0.7136	0.9757	0.6518	0.9088	0.9294	0.9753

Original models and thresholds are retained. Family balance and false-positive rates appear in Table B.12.

Table B.12. L-Town: frozen-rule diagnostic balance and false-positive rates

Alarm rule	Macro F1	Worst F1	Clean FPR (%)	All-negative FPR (%)
Full frozen rule	0.9083	0.7312	0.0550	0.0651
General alarms only	0.8655	0.7408	0.0497	0.0548
General + Drift alarms	0.8826	0.7416	0.0547	0.0607
General + Noise alarms	0.8915	0.7305	0.0499	0.0592
Drift + Noise alarms	0.3978	0.0123	0.0058	0.0122
Verifier bypassed	0.8989	0.7082	0.3409	0.3358
External replay veto disabled	0.9083	0.7281	0.0557	0.0670
Verifier and external replay veto disabled	0.8882	0.6518	0.3610	0.3736

Both FPR columns are percentages. Macro and Worst average cellwise summaries. Individual-family F1 appears in Table B.11.

Appendix C

Supplementary reference-factorial results

This appendix gives all four calibration policies for the reference experiment in Section 6.10. The factors are target-group exclusion (E) and iterative residual reweighting (R); one enables the operation and zero disables it. The retained reference representation, normalization and downstream learning recipe are shared across conditions. Each condition selects its own thresholds on calibration data.

The primary policy maximizes pooled F1 under a calibration clean-FPR cap of 0.05%. The secondary policies maximize worst-family F1 at that cap and use both objectives at 0.50%. Evaluation means first average the three fitted-model seeds within each source and then average the six sources equally. Worst-family F1 is computed within each model-seed/source cell before averaging. Achieved calibration and evaluation false-positive rates are reported separately; the calibration cap does not guarantee the same evaluation rate.

C.1 Modena

Table C.1 reports pooled and family F1 for every condition and policy. Table C.2 gives family balance and the achieved false-positive rates for the same selections.

Table C.1. Modena: all factorial policies, detection. Conditional F1 means over three model seeds and six shared evaluation sources. All four conditions and four prespecified policies are shown; the primary policy is marked *.

Cal. cap (%)	Objective	E	R	Pooled	Random	Replay	Drift	Noise	Targeted
0.0500	Pooled*	0	0	0.8168	0.9142	0.6395	0.7934	0.7892	0.9283
0.0500	Pooled*	0	1	0.8752	0.9438	0.9425	0.8150	0.8507	0.9577
0.0500	Pooled*	1	0	0.8104	0.9038	0.6282	0.7873	0.7860	0.9279
0.0500	Pooled*	1	1	0.8709	0.9429	0.9335	0.8144	0.8334	0.9573
0.0500	Balanced	0	0	0.7984	0.8762	0.7196	0.7519	0.7180	0.8910
0.0500	Balanced	0	1	0.8717	0.9520	0.9379	0.8228	0.8582	0.9650
0.0500	Balanced	1	0	0.7916	0.8647	0.7151	0.7410	0.7128	0.8873
0.0500	Balanced	1	1	0.8693	0.9514	0.9286	0.8226	0.8414	0.9654
0.5000	Pooled	0	0	0.8168	0.9142	0.6395	0.7934	0.7892	0.9283
0.5000	Pooled	0	1	0.8752	0.9438	0.9425	0.8150	0.8507	0.9577
0.5000	Pooled	1	0	0.8104	0.9038	0.6282	0.7873	0.7860	0.9279
0.5000	Pooled	1	1	0.8709	0.9429	0.9335	0.8144	0.8334	0.9573
0.5000	Balanced	0	0	0.7984	0.8762	0.7196	0.7519	0.7180	0.8910
0.5000	Balanced	0	1	0.8649	0.9585	0.9305	0.8307	0.8621	0.9699
0.5000	Balanced	1	0	0.7916	0.8647	0.7151	0.7410	0.7128	0.8873
0.5000	Balanced	1	1	0.8501	0.9576	0.9149	0.8322	0.8466	0.9685

E denotes target-group exclusion and R denotes iterative residual reweighting; 1 is enabled and 0 is disabled. The shared reference representation and scales remain fixed. Pooled maximizes pooled F1; balanced maximizes worst-family F1. Family balance and achieved false-positive rates appear in Table C.2.

Table C.2. Modena: all factorial policies, balance and false positives. Conditional evaluation means over three model seeds and six shared sources; calibration clean FPR averages the three fitted models. All FPR columns and caps are percentages. The primary policy is marked *.

Cal. cap (%)	Objective	E	R	Macro	Worst	Cal. clean (%)	Eval. clean (%)	Eval. neg. (%)
0.0500	Pooled*	0	0	0.8129	0.6395	0.0173	0.0170	0.0744
0.0500	Pooled*	0	1	0.9020	0.8093	0.0334	0.0380	0.0422
0.0500	Pooled*	1	0	0.8066	0.6282	0.0184	0.0179	0.0807
0.0500	Pooled*	1	1	0.8963	0.8036	0.0351	0.0378	0.0428

Continued on next page

Table C.2. Modena: all factorial policies, balance and false positives (continued)

Cal. cap (%)	Objective	E	R	Macro	Worst	Cal. clean (%)	Eval. clean (%)	Eval. neg. (%)
0.0500	Balanced	0	0	0.7913	0.6935	0.0082	0.0073	0.0426
0.0500	Balanced	0	1	0.9072	0.8187	0.0474	0.0526	0.0594
0.0500	Balanced	1	0	0.7842	0.6863	0.0079	0.0068	0.0433
0.0500	Balanced	1	1	0.9019	0.8126	0.0478	0.0503	0.0578
0.5000	Pooled	0	0	0.8129	0.6395	0.0173	0.0170	0.0744
0.5000	Pooled	0	1	0.9020	0.8093	0.0334	0.0380	0.0422
0.5000	Pooled	1	0	0.8066	0.6282	0.0184	0.0179	0.0807
0.5000	Pooled	1	1	0.8963	0.8036	0.0351	0.0378	0.0428
0.5000	Balanced	0	0	0.7913	0.6935	0.0082	0.0073	0.0426
0.5000	Balanced	0	1	0.9103	0.8253	0.0630	0.0703	0.0791
0.5000	Balanced	1	0	0.7842	0.6863	0.0079	0.0068	0.0433
0.5000	Balanced	1	1	0.9040	0.8218	0.0864	0.0888	0.0996

Macro is family-macro F1; worst is the mean of each cell's minimum family F1. Clean FPR uses family-0 rows; negative FPR uses all negative endpoints. The calibration cap and achieved evaluation rates have different populations. Family detection results appear in Table C.1.

C.2 L-Town

Table C.3 reports pooled and family F1 for every condition and policy. Table C.4 gives family balance and the achieved false-positive rates for the same selections.

Table C.3. L-Town: all factorial policies, detection. Conditional F1 means over three model seeds and six shared evaluation sources. All four conditions and four prespecified policies are shown; the primary policy is marked *.

Cal. cap (%)	Objective	E	R	Pooled	Random	Replay	Drift	Noise	Targeted
0.0500	Pooled*	0	0	0.9011	0.9806	0.6801	0.8778	0.8921	0.9784
0.0500	Pooled*	0	1	0.9152	0.9941	0.6989	0.8903	0.9038	0.9945
0.0500	Pooled*	1	0	0.9017	0.9834	0.6878	0.8859	0.9016	0.9814
0.0500	Pooled*	1	1	0.9129	0.9938	0.6876	0.8900	0.9028	0.9942
0.0500	Balanced	0	0	0.8918	0.9795	0.7248	0.8906	0.9214	0.9774
0.0500	Balanced	0	1	0.9043	0.9936	0.7381	0.8994	0.9307	0.9938
0.0500	Balanced	1	0	0.8894	0.9762	0.7077	0.8923	0.9210	0.9749
0.0500	Balanced	1	1	0.9015	0.9924	0.7293	0.8984	0.9275	0.9922
0.5000	Pooled	0	0	0.9011	0.9806	0.6801	0.8778	0.8921	0.9784
0.5000	Pooled	0	1	0.9152	0.9941	0.6989	0.8903	0.9038	0.9945
0.5000	Pooled	1	0	0.9017	0.9834	0.6878	0.8859	0.9016	0.9814
0.5000	Pooled	1	1	0.9129	0.9938	0.6876	0.8900	0.9028	0.9942
0.5000	Balanced	0	0	0.8855	0.9781	0.7228	0.8924	0.9238	0.9765
0.5000	Balanced	0	1	0.9004	0.9935	0.7373	0.9007	0.9314	0.9934
0.5000	Balanced	1	0	0.8865	0.9751	0.7075	0.8919	0.9215	0.9734
0.5000	Balanced	1	1	0.8961	0.9919	0.7303	0.8987	0.9282	0.9918

E denotes target-group exclusion and R denotes iterative residual reweighting; 1 is enabled and 0 is disabled. The shared reference representation and scales remain fixed. Pooled maximizes pooled F1; balanced maximizes worst-family F1. Family balance and achieved false-positive rates appear in Table C.4.

Table C.4. L-Town: all factorial policies, balance and false positives. Conditional evaluation means over three model seeds and six shared sources; calibration clean FPR averages the three fitted models. All FPR columns and caps are percentages. The primary policy is marked *.

Cal. cap (%)	Objective	E	R	Macro	Worst	Cal. clean (%)	Eval. clean (%)	Eval. neg. (%)
0.0500	Pooled*	0	0	0.8818	0.6801	0.0174	0.0177	0.0259
0.0500	Pooled*	0	1	0.8963	0.6989	0.0176	0.0179	0.0195
0.0500	Pooled*	1	0	0.8880	0.6878	0.0233	0.0234	0.0331
0.0500	Pooled*	1	1	0.8937	0.6876	0.0193	0.0191	0.0215
0.0500	Balanced	0	0	0.8987	0.7248	0.0477	0.0459	0.0625
0.0500	Balanced	0	1	0.9111	0.7381	0.0474	0.0453	0.0515
0.0500	Balanced	1	0	0.8944	0.7077	0.0459	0.0449	0.0655
0.0500	Balanced	1	1	0.9080	0.7293	0.0479	0.0462	0.0537
0.5000	Pooled	0	0	0.8818	0.6801	0.0174	0.0177	0.0259
0.5000	Pooled	0	1	0.8963	0.6989	0.0176	0.0179	0.0195
0.5000	Pooled	1	0	0.8880	0.6878	0.0233	0.0234	0.0331
0.5000	Pooled	1	1	0.8937	0.6876	0.0193	0.0191	0.0215
0.5000	Balanced	0	0	0.8987	0.7228	0.0571	0.0547	0.0746
0.5000	Balanced	0	1	0.9113	0.7373	0.0527	0.0506	0.0584
0.5000	Balanced	1	0	0.8939	0.7075	0.0495	0.0482	0.0705
0.5000	Balanced	1	1	0.9082	0.7303	0.0552	0.0535	0.0627

Macro is family-macro F1; worst is the mean of each cell's minimum family F1. Clean FPR uses family-0 rows; negative FPR uses all negative endpoints. The calibration cap and achieved evaluation rates have different populations. Family detection results appear in Table C.3.

Bibliography

[1] Riccardo Taormina, Stefano Galelli, Nils Ole Tippenhauer, Elad Salomons, Avi Ostfeld, Demetrios G. Eliades, et al. Battle of the attack detection algorithms: Disclosing cyber attacks on water distribution networks. *Journal of Water Resources Planning and Management*, 144(8):04018048, 2018. doi: 10.1061/(ASCE)WR.1943-5452.0000969.

[2] Katherine A. Klise, David Hart, Dylan M. Moriarty, Michael L. Bynum, Regan Murray, Jonathan Burkhardt, and Terra Haxton. Water network tool for resilience (WNTR) user manual. Technical Report SAND2017-8883R; EPA/600/R-17/264, Sandia National Laboratories and U.S. Environmental Protection Agency, August 2017. URL `https://www.osti.gov/biblio/1376816`.

[3] Katherine A. Klise, Michael Bynum, Dylan Moriarty, and Regan Murray. A software framework for assessing the resilience of drinking water systems to disasters with an example earthquake case study. *Environmental Modelling & Software*, 95:420–431, 2017. doi: 10.1016/j.envsoft.2017.06.022.

[4] Lewis A. Rossman, Hyoungmin Woo, Michael Tryby, Feng Shang, Robert Janke, and Terranna Haxton. EPANET 2.2 user manual. Technical Report EPA/600/R-20/133, U.S. Environmental Protection Agency, 2020. URL `https://cfpub.epa.gov/si/si_public_record_Report.cfm?LAB=CESER&dirEntryID=348882`.

[5] Riccardo Taormina, Stefano Galelli, Hunter C. Douglas, Nils Ole Tippenhauer, Elad Salomons, and Avi Ostfeld. A toolbox for assessing the impacts of cyber-physical attacks on water distribution systems. *Environmental Modelling & Software*, 112:46–51, 2019. doi: 10.1016/j.envsoft.2018.11.008.

[6] Aditya P. Mathur and Nils Ole Tippenhauer. SWaT: a water treatment testbed for research and training on ICS security. In *2016 International Workshop on Cyber-physical Systems for Smart Water Networks (CySWater)*, pages 31–36. IEEE, 2016. doi: 10.1109/CySWater.2016.7469060.

[7] Jonathan Goh, Sridhar Adepu, Khurum Nazir Junejo, and Aditya Mathur. A dataset to support research in the design of secure water treatment systems. In Grigore Havarneanu, Roberto Setola, Hypatia Nassopoulos, and Stephen Wolthusen, editors, *Critical Information Infrastructures Security*, volume 10242 of *Lecture Notes in Computer Science*, pages 88–99. Springer, Cham, 2017. doi: 10.1007/978-3-319-71368-7_8.

[8] Chuadhry Mujeeb Ahmed, Venkata Reddy Palleti, and Aditya P. Mathur. WADI: a water distribution testbed for research in the design of secure cyber physical systems. In *Proceedings of the 3rd International Workshop on Cyber-Physical Systems for Smart Water Networks*, pages 25–28. Association for Computing Machinery, 2017. doi: 10.1145/3055366.3055375.

[9] Stelios G. Vrachimis, Demetrios G. Eliades, Riccardo Taormina, Zoran Kapelan, Avi Ostfeld, Shuming Liu, Marios Kyriakou, Pavlos Pavlou, Mengning Qiu, and Marios M. Polycarpou. Battle of the leakage detection and isolation methods. *Journal of Water Resources Planning and Management*, 148(12):04022068, 2022. doi: 10.1061/(ASCE)WR.1943-5452.0001601.

[10] Cristiana Bragalli, Claudia D'Ambrosio, Jon Lee, Andrea Lodi, and Paolo Toth. On the optimal design of water distribution networks: a practical MINLP approach. *Optimization and Engineering*, 13(2):219–246, 2012. doi: 10.1007/s11081-011-9141-7.

[11] Riccardo Taormina and Stefano Galelli. Deep-learning approach to the detection and localization of cyber-physical attacks on water distribution systems. *Journal of Water Resources Planning and Management*, 144(10):04018065, 2018. doi: 10.1061/(ASCE)WR.1943-5452.0000983.

[12] Mashor Housh and Ziv Ohar. Model-based approach for cyber-physical attack detection in water distribution systems. *Water Research*, 139:132–143, 2018. doi: 10.1016/j.watres.2018.03.039.

[13] Sarin E. Chandy, Amin Rasekh, Zachary A. Barker, and M. Ehsan Shafiee. Cyberattack detection using deep generative models with variational inference. *Journal of Water Resources Planning and Management*, 145(2):04018093, 2019. doi: 10.1061/(ASCE)WR.1943-5452.0001007.

[14] Faegheh Moazeni and Javad Khazaei. Detection of random false data injection cyberattacks in smart water systems using optimized deep neural networks. *Energies*, 15(13):4832, 2022. doi: 10.3390/en15134832.

[15] Md Nazmul Kabir Sikder, Minh B. T. Nguyen, E. Donald Elliott, and Feras A. Batarseh. Deep H2O: Cyber attacks detection in water distribution systems using deep learning. *Journal of Water Process Engineering*, 52:103568, 2023. doi: 10.1016/j.jwpe.2023.103568.

[16] William L. Hamilton, Rex Ying, and Jure Leskovec. Inductive representation learning on large graphs. In *Advances in Neural Information Processing Systems*, volume 30, 2017. URL `https://proceedings.neurips.cc/paper/2017/hash/5dd9db5e033da9c6fb5ba83c7a7ebea9-Abstract.html`.

[17] Lu Xing and Lina Sela. Graph neural networks for state estimation in water distribution systems: Application of supervised and semisupervised learning. *Journal of Water Resources Planning and Management*, 148(5):04022018, 2022. doi: 10.1061/(ASCE)WR.1943-5452.0001550.

[18] Huy Truong, Andrés Tello, Alexander Lazovik, and Victoria Degeler. Graph neural networks for pressure estimation in water distribution systems. *Water Resources Research*, 60(7):e2023WR036741, 2024. doi: 10.1029/2023WR036741.

[19] Ailin Deng and Bryan Hooi. Graph neural network-based anomaly detection in multivariate time series. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 4027–4035, 2021. doi: 10.1609/aaai.v35i5.16523.

[20] Kyunghyun Cho, Bart van Merriënboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. Learning phrase representations using RNN encoder–decoder for statistical machine translation. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing*, pages 1724–1734. Association for Computational Linguistics, 2014. doi: 10.3115/v1/D14-1179. URL `https://aclanthology.org/D14-1179/`.

[21] Shreshth Tuli, Giuliano Casale, and Nicholas R. Jennings. TranAD: Deep transformer networks for anomaly detection in multivariate time series data. *Proceedings of the VLDB Endowment*, 15(6):1201–1214, 2022. doi: 10.14778/3514061.3514067.

[22] Robert A. Jacobs, Michael I. Jordan, Steven J. Nowlan, and Geoffrey E. Hinton. Adaptive mixtures of local experts. *Neural Computation*, 3(1):79–87, 1991. doi: 10.1162/neco.1991.3.1.79.

[23] Michael I. Jordan and Robert A. Jacobs. Hierarchical mixtures of experts and the EM algorithm. *Neural Computation*, 6(2):181–214, 1994. doi: 10.1162/neco.1994.6.2.181.

[24] Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarz, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. arXiv preprint arXiv:1701.06538, 2017. URL `https://arxiv.org/abs/1701.06538`.

[25] Xiaoyu Huang, Weidong Chen, Bo Hu, and Zhendong Mao. Graph mixture of experts and memory-augmented routers for multivariate time series anomaly detection. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 17476–17484, 2025. doi: 10.1609/aaai.v39i16.33921.

[26] Joshua Batson and Loic Royer. Noise2Self: Blind denoising by self-supervision. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 524–533, 2019. URL `https://proceedings.mlr.press/v97/batson19a.html`.

[27] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, volume 30, 2017. URL `https://proceedings.neurips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html`.

[28] Takaya Saito and Marc Rehmsmeier. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE*, 10(3):e0118432, 2015. doi: 10.1371/journal.pone.0118432.

[29] Gavin C. Cawley and Nicola L. C. Talbot. On over-fitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research*, 11(70):2079–2107, 2010. URL `https://www.jmlr.org/papers/v11/cawley10a.html`.

[30] Siwon Kim, Kukjin Choi, Hyun-Soo Choi, Byunghan Lee, and Sungroh Yoon. Towards a rigorous evaluation of time-series anomaly detection. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, pages 7194–7201, 2022. doi: 10.1609/aaai.v36i7.20680.